

# The Baker–Campbell–Hausdorff Formula

A Unified Treatment with Complete Proofs,  
All-Order Expansions, and Analytic Qualifications

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## Abstract

This article consolidates three independent proof-based treatments of the mathematical content of the Wikipedia article on the Baker–Campbell–Hausdorff formula into one self-contained account, keeping the strongest argument available for each result. The formal logarithm of  $e^Xe^Y$  is first computed coefficient by coefficient in a completed free associative algebra, giving a finite nonrecursive formula for every associative word coefficient. A direct proof of the Dynkin–Specht–Wever identity establishes Friedrichs’ characterization of primitive elements and yields Dynkin’s complete commutator expansion together with a finite formula for every bidegree. An independent differential route proves the Campbell conjugation identity, the differential of the exponential, the Poincaré integral formula, complete Bernoulli recurrences and partial fractions, an all-orders expansion in the number of occurrences of one variable with an explicit bivariate kernel for its quadratic part, and a finite-at-each-order analytic expansion around a regular base point. We prove quantitative convergence and remainder bounds, the trace identity with its branch restriction, and a complete classification of the logarithms in the article’s nonconvergence example. Lie-group consequences include exponential coordinates, the local Lie correspondence, integration from simply connected groups, and the polynomial group law of nilpotent algebras. Central and eigenvector commutator relations are resummed globally. For the Zassenhaus factorization we prove existence and uniqueness, three recursive constructions, a finite nonrecursive weighted-tree formula for every exponent, a globally convergent ordered-simplex expansion, and a quantitative convergence criterion. Trotter, symmetric, and recursively higher-order product formulas follow with complete logarithmic error series. The infinitesimal formulas are developed into all-order coframe, pullback-metric, and volume-density expansions, with the corrections needed in the source’s geometric paragraph. Finally the Weyl and displacement-operator identities are proved in the Schrödinger and Bargmann–Fock representations, with operator domains made explicit and a counterexample showing why a dense-domain commutator alone does not suffice. Exact rational computations through degree twelve accompany the proofs.

**Keywords.** Free Lie algebra; noncommutative power series; Dynkin formula; Bernoulli numbers; Zassenhaus factorization; Trotter and Suzuki product formulas; exponential coordinates; Killing form; Weyl relations; displacement operators.

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# 1 Scope, conventions, and the three meanings of BCH

The question that organizes this article is elementary to state: given two quantities  $X, Y$  whose products need not commute, can the product  $e^X e^Y$  be written as a single exponential, and what exactly is its exponent? The central answer is

$$e^X e^Y = e^{Z(X,Y)}, \quad Z(X,Y) = X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}[X, [X, Y]] + \frac{1}{12}[Y, [Y, X]] + \cdots \quad (1.1)$$

The substantive assertion is not the existence of a formal logarithm, which is immediate in a completed free associative algebra. It is that every homogeneous part of that logarithm is a *Lie polynomial*, a linear combination of nested commutators, and that its coefficients can be specified completely.

The equation  $e^X e^Y = e^Z$  is not, by itself, a definition of a unique logarithm. The exponential can fail to be injective, and a product of exponentials need not be an exponential *in the specified Lie algebra*. What is universal is the formal series

$$Z(X, Y) = \text{BCH}(X, Y) = \log(e^X e^Y) = \sum_{n \geq 1} Z_n(X, Y), \quad (1.2)$$

where  $Z_n$  has total degree  $n$ . A principal matrix logarithm, a continued branch of a logarithm, and a convergent BCH series are related but distinct objects.

## 1.1 Coverage target and what is not claimed

The coverage target is the English Wikipedia article *Baker–Campbell–Hausdorff formula* in the revision with identifier 1368909113, last edited on 11 August 2026 and consulted on 16 September 2026 [1]. Every displayed mathematical identity and every mathematical assertion stated in its main text or its mathematical notes is addressed in this article; section C gives a claim-by-claim concordance. The article is used as a list of questions, not as a substitute for proofs. Historical priority statements and application surveys are bibliographical claims rather than theorems. A theorem merely named in a link, such as the Stone–von Neumann uniqueness theorem or the Golden–Thompson inequality, is not thereby stated on the page and is not part of the coverage target; the actual quantum-mechanical exponential formulas on the page are proved directly in section 13.

Several assertions on the page require correction or explicit hypotheses rather than proof as written. The coframe matrix in exponential coordinates is  $W = \phi(\text{ad}_X)$  and not the adjoint matrix  $\text{ad}_X$ , which is always singular in positive dimension; a Killing form need not be a metric; an arbitrary pullback can be degenerate; the trace identity concerns the BCH branch of the logarithm; and a commutator relation on a dense domain does not by itself imply exponentiated Weyl relations. In each case we prove the valid statement and exhibit a counterexample to the unqualified one.

The present article combines three earlier treatments of the same coverage target that were prepared independently. Where those treatments proved the same result by different routes, we keep the route that is most complete or most elementary; where one contained a construction absent from the others, that construction is included here. All proofs are written out in full.

## 1.2 Three distinct settings

Conflating the following settings is the source of most apparent paradoxes surrounding the formula.

### Formal algebra.

$X$  and  $Y$  have positive degree in a graded algebra over a field  $\mathbb{K}$  of characteristic zero. Every

identity is an identity of formal series, and every fixed-degree coefficient is a finite sum. No numerical convergence is intended or asserted.

### Local analysis.

$X, Y$  belong to a real or complex unital Banach algebra  $\mathcal{A}$  with a submultiplicative norm, or to a finite-dimensional Lie algebra, and are sufficiently small. The series converges and specifies the logarithm branch issuing from zero. Finite matrices are the principal example.

### Global identities under special relations.

Commutation relations may allow the series to terminate or to be replaced by a closed expression. An independent differential argument can then prove an exponential identity far beyond the convergence domain of its Taylor series. Such an identity need not identify a principal logarithm, and for unbounded operators domains cannot be suppressed.

A matrix may have logarithms even though its BCH series diverges; a closed expression may continue the local BCH logarithm without making the original series convergent; and a logarithm need not belong to the Lie algebra generated by its two inputs. Each of these phenomena is exhibited by an explicit example below.

## 1.3 Conventions

Throughout,  $[A, B] = AB - BA$  in an associative algebra,  $\text{ad}_A B = [A, B]$ , and compositions of adjoint maps are read from right to left. Associativity alone gives antisymmetry and the Jacobi identity: expanding  $[A, [B, C]] + [B, [C, A]] + [C, [A, B]]$  produces each of the six ordered products  $ABC, ACB, BAC, BCA, CAB, CBA$  twice with opposite signs. Two consequences used constantly are the derivation rule and its adjoint form,

$$[A, [B, C]] = [[A, B], C] + [B, [A, C]], \quad [\text{ad}_A, \text{ad}_B] = \text{ad}_{[A, B]}. \quad (1.3)$$

For a nonempty word  $w = a_1 \cdots a_n$  in the letters  $X, Y$  its *right-nested bracket* is

$$R(a_1 \cdots a_n) = [a_1, [a_2, \dots, [a_{n-1}, a_n] \dots]], \quad R(a) = a, \quad R(1) = 0. \quad (1.4)$$

Thus  $R(XXY) = [X, [X, Y]] = \text{ad}_X^2 Y$  and  $R(YYX) = \text{ad}_Y^2 X$ . A symbol such as  $X^r Y^s$  inside  $R(\cdot)$  describes a word with  $r$  letters  $X$  followed by  $s$  letters  $Y$ , not a commutator of associative powers; empty runs are omitted. We write  $Z_n$  for the part of  $Z$  of total degree  $n$ , and  $Z_{p,q}$  for the part of degree  $p$  in  $X$  and  $q$  in  $Y$ . For an associative polynomial  $P$  and a word  $w$ ,  $[w]P$  denotes the coefficient of  $w$  in  $P$ ; it is never a commutator. The notation  $O_{\deg}(m)$  denotes terms of total degree at least  $m$ , and  $O(Y^m)$  denotes terms containing at least  $m$  occurrences of  $Y$ , with no claim of uniformity as  $X$  varies.

The scalar functions that organize the entire subject are

$$\begin{aligned} \phi(z) &= \frac{1 - e^{-z}}{z}, & \phi(-z) &= \frac{e^z - 1}{z}, & \beta(z) &= \frac{z}{1 - e^{-z}}, \\ \gamma(z) &= \beta(-z) = \frac{z}{e^z - 1}, & \psi(x) &= \frac{x \log x}{x - 1}. \end{aligned} \quad (1.5)$$

The first two are entire, with value 1 at zero. The functions  $\beta$  and  $\gamma$  are meromorphic with a removable singularity at zero and simple poles at the nonzero points of  $2\pi i\mathbb{Z}$ . The function  $\psi$  is used near  $x = 1$ , with the logarithm branch analytic there and  $\psi(1) = 1$ . An expression such as  $\phi(M)$  or  $\beta(\text{ad}_X)$  always denotes functional calculus, formal or analytic, and never presupposes that  $M$  or  $\text{ad}_X$  is invertible.

Bernoulli numbers appear in the convention  $B_1^+ = +\frac{1}{2}$  of the source page; we write  $b_n^+ = B_n^+/n!$  for the Taylor coefficients of  $\beta$  and  $b_n^- = (-1)^n b_n^+ = B_n^-/n!$  for those of  $\gamma$ .

## 1.4 A guide to the results

Sections 2–4 establish the algebraic theorem and every coefficient: a finite formula for each associative word coefficient, Dynkin’s commutator formula, a finite formula for each bidegree, and the complete polynomial through degree six with a finite certificate. Sections 5–6 give the differential constructions: the Campbell identity, the differential of the exponential, Bernoulli coefficients, the Poincaré integral, every order in the number of occurrences of  $Y$ , and two total-degree recursions. Section 7 proves quantitative convergence estimates, the trace identity, and the nonconvergence example. Section 8 treats Lie groups, and section 9 the exactly summable cases. Sections 10 and 11 treat the Zassenhaus factorization and the product formulas. Sections 12 and 13 treat exponential coordinates and the quantum-mechanical identities. Section 14 describes the exact computations. The appendices contain the ordered Poincaré–Birkhoff–Witt theorem, coefficient certificates and further complete degrees, and the coverage concordance.

A reader interested chiefly in coefficients can use the nonrecursive formulas (2.5), (4.1), (6.4), (6.3), and (10.16) directly.

## 2 Formal exponentials and the complete word expansion

### 2.1 The degree completion

Let  $V$  be the vector space over  $\mathbb{K}$  with basis  $X, Y$ , and put

$$T(V) = \bigoplus_{n \geq 0} V^{\otimes n} = \mathbb{K}\langle X, Y \rangle, \quad \widehat{T}(V) = \prod_{n \geq 0} V^{\otimes n} = \mathbb{K}\langle\langle X, Y \rangle\rangle.$$

The space  $V^{\otimes n} = T_n$  has as basis the  $2^n$  words of length  $n$ , the empty word being the unit 1. Multiplication concatenates words and is extended to  $\widehat{T}$  by the Cauchy product: the degree- $n$  part of a product  $AB$  is  $\sum_{p+q=n} A_p B_q$ , a finite sum. The filtration  $F^m \widehat{T} = \prod_{n \geq m} V^{\otimes n}$  satisfies  $F^m F^n \subseteq F^{m+n}$ . A sum  $\sum_j A_j$  of elements of  $\widehat{T}$  *converges formally* if for every degree  $n$  only finitely many  $A_j$  have a nonzero degree- $n$  part; its value is then defined degree by degree. No norm is involved. In particular, if  $A \in F^1 \widehat{T}$ , then  $A^k \in F^k \widehat{T}$ , so for any scalar sequence  $(c_k)$  the series  $\sum_{k \geq 0} c_k A^k$  converges formally. Consequently substitution of positive-degree series into any one-variable formal power series is well defined, and

$$\exp A = \sum_{k \geq 0} \frac{A^k}{k!}, \quad \log(1 + U) = \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} U^k \quad (A, U \in F^1 \widehat{T}) \quad (2.1)$$

are well-defined elements of  $\widehat{T}$ .

**Lemma 2.1** (Formal exponential and logarithm). *The maps  $\exp : F^1 \widehat{T} \rightarrow 1 + F^1 \widehat{T}$  and  $\log : 1 + F^1 \widehat{T} \rightarrow F^1 \widehat{T}$  of (2.1) are mutually inverse bijections. If  $A, B \in F^1 \widehat{T}$  commute, then  $e^{A+B} = e^A e^B$ . If  $g, h \in 1 + F^1 \widehat{T}$  commute, then  $\log(gh) = \log g + \log h$  and  $\log(g^{-1}) = -\log g$ .*

*Proof.* We first prove the scalar identities  $\log(e^t) = t$  and  $e^{\log(1+t)} = 1 + t$  in  $\mathbb{Q}[[t]]$ . Put  $E(t) = \sum_{k \geq 0} t^k/k!$  and  $L(u) = \sum_{k \geq 1} (-1)^{k-1} u^k/k$ . Formal differentiation gives  $E' = E$  and  $L'(u) = \sum_{k \geq 1} (-1)^{k-1} u^{k-1} = (1+u)^{-1}$ . The composition  $L(E(t) - 1)$  has derivative  $E'(t)/(1 + E(t) - 1) = E(t)/E(t) = 1$  and vanishes at  $t = 0$ ; a formal power series is determined by its derivative and constant term, so  $L(E(t) - 1) = t$ . Likewise  $F(u) = E(L(u))$  satisfies  $F'(u) = E(L(u))L'(u) = F(u)/(1+u)$ ; the series  $1+u$  satisfies the same first-order equation with the same constant term 1, and a solution of  $(1+u)F' = F$  is determined by its constant term because comparing the coefficient of  $u^n$  gives  $(n+1)F_{n+1} = (1-n)F_n$  recursively. Hence  $E(L(u)) = 1 + u$ .

Now let  $A \in F^1\hat{T}$ . The powers of the single element  $A$  commute with each other, so the subalgebra  $\mathbb{K}[[A]]$  of  $\hat{T}$  generated by  $A$  is commutative, and the substitution  $t \mapsto A$  defines a ring homomorphism from  $\mathbb{Q}[[t]]$  into  $\hat{T}$ , well defined because each coefficient of the image is a finite sum. Applying this homomorphism to  $L(E(t) - 1) = t$  gives  $\log(\exp A) = A$ ; applying the analogous homomorphism  $u \mapsto U$  for  $U \in F^1\hat{T}$  to  $E(L(u)) = 1 + u$  gives  $\exp(\log(1 + U)) = 1 + U$ . This proves that the two maps are mutually inverse bijections.

If  $A$  and  $B$  commute, the binomial theorem holds for  $(A + B)^n$ , and

$$\sum_{n \geq 0} \frac{(A + B)^n}{n!} = \sum_{n \geq 0} \sum_{r+s=n} \frac{A^r B^s}{r!s!} = \sum_{r,s \geq 0} \frac{A^r B^s}{r!s!} = e^A e^B,$$

all rearrangements being finite in each degree. If  $g, h \in 1 + F^1\hat{T}$  commute, then  $\log g$  and  $\log h$  commute, since each is a series in one of the commuting elements  $g - 1$ ,  $h - 1$ ; hence  $e^{\log g + \log h} = e^{\log g} e^{\log h} = gh$ , and applying  $\log$  gives  $\log(gh) = \log g + \log h$ . Taking  $h = g^{-1}$ , which exists as the geometric series  $\sum_{k \geq 0} (1 - g)^k$  and commutes with  $g$ , gives  $\log(g^{-1}) = -\log g$ . No commutativity of unrelated elements was used.  $\square$

In a Banach algebra the exponential series converges absolutely for every  $A$ , since  $\|A^n/n!\| \leq \|A\|^n/n!$ , and the logarithm series converges absolutely whenever  $\|U\| < 1$ . The scalar identities of the proof then hold in the closed commutative subalgebra generated by  $A$  or  $U$ , by the same substitution argument with absolutely convergent series, so the analytic statements corresponding to [theorem 2.1](#) hold on those domains.

## 2.2 Definition of the BCH series and the block expansion

Define

$$Z(X, Y) = \text{BCH}(X, Y) = \log(e^X e^Y) = \sum_{n \geq 1} Z_n(X, Y), \quad Z_n \in V^{\otimes n}. \quad (2.2)$$

By [theorem 2.1](#),  $Z$  is the unique element of  $F^1\hat{T}$  with  $e^Z = e^X e^Y$ ; thus (1.1) already holds as an associative formal identity. Expanding the two exponentials first and then the logarithm gives

$$Z(X, Y) = \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} \sum_{\substack{r_i, s_i \geq 0 \\ r_i + s_i > 0 \ (1 \leq i \leq k)}} \frac{X^{r_1} Y^{s_1} \dots X^{r_k} Y^{s_k}}{\prod_{i=1}^k r_i! s_i!}. \quad (2.3)$$

Indeed  $e^X e^Y - 1 = \sum_{r+s>0} X^r Y^s / (r!s!)$ , and the  $k$ th power of this sum is the sum over all  $k$ -tuples of nonempty blocks  $X^{r_i} Y^{s_i}$ . This proves the associative expansion displayed on the source page. At this stage the inner sum is noncommutative multiplication, not a sum over independent commutators, and nothing has yet been proved about the Lie nature of the homogeneous parts.

## 2.3 A finite nonrecursive formula for every word

For a nonempty word  $w$  define its *block weight*

$$a(w) = \begin{cases} \frac{1}{r!s!}, & w = X^r Y^s, \quad r, s \geq 0, \quad r + s > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (2.4)$$

Thus  $a(w) = 0$  precisely when  $w$  contains the adjacent pattern  $YX$ ; pure powers of one letter are permitted. A *cut* of  $w$  into  $k$  pieces is a factorization  $w = w_1 \dots w_k$  into consecutive nonempty subwords.

**Theorem 2.2** (Every associative BCH coefficient). *For every nonempty word  $w = a_1 \cdots a_n$ ,*

$$\begin{aligned} c(w) &:= [w]Z(X, Y) = \sum_{k=1}^n \frac{(-1)^{k-1}}{k} \sum_{\substack{w=w_1 \cdots w_k \\ |w_i| > 0}} \prod_{i=1}^k a(w_i) \\ &= \sum_{k=1}^n \frac{(-1)^{k-1}}{k} \sum_{0=i_0 < i_1 < \cdots < i_k=n} \prod_{j=1}^k a(a_{i_{j-1}+1} \cdots a_{i_j}). \end{aligned} \quad (2.5)$$

Consequently

$$Z_n = \sum_{w \in \{X, Y\}^n} c(w) w = \sum_{k=1}^n \frac{(-1)^{k-1}}{k} \sum_{\substack{r_i + s_i > 0 \\ \sum_i (r_i + s_i) = n}} \frac{X^{r_1} Y^{s_1} \cdots X^{r_k} Y^{s_k}}{\prod_i r_i! s_i!}. \quad (2.6)$$

Both formulas are finite and nonrecursive, and every coefficient is rational. For a word of length  $n$  there are exactly  $2^{n-1}$  cut patterns before zero terms are discarded.

*Proof.* Write  $U = e^X e^Y - 1 \in F^1 \hat{T}$ . Its coefficient on a nonempty word  $v$  is  $a(v)$ : the only words with nonzero coefficient in  $e^X e^Y$  are the products  $X^r Y^s$ , which arise once each, with coefficient  $1/(r!s!)$ . The coefficient of  $w$  in  $U^k$  is obtained by expanding the product of  $k$  copies of  $U$ : a word appears in the product exactly when it is the concatenation  $w_1 \cdots w_k$  of one word from each factor, and its coefficient is the sum over all such factorizations of the product of the factor coefficients. Since every word in  $U$  is nonempty, these factorizations are precisely the cuts of  $w$  into  $k$  nonempty pieces, and the coefficient is  $\sum_{w=w_1 \cdots w_k} \prod_i a(w_i)$ . Because  $w$  has length  $n$ , no cut into more than  $n$  nonempty pieces exists, so  $U^k$  has zero coefficient on  $w$  for  $k > n$ . Substituting into  $Z = \sum_{k \geq 1} (-1)^{k-1} U^k / k$  gives (2.5); the second form just parametrizes a cut by the positions  $i_1 < \cdots < i_{k-1}$  at which it breaks. Choosing a subset of the  $n-1$  gaps between consecutive letters is the same as choosing a cut pattern, which gives the count  $2^{n-1}$ . Expanding each permitted block as  $X^{r_i} Y^{s_i}$  in (2.5) and summing over words of length  $n$  gives (2.6), which is the degree- $n$  part of (2.3).  $\square$

**Example 2.3.** For  $w = XY$  the one-piece contribution is  $a(XY) = 1$  and the two-piece contribution is  $-\frac{1}{2}a(X)a(Y) = -\frac{1}{2}$ , so  $c(XY) = \frac{1}{2}$ . For  $w = YX$  the one-piece contribution is  $a(YX) = 0$ , so  $c(YX) = -\frac{1}{2}$ . For  $w = XXY$  the one-piece contribution is  $a(XXY) = \frac{1}{2}$ ; the two cuts  $X | XY$  and  $XX | Y$  contribute  $-\frac{1}{2}(1 \cdot 1 + \frac{1}{2} \cdot 1) = -\frac{3}{4}$ ; the three-piece cut contributes  $\frac{1}{3}$ . Their sum is  $\frac{1}{2} - \frac{3}{4} + \frac{1}{3} = \frac{1}{12}$ . For  $w = XYX$  the one-piece contribution is zero, the two cuts  $X | YX$  and  $XY | X$  contribute  $-\frac{1}{2}(0 + 1) = -\frac{1}{2}$ , and the three-piece cut contributes  $\frac{1}{3}$ , giving  $c(XYX) = -\frac{1}{6}$ . These small computations already show why the coefficients are rational and why cancellation between different cut patterns matters.

## 2.4 Any finite number of exponential factors

The same construction works without change for several factors.

**Corollary 2.4** (Several exponentials). *Let  $A_1, \dots, A_m$  be letters. For a  $k \times m$  array  $\rho = (r_{ji})$  of nonnegative integers in which every row has positive sum, let  $w(\rho)$  be the word obtained by concatenating the row words  $A_1^{r_{j1}} \cdots A_m^{r_{jm}}$  for  $j = 1, \dots, k$ , and put  $|\rho| = \sum_{j,i} r_{ji}$ . Then*

$$\log(e^{A_1} \cdots e^{A_m}) = \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} \sum_{\rho} \frac{w(\rho)}{\prod_{j,i} r_{ji}!}, \quad (2.7)$$

and the coefficient of a given word  $w$  is the finite sum over cuts of  $w$  into  $k$  nonempty pieces of the product of the block weights  $a_m(w_i)$ , where  $a_m(A_1^{r_1} \cdots A_m^{r_m}) = (r_1! \cdots r_m!)^{-1}$  and  $a_m$  vanishes

on every other word. Equivalently, with  $t$  a formal scalar,

$$\log\left(\prod_{i=1}^m e^{tA_i}\right) = \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} \sum_{\rho} \frac{t^{|\rho|}}{\prod_{j,i} r_{ji}!} w(\rho). \quad (2.8)$$

*Proof.* Repeat the proof of [theorem 2.2](#) with  $U = e^{A_1} \cdots e^{A_m} - 1$ , whose nonempty words are exactly the products  $A_1^{r_1} \cdots A_m^{r_m}$  with coefficient  $(r_1! \cdots r_m!)^{-1}$ . The scalar  $t$  simply counts total degree.  $\square$

If the factors are of the form  $e^{\alpha_j X}$  or  $e^{\beta_j Y}$  with the letters  $X, Y$  repeated in different positions, one first uses a distinct letter for each factor and then substitutes; the resulting coefficients are finite polynomials in the scalar weights  $\alpha_j, \beta_j$ . After [section 3](#) it will follow that each homogeneous part of (2.7) is a Lie polynomial and that each word  $w(\rho)$  may be replaced by  $R(w(\rho))/|\rho|$ . This observation supplies every order, not only the leading error, for the splitting formulas of [section 11](#).

### 3 Hopf structure, primitive elements, and why the logarithm is a Lie series

We prove the primitive-element characterization used by the source page rather than treating it as a black box. The proof simultaneously constructs the projection that produces Dynkin's formula.

#### 3.1 The coproduct must take values in a completion

The relevant target of the coproduct on infinite series is the *completed* tensor product

$$\widehat{T} \widehat{\otimes} \widehat{T} = \prod_{n \geq 0} \bigoplus_{p+q=n} V^{\otimes p} \otimes V^{\otimes q},$$

with multiplication and infinite sums interpreted by total degree exactly as in  $\widehat{T}$ . The ordinary algebraic tensor product  $\widehat{T} \otimes \widehat{T}$  is generally too small: we show in [section A](#) that already the coproduct of the one-variable series  $\sum_{n \geq 0} X^n$  is not a finite sum of separated tensors. Define the continuous algebra homomorphism

$$\Delta : \widehat{T} \longrightarrow \widehat{T} \widehat{\otimes} \widehat{T}, \quad \Delta X = X \otimes 1 + 1 \otimes X, \quad \Delta Y = Y \otimes 1 + 1 \otimes Y. \quad (3.1)$$

Since  $\widehat{T}$  is the completion of the free algebra on  $X, Y$ , an algebra homomorphism on  $T(V)$  is determined freely by the images of  $X$  and  $Y$ ; and because  $\Delta$  raises neither total degree nor lowers it, it extends uniquely by continuity to  $\widehat{T}$ . For a word  $w = a_1 \cdots a_n$  and a subset  $I \subseteq \{1, \dots, n\}$  let  $w_I$  be the subword formed by the letters at the positions in  $I$ , in their original order, with  $w_{\emptyset} = 1$ . Then

$$\Delta w = \sum_{I \subseteq \{1, \dots, n\}} w_I \otimes w_{I^c}. \quad (3.2)$$

This *unshuffle* formula follows by multiplying out the  $n$  factors  $a_j \otimes 1 + 1 \otimes a_j$ : each choice of one term from each factor assigns every letter to the left or right tensor factor, the letters assigned to the left concatenate in their original order to  $w_I$ , and those assigned to the right concatenate to  $w_{I^c}$ . Coassociativity,  $(\Delta \otimes \text{id})\Delta = (\text{id} \otimes \Delta)\Delta$ , follows in the same way: either iterated coproduct assigns each letter independently to one of three tensor factors, and the resulting sum over assignments does not depend on the order in which the two splittings are performed. The counit  $\varepsilon : \widehat{T} \rightarrow \mathbb{K}$  extracts the constant term; the identities  $(\varepsilon \otimes \text{id})\Delta = \text{id} = (\text{id} \otimes \varepsilon)\Delta$  hold on words by (3.2), since the only term with  $w_I = 1$  is  $I = \emptyset$ .

Define the antipode on words by

$$S(a_1 \cdots a_n) = (-1)^n a_n \cdots a_1, \quad S(1) = 1, \quad (3.3)$$

extended linearly and continuously. It is an antiautomorphism:  $S(uv) = S(v)S(u)$ , since reversing a concatenation reverses the factors in the opposite order and the signs multiply. For linear maps  $f, g$  on  $\widehat{T}$  the *convolution* is  $f * g = m(f \otimes g)\Delta$ , where  $m$  is multiplication; by coassociativity convolution is associative, with unit  $\eta\varepsilon$ , where  $\eta$  inserts scalars.

**Lemma 3.1** (Antipode identities).  $\text{id} * S = S * \text{id} = \eta\varepsilon$ , that is, for every word  $w$ ,

$$m(\text{id} \otimes S)\Delta w = m(S \otimes \text{id})\Delta w = \varepsilon(w)1. \quad (3.4)$$

*Proof.* Write  $\Delta v = \sum v_{(1)} \otimes v_{(2)}$  in the finite notation of (3.2) and put  $K(v) = \sum v_{(1)} S(v_{(2)})$ . For a letter  $a$  and a word  $v$ , the unshuffle formula for  $av$  splits according to whether the first letter is assigned to the left or the right factor:

$$\Delta(av) = \sum av_{(1)} \otimes v_{(2)} + \sum v_{(1)} \otimes av_{(2)}.$$

Using  $S(av_{(2)}) = -S(v_{(2)})a$  we obtain

$$K(av) = a \sum v_{(1)} S(v_{(2)}) - \sum v_{(1)} S(v_{(2)})a = aK(v) - K(v)a.$$

Since  $K(1) = 1 \cdot S(1) = 1$ , we get  $K(a) = a - a = 0$  for every letter, and then inductively  $K(w) = 0$  for every nonempty word. This proves the first identity. For the second, put  $K'(v) = \sum S(v_{(1)})v_{(2)}$  and split the coproduct of  $va$  according to the position of the last letter  $a$ :  $\Delta(va) = \sum v_{(1)}a \otimes v_{(2)} + \sum v_{(1)} \otimes v_{(2)}a$ , whence  $K'(va) = -aK'(v) + K'(v)a$ , and the same induction applies.  $\square$

Thus  $T(V)$ , and continuously its completion, carries the asserted Hopf algebra structure, with no appeal to an external structure theorem.

**Definition 3.2.** A series  $P \in \widehat{T}$  is *primitive* if  $\Delta P = P \otimes 1 + 1 \otimes P$ . A series  $G \in 1 + F^1\widehat{T}$  is *group-like* if  $\Delta G = G \otimes G$ .

The normalization of the constant term is part of the definition of a group-like element: the zero series satisfies  $\Delta 0 = 0 \otimes 0$  but does not belong to a group. A primitive element has zero constant term, since a primitive scalar  $c$  would satisfy  $c \otimes 1 = 2c \otimes 1$ .

**Lemma 3.3** (Exponentials of primitives). *Let  $P \in F^1\widehat{T}$ . Then  $P$  is primitive if and only if  $e^P$  is group-like. The group-like elements form a group under multiplication.*

*Proof.* Since  $\Delta$  is a continuous algebra homomorphism, it commutes with the exponential series:  $\Delta(e^P) = e^{\Delta P}$ . If  $P$  is primitive, then  $\Delta P = P \otimes 1 + 1 \otimes P$  is a sum of two commuting elements, because  $(P \otimes 1)(1 \otimes P) = P \otimes P = (1 \otimes P)(P \otimes 1)$ . By [theorem 2.1](#) applied in  $\widehat{T} \widehat{\otimes} \widehat{T}$ ,

$$\Delta(e^P) = e^{P \otimes 1 + 1 \otimes P} = e^{P \otimes 1} e^{1 \otimes P} = (e^P \otimes 1)(1 \otimes e^P) = e^P \otimes e^P.$$

Conversely, if  $G = e^P$  is group-like, then  $\Delta G = G \otimes G = (G \otimes 1)(1 \otimes G)$  is a product of two commuting elements of  $1 + F^1(\widehat{T} \widehat{\otimes} \widehat{T})$ , and the commuting-logarithm rule of [theorem 2.1](#) gives

$$\Delta P = \log(\Delta G) = \log(G \otimes 1) + \log(1 \otimes G) = P \otimes 1 + 1 \otimes P,$$

where  $\log(G \otimes 1) = (\log G) \otimes 1$  because  $A \mapsto A \otimes 1$  is a continuous algebra homomorphism. Finally, if  $G, H$  are group-like then  $\Delta(GH) = (G \otimes G)(H \otimes H) = GH \otimes GH$ , and  $G$  is invertible by its geometric series, with  $\Delta(G^{-1}) = (\Delta G)^{-1} = (G \otimes G)^{-1} = G^{-1} \otimes G^{-1}$ .  $\square$

### 3.2 An explicit projection: the Dynkin–Specht–Wever identity

Let  $N$  be the degree operator,  $N(w) = |w|w$  and  $N(1) = 0$ . It is a derivation of  $T(V)$ , because degrees add under concatenation. Consider the convolution

$$D = N * S = m(N \otimes S)\Delta. \quad (3.5)$$

The order of the two convolution factors is what produces *right*-nested rather than left-nested brackets. This convolution operator is a special case of the generalized Dynkin operators studied in [4]; the elementary computation below contains the entire argument required here.

**Lemma 3.4** (Right Dynkin operator). *For every nonempty word  $w$ ,  $D(w) = R(w)$ . If  $P$  is a homogeneous primitive element of degree  $n \geq 1$ , then*

$$R(P) = D(P) = nP, \quad \text{that is,} \quad P = \frac{1}{n} \sum_{|w|=n} [w]P R(w). \quad (3.6)$$

*Proof.* From the definitions,  $D(1) = N(1)S(1) = 0$  and, for a letter  $a$ ,  $D(a) = N(a)S(1) + N(1)S(a) = a$ . Let  $a$  be a letter and  $w$  a nonempty word, and split  $\Delta(aw)$  as in the proof of theorem 3.1:  $\Delta(aw) = \sum aw_{(1)} \otimes w_{(2)} + \sum w_{(1)} \otimes aw_{(2)}$ . Using  $N(aw_{(1)}) = aw_{(1)} + aN(w_{(1)})$  (derivation property with  $N(a) = a$ ) and  $S(aw_{(2)}) = -S(w_{(2)})a$ , we get

$$\begin{aligned} D(aw) &= \sum aw_{(1)}S(w_{(2)}) + \sum aN(w_{(1)})S(w_{(2)}) - \sum N(w_{(1)})S(w_{(2)})a \\ &= aK(w) + aD(w) - D(w)a, \end{aligned}$$

where  $K(w) = \sum w_{(1)}S(w_{(2)}) = \varepsilon(w)1 = 0$  for nonempty  $w$  by theorem 3.1. Hence  $D(aw) = [a, D(w)]$ , and induction on the length starting from  $D(a) = a$  gives  $D(a_1 \cdots a_n) = [a_1, D(a_2 \cdots a_n)] = \cdots = R(a_1 \cdots a_n)$ .

If  $P$  is homogeneous primitive of degree  $n$ , then its coproduct has only the two terms  $P \otimes 1$  and  $1 \otimes P$ , and

$$D(P) = m(N \otimes S)(P \otimes 1 + 1 \otimes P) = N(P)S(1) + N(1)S(P) = nP + 0.$$

Since  $D$  is linear and  $D(w) = R(w)$  on words,  $D(P) = \sum_w [w]P R(w)$ , which gives the second form of (3.6) after division by  $n$ .  $\square$

Let  $\mathcal{L} = \mathcal{L}(X, Y) \subseteq T(V)$  be the Lie subalgebra generated by  $X, Y$  under the commutator bracket, and  $\mathcal{L}_n = \mathcal{L} \cap V^{\otimes n}$ . Since  $X, Y$  are homogeneous,  $\mathcal{L}$  is spanned by homogeneous bracket monomials, so  $\mathcal{L} = \bigoplus_n \mathcal{L}_n$ ; we write  $\hat{\mathcal{L}} = \prod_{n \geq 1} \mathcal{L}_n$ .

**Theorem 3.5** (Friedrichs’ characterization; Dynkin–Specht–Wever). *A homogeneous element  $P \in V^{\otimes n}$ ,  $n \geq 1$ , is primitive if and only if  $P \in \mathcal{L}_n$ . For such  $P$ , (3.6) holds, so  $D/n$  is an idempotent projection of  $V^{\otimes n}$  onto  $\mathcal{L}_n$  whose kernel consists of the degree- $n$  elements annihilated by  $D$ . The primitive elements of  $\hat{T}$  are exactly the elements of  $\hat{\mathcal{L}}$ , the formal degree-wise sums of Lie polynomials:*

$$\text{Prim}(\hat{T}) = \prod_{n \geq 1} \mathcal{L}_n. \quad (3.7)$$

*Proof.* The generators  $X, Y$  are primitive by definition of  $\Delta$ . If  $P, Q$  are primitive, then

$$\Delta[P, Q] = [\Delta P, \Delta Q] = [P \otimes 1 + 1 \otimes P, Q \otimes 1 + 1 \otimes Q] = [P, Q] \otimes 1 + 1 \otimes [P, Q],$$

because the mixed terms cancel:  $(P \otimes 1)(1 \otimes Q) = P \otimes Q = (1 \otimes Q)(P \otimes 1)$ , and similarly with  $P$  and  $Q$  interchanged. Hence every bracket monomial in  $X, Y$  is primitive, every Lie polynomial is

primitive by linearity, and every degreewise-convergent sum of Lie polynomials is primitive by continuity of  $\Delta$ .

Conversely, let  $P \in \widehat{T}$  be primitive. Its constant term vanishes, as noted above. Since  $\Delta$  preserves total degree, comparing the degree- $n$  parts of  $\Delta P = P \otimes 1 + 1 \otimes P$  shows that each homogeneous component  $P_n$  is primitive. By theorem 3.4,  $P_n = \frac{1}{n} \sum_{|w|=n} [w] P_n R(w)$  is a linear combination of right-nested brackets, hence an element of  $\mathcal{L}_n$ . Division by  $n$  is exactly where characteristic zero is used. This proves the homogeneous characterization and (3.7).

Finally, since  $D$  always takes values in  $\mathcal{L}$  (its value on any word is a nested bracket) and every element of  $\mathcal{L}_n$  is primitive with  $D(P) = nP$ , we have  $D(D(Q)) = nD(Q)$  for every  $Q \in V^{\otimes n}$ ; so  $D/n$  is idempotent with image  $\mathcal{L}_n$ .  $\square$

**Theorem 3.6** (Formal BCH theorem). *There is a unique  $Z \in F^1 \widehat{T}$  with  $e^Z = e^X e^Y$ . Every homogeneous component  $Z_n$  is a Lie polynomial with rational coefficients,  $Z_1 = X + Y$ , and  $Z - X - Y$  belongs degreewise to the Lie ideal of  $\mathcal{L}$  generated by  $[X, Y]$ .*

*Proof.* Existence and uniqueness follow from theorem 2.1. Since  $X$  and  $Y$  are primitive,  $e^X$  and  $e^Y$  are group-like by theorem 3.3, their product  $e^X e^Y$  is group-like because group-like elements form a group, and  $Z = \log(e^X e^Y)$  is primitive by theorem 3.3 again. By theorem 3.5 each  $Z_n$  lies in  $\mathcal{L}_n$ . Rationality follows from (2.5), whose ingredients are factorials and the coefficients  $(-1)^{k-1}/k$ . Taking degree one in (2.3) gives  $Z_1 = X + Y$ .

For the ideal assertion, let  $J$  be the Lie ideal of  $\mathcal{L}$  generated by  $[X, Y]$ , and consider the quotient Lie algebra  $\mathcal{L}/J$ . Its two generators commute, so every bracket of length at least two of elements of  $\mathcal{L}/J$  vanishes and  $\mathcal{L}/J$  is abelian. Alternatively and more concretely: a right-nested bracket  $R(w)$  with  $|w| \geq 2$  has innermost bracket  $[a_{n-1}, a_n]$ , which is either zero or  $\pm[X, Y]$ , and an iterated bracket whose innermost factor lies in the ideal  $J$  lies in  $J$ ; since each  $Z_n$  with  $n \geq 2$  is a combination of such brackets by (3.6), it lies in  $J$ .  $\square$

The phrase “every term beyond  $X + Y$  contains  $[X, Y]$ ” is thus an ideal-theoretic statement; it does not mean that an arbitrary display of  $Z_n$  must visibly show  $[X, Y]$  as the innermost bracket of every summand.

### 3.3 Universality and enveloping algebras

A homogeneous Lie polynomial can be evaluated at any pair of elements of any Lie algebra  $\mathfrak{g}$  over  $\mathbb{K}$ . The identification of  $\mathcal{L}(X, Y)$  with the abstract free Lie algebra, and the fact that the free associative algebra is its universal enveloping algebra, rest on the injectivity of  $\mathfrak{g} \rightarrow U(\mathfrak{g})$ , which is proved in section A by the ordered Poincaré–Birkhoff–Witt theorem (theorem A.1). The argument is as follows. Given  $x, y \in \mathfrak{g}$ , the associative homomorphism  $T(V) \rightarrow U(\mathfrak{g})$  determined by  $X \mapsto x$ ,  $Y \mapsto y$  sends every iterated commutator of  $X, Y$  to the corresponding iterated bracket of  $x, y$ , which lies in the image of  $\mathfrak{g}$ . Its restriction to  $\mathcal{L}$  is therefore a Lie homomorphism  $\mathcal{L} \rightarrow \mathfrak{g}$ , and it is the unique one sending  $X, Y$  to  $x, y$  because  $X, Y$  generate  $\mathcal{L}$ . This is exactly the universal property of the free Lie algebra. Consequently an associative polynomial identity between Lie polynomials in  $X, Y$ , verified as an equality of word coefficients in  $T(V)$ , is an identity valid under substitution in *every* Lie algebra.

**Proposition 3.7** (Enveloping algebra of the free Lie algebra). *The universal enveloping algebra of  $\mathcal{L}(X, Y)$  is canonically  $T(V)$ , and the Hopf structure of section 3 is the enveloping Hopf structure determined by  $\Delta u = u \otimes 1 + 1 \otimes u$ ,  $\varepsilon(u) = 0$ ,  $S(u) = -u$  for  $u \in \mathcal{L}$ . The same formulas define a Hopf algebra structure on  $U(\mathfrak{g})$  for every Lie algebra  $\mathfrak{g}$ .*

*Proof.* The inclusion  $\mathcal{L} \rightarrow T(V)$  is a Lie homomorphism into the commutator algebra of  $T(V)$ , so by the universal property of  $U(\mathcal{L})$  it extends to an associative homomorphism  $\alpha : U(\mathcal{L}) \rightarrow T(V)$ .

The assignment  $X \mapsto X$ ,  $Y \mapsto Y$  into  $U(\mathcal{L})$  extends, by freeness of  $T(V)$ , to an associative homomorphism  $\beta : T(V) \rightarrow U(\mathcal{L})$ . Both composites  $\alpha\beta$  and  $\beta\alpha$  fix  $X$  and  $Y$ ; since  $X, Y$  generate  $T(V)$  as an algebra and generate  $U(\mathcal{L})$  as an algebra (because they generate  $\mathcal{L}$  as a Lie algebra and  $\mathcal{L}$  generates  $U(\mathcal{L})$ ), both composites are identities. The Hopf assertion for general  $\mathfrak{g}$  is proved in [section A](#); on generators the two structures agree, and each is determined by its values on generators.  $\square$

For an arbitrary Lie algebra  $\mathfrak{g}$  without a topology, the unambiguous universal statement is

$$\text{BCH}(tx, ty) = \sum_{n \geq 1} t^n Z_n(x, y) \in t \mathfrak{g}[[t]],$$

obtained by evaluating in  $U(\mathfrak{g})[[t]]$ . An infinite sum at  $t = 1$  in the untopologized vector space  $\mathfrak{g}$  has no meaning until nilpotence ([section 8.5](#)), a complete filtration, or a convergence argument ([section 7](#)) has been supplied.

## 4 Dynkin’s formula, every bidegree, and the polynomial through degree six

### 4.1 Dynkin’s complete commutator formula

The structural theorem becomes completely explicit by applying the Dynkin operator to the associative logarithm. The classical coefficient formula is due to Dynkin [2]; modern computational treatments include [5]. The proof below derives its coefficients directly.

**Theorem 4.1** (Dynkin’s all-terms formula). *With  $d = \sum_{i=1}^k (r_i + s_i)$ ,*

$$Z(X, Y) = \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} \sum_{\substack{r_i, s_i \geq 0 \\ r_i + s_i > 0 \ (1 \leq i \leq k)}} \frac{R(X^{r_1} Y^{s_1} \dots X^{r_k} Y^{s_k})}{d \prod_{i=1}^k r_i! s_i!}. \quad (4.1)$$

*Equivalently, in terms of the word coefficients (2.5),*

$$Z_n = \frac{1}{n} \sum_{|w|=n} c(w) R(w), \quad (4.2)$$

*and, separating the bidegrees,*

$$Z_{p,q} = \frac{1}{p+q} \sum_{\substack{w \in \{X,Y\}^{p+q} \\ \#_X w = p, \#_Y w = q}} c(w) R(w), \quad p+q \geq 1. \quad (4.3)$$

*In each total degree the sums are finite. A summand of (4.1) vanishes if  $s_k > 1$ , or if  $s_k = 0$  and  $r_k > 1$ .*

*Proof.* By [theorem 3.6](#), each  $Z_n$  is a homogeneous primitive element of degree  $n$ . [Theorem 3.4](#) therefore gives  $Z_n = \frac{1}{n} D(Z_n)$ . Since  $D$  is linear and  $D(w) = R(w)$  on words, applying  $D/n$  to the word expansion  $Z_n = \sum_{|w|=n} c(w)w$  of (2.6) gives (4.2). Applying it instead to the block form of (2.6) replaces each block word  $X^{r_1} Y^{s_1} \dots X^{r_k} Y^{s_k}$  by its right-nested bracket and divides by its length  $d = n$ ; summing over  $n$  gives (4.1). Finally,  $Z_n = \sum_{p+q=n} Z_{p,q}$  and the operator  $D$  preserves bidegree (it maps a word to a combination of words with the same letters), so the bidegree- $(p, q)$  part of (4.2) is (4.3). If  $s_k > 1$  the two innermost letters of the bracketed word are  $Y, Y$ , and if  $s_k = 0$ ,  $r_k > 1$  they are  $X, X$ ; in either case the innermost bracket  $[a_{d-1}, a_d]$  vanishes and so does the whole nested bracket.  $\square$

The two vanishing tests are sufficient but not necessary: a run of equal letters may continue across a block boundary, for instance in the block word  $X \cdot XY$  followed by the block  $Y$ . Notice also that one must project a *whole homogeneous Lie polynomial*; replacing an arbitrary associative polynomial by  $D/n$  generally changes it. The right-nested brackets form a spanning family of  $\mathcal{L}_n$ , not a basis, so different displays can represent the same Lie element by Jacobi identities. This is redundancy in the representation, not ambiguity in  $Z$ . Formula (4.3), combined with the finite cut formula (2.5), is a particularly compact answer to the request for *all* terms: there are no unspecified coefficients or auxiliary recurrences.

## 4.2 Symmetries, associativity, and functoriality

**Proposition 4.2.** *As universal formal series, with  $T$  a third letter,*

$$Z(Y, X) = -Z(-X, -Y), \quad (4.4)$$

$$Z_n(Y, X) = (-1)^{n+1} Z_n(X, Y), \quad (4.5)$$

$$\text{rev}(Z_n(X, Y)) = (-1)^{n+1} Z_n(X, Y), \quad \text{that is,} \quad c(\text{rev } w) = (-1)^{|w|+1} c(w), \quad (4.6)$$

$$Z(Z(X, Y), T) = Z(X, Z(Y, T)), \quad (4.7)$$

$$Z(X, 0) = Z(0, X) = X, \quad Z(X, -X) = 0. \quad (4.8)$$

Here  $\text{rev}$  reverses every word. Moreover, every Lie-algebra homomorphism  $f : \mathfrak{g} \rightarrow \mathfrak{h}$  satisfies  $f(Z_n(x, y)) = Z_n(fx, fy)$  for all  $n$ , and hence commutes with the BCH series wherever formal substitution or analytic convergence is valid.

*Proof.* Taking inverses in  $e^{-X}e^{-Y} = e^{Z(-X, -Y)}$  gives  $e^Y e^X = (e^{Z(-X, -Y)})^{-1} = e^{-Z(-X, -Y)}$ , using  $(e^A)^{-1} = e^{-A}$  from theorem 2.1. Since also  $e^Y e^X = e^{Z(Y, X)}$ , uniqueness of the formal logarithm gives (4.4). Replacing  $(X, Y)$  by  $(-X, -Y)$  multiplies the degree- $n$  part by  $(-1)^n$ , so comparing homogeneous parts in (4.4) gives (4.5). Word reversal  $\text{rev}$  is an antiautomorphism of  $\widehat{T}$ :  $\text{rev}(uv) = \text{rev}(v)\text{rev}(u)$ . It fixes  $X$  and  $Y$ , hence  $\text{rev}(e^X) = e^X$ ,  $\text{rev}(e^Y) = e^Y$ , and  $\text{rev}(e^X e^Y) = e^Y e^X$ . Because  $\text{rev}$  is linear, continuous, and multiplicative on powers of a single element, it commutes with the logarithm series:  $\text{rev}(\log(1 + U)) = \log(1 + \text{rev } U)$ . Therefore  $\text{rev}(Z(X, Y)) = \log(e^Y e^X) = Z(Y, X)$ , and (4.5) gives (4.6); the coefficient statement is its coordinate form.

For (4.7), work in the completion of the free algebra on three letters  $X, Y, T$ . Both sides exponentiate to  $e^X e^Y e^T$ : the left side because  $e^{Z(Z(X, Y), T)} = e^{Z(X, Y)} e^T = e^X e^Y e^T$ , the right side similarly; here we use that  $Z(A, B)$  for positive-degree  $A, B$  is defined by substitution into the universal series and satisfies  $e^{Z(A, B)} = e^A e^B$  by the substitution homomorphism. Uniqueness of the logarithm proves (4.7). The identities (4.8) follow in the same way from  $e^X e^0 = e^X$  and  $e^X e^{-X} = 1$ .

A Lie homomorphism  $f$  preserves every iterated bracket, hence every right-nested bracket  $R(w)$  evaluated at  $(x, y)$ , hence every  $Z_n$  by (4.2). In a convergent analytic setting a continuous linear map preserves the limit of the series; a linear map between finite-dimensional spaces is automatically continuous.  $\square$

Both the swap symmetry and the reversal symmetry provide useful independent checks on large coefficient tables, and they are used as such in section 14.

### 4.3 Every term through degree six

With the right-nesting convention (1.4), the terms printed on the source page are exactly

$$Z_1 = X + Y, \quad (4.9)$$

$$Z_2 = \frac{1}{2}R(XY) = \frac{1}{2}[X, Y], \quad (4.10)$$

$$Z_3 = \frac{1}{12}(R(XXY) + R(YYX)) = \frac{1}{12}[X, [X, Y]] + \frac{1}{12}[Y, [Y, X]], \quad (4.11)$$

$$Z_4 = -\frac{1}{24}R(YXXY) = -\frac{1}{24}[Y, [X, [X, Y]]], \quad (4.12)$$

$$Z_5 = -\frac{1}{720}(R(YYYYX) + R(XXXXY)) + \frac{1}{360}(R(XYYYYX) + R(YXXXXY)) \\ + \frac{1}{120}(R(YXYXY) + R(XYXYX)), \quad (4.13)$$

$$Z_6 = \frac{1}{240}R(XYXYXY) + \frac{1}{720}(R(XYXXXXY) - R(XXYYXY)) \\ + \frac{1}{1440}(R(XYYYYXY) - R(XXYXXY)). \quad (4.14)$$

For example  $R(XYXXXXY) = [X, [Y, [X, [X, [X, Y]]]]$ . There is no omitted term of any of these degrees; equations (2.5) and (4.1) define every subsequent degree.

For direct comparison with associative matrix multiplication, the first four degrees are

$$Z_2 = \frac{1}{2}(XY - YX), \quad (4.15)$$

$$Z_3 = \frac{1}{12}(X^2Y + XY^2 - 2XYX + Y^2X + YX^2 - 2YXY), \quad (4.16)$$

$$Z_4 = \frac{1}{24}(X^2Y^2 - 2XYXY - Y^2X^2 + 2YXYX). \quad (4.17)$$

To turn displayed brackets into word coefficients we use the following explicit expansion. For  $w = a_1 \cdots a_n$  and a subset  $S \subseteq \{1, \dots, n-1\}$ , let  $w_S$  be the subword of  $a_1 \cdots a_{n-1}$  at the positions in  $S$  and  $w_{S^c}$  the subword at the remaining positions of  $\{1, \dots, n-1\}$ . Then

$$R(w) = \sum_{S \subseteq \{1, \dots, n-1\}} (-1)^{n-1-|S|} w_S a_n \text{rev}(w_{S^c}). \quad (4.18)$$

*Proof of (4.18).* Induct on  $n$ ; the case  $n = 1$  reads  $R(a_1) = a_1$ . For  $n \geq 2$ ,  $R(w) = a_1 R(a_2 \cdots a_n) - R(a_2 \cdots a_n) a_1$ . By the induction hypothesis each term of  $R(a_2 \cdots a_n)$  has the form  $\pm(\text{subword of } a_2 \cdots a_{n-1}) a_n$  (reversed complementary subword). Placing  $a_1$  on the left adds position 1 to  $S$  and keeps the sign; placing  $a_1$  on the right, with a minus sign, appends  $a_1$  at the end of the reversed complement, which is exactly where position 1 belongs in  $\text{rev}(w_{S^c})$  when  $1 \notin S$ , and changes the sign, matching the factor  $(-1)^{n-1-|S|}$  because  $|S^c|$  has increased by one.  $\square$

**Proposition 4.3.** *Equations (4.9)–(4.14) are the homogeneous components of degrees one through six of  $Z(X, Y)$ , and (4.15)–(4.17) are their associative expansions.*

*Proof.* Degree one is theorem 3.6. For degree two, (2.5) gives  $c(XY) = \frac{1}{2}$  and  $c(YX) = -\frac{1}{2}$ , while  $c(XX) = a(XX) - \frac{1}{2}a(X)a(X) = \frac{1}{2} - \frac{1}{2} = 0$  and likewise  $c(YY) = 0$ . Hence  $Z_2 = \frac{1}{2}(XY - YX) = \frac{1}{2}[X, Y]$ , which is (4.10) and (4.15).

For degree three the cut formula gives the following values. For  $XXY$  we computed  $\frac{1}{12}$ ; by reversal symmetry (4.6) with the sign  $(-1)^4 = +1$ ,  $c(YXX) = \frac{1}{12}$ , and by swap symmetry (4.5),  $c(YYX) = c(XYY) = \frac{1}{12}$ . For  $XYX$  we computed  $-\frac{1}{6}$ , and by swap,  $c(YXY) = -\frac{1}{6}$ . For  $XXX$ : the one-piece contribution is  $\frac{1}{6}$ , the two cuts contribute  $-\frac{1}{2}(\frac{1}{2} + \frac{1}{2}) = -\frac{1}{2}$ , and the three-piece cut contributes  $\frac{1}{3}$ ; the sum is zero, and likewise  $c(YYY) = 0$ . Thus

$$Z_3 = \frac{1}{12}(XXY + YXX + YYX + XYY - 2XYX - 2YXY),$$

which is (4.16). On the other hand, expanding the brackets,

$$[X, [X, Y]] = X^2Y - 2XYX + YX^2, \quad [Y, [Y, X]] = Y^2X - 2YXY + XY^2,$$

and  $\frac{1}{12}$  times their sum is the same polynomial. This proves (4.11).

For degree four, direct evaluation of (2.5) on the sixteen words gives nonzero values only for

$$c(XXY Y) = \frac{1}{24}, \quad c(XYXY) = -\frac{1}{12}, \quad c(YXYX) = \frac{1}{12}, \quad c(YYXX) = -\frac{1}{24},$$

and zero on the remaining twelve words. We verify one representative of each type. For  $XXYY$ : one piece gives  $a(XXYY) = \frac{1}{4}$ ; the three two-piece cuts  $X | XYY$ ,  $XX | YY$ ,  $XXY | Y$  give  $-\frac{1}{2}(\frac{1}{2} + \frac{1}{4} + \frac{1}{2}) = -\frac{5}{8}$ ; the three three-piece cuts  $X | X | YY$ ,  $X | XY | Y$ ,  $XX | Y | Y$  give  $\frac{1}{3}(\frac{1}{2} + 1 + \frac{1}{2}) = \frac{2}{3}$ ; the four-piece cut gives  $-\frac{1}{4}$ ; the sum is  $\frac{1}{4} - \frac{5}{8} + \frac{2}{3} - \frac{1}{4} = \frac{1}{24}$ . For  $XYXY$ : one piece gives 0; two-piece cuts  $X | YXY$  (0),  $XY | XY$  (1),  $XYX | Y$  (0) give  $-\frac{1}{2}$ ; three-piece cuts  $X | Y | XY$ ,  $X | YX | Y$ ,  $XY | X | Y$  give  $\frac{1}{3}(1 + 0 + 1) = \frac{2}{3}$ ; four pieces give  $-\frac{1}{4}$ ; the sum is  $-\frac{1}{2} + \frac{2}{3} - \frac{1}{4} = -\frac{1}{12}$ . The values at  $YYXX$  and  $YXYX$  follow by reversal symmetry with sign  $(-1)^5 = -1$ . For a word with zero coefficient, take  $XXXY$ : one piece gives  $a(XXXY) = \frac{1}{6}$ ; the two-piece cuts  $X | XXY$ ,  $XX | XY$ ,  $XXX | Y$  give  $-\frac{1}{2}(\frac{1}{2} + \frac{1}{2} + \frac{1}{6}) = -\frac{7}{12}$ ; the three-piece cuts  $X | X | XY$  (1),  $X | XX | Y$  ( $\frac{1}{2}$ ),  $XX | X | Y$  ( $\frac{1}{2}$ ) give  $\frac{1}{3} \cdot 2 = \frac{2}{3}$ ; four pieces give  $-\frac{1}{4}$ ; the sum is  $\frac{1}{6} - \frac{7}{12} + \frac{2}{3} - \frac{1}{4} = 0$ . The remaining zero coefficients are checked in the same way; the eleven other words are  $XXXX$ ,  $YYYY$ ,  $XXYX$ ,  $XYXX$ ,  $YXXX$ ,  $XYYY$ ,  $YXY Y$ ,  $YYXY$ ,  $YYXX$ ,  $XYXX$ ,  $YXXY$ , and each cut sum vanishes (for the last two, for example, the contributions are  $0 - \frac{1}{2}(0 + 0 + \frac{1}{2}) + \frac{1}{3}(0 + \frac{1}{2} + 1) - \frac{1}{4} = 0$  and  $0 - \frac{1}{2}(\frac{1}{2} + 0 + 0) + \frac{1}{3}(1 + \frac{1}{2} + 0) - \frac{1}{4} = 0$ , the cuts being listed from left to right by the position of the first break). Thus (4.17) holds. Expanding the bracket,

$$[Y, [X, [X, Y]]] = [Y, X^2Y - 2XYX + YX^2] = 2XYXY - 2YXYX + Y^2X^2 - X^2Y^2,$$

and  $-\frac{1}{24}$  times this is (4.17). This proves (4.12).

For degrees five and six we give a finite coefficientwise certificate. Expand each right-nested bracket in (4.13) and (4.14) by (4.18) and collect the coefficient of every word  $w$  of length five or six. Independently, evaluate the finite cut sum (2.5) for every such word. Section B prints all nonzero values of  $c(w)$  through degree six (there are 30 nonzero coefficients in degree five and 28 in degree six), and both computations give zero on every word absent from the table. Thus the two polynomials agree in every one of the 32 and 64 coordinates respectively. An equality of noncommutative polynomials is fully determined by these word coefficients, so the displayed formulas are identities in the free associative algebra, hence in every associative specialization. The accompanying exact-arithmetic program performs this comparison, including the zero coefficients, and additionally recomputes both sides from the differential recursions of section 6.  $\square$

These displays do not select a linearly independent basis of the free Lie algebra; for instance  $[X, [Y, [X, Y]]] = [Y, [X, [X, Y]]]$ , since the difference is  $[[X, Y], [X, Y]] = 0$  by (1.3). The coefficient of  $X^4Y$  in  $Z_5$  is  $-\frac{1}{720} = B_4^+/4!$ , and that of  $X^5Y$  in  $Z_6$  is  $0 = B_5^+/5!$ , as the single- $Y$  Bernoulli coefficients of theorem 6.4 predict.

## 5 Conjugation, the differential of the exponential, and Bernoulli series

This section opens a second route to the BCH series, independent of the Hopf algebra argument. It also proves the Campbell identity and every infinitesimal identity used later. Parameter differentiation of exponentials is treated systematically in [9]; all formulas required here are derived in full.

## 5.1 Campbell’s conjugation identity and the braiding formulas

**Theorem 5.1** (Campbell–Hadamard identity). *In the formal setting, where  $X, Y$  have positive degree and  $s$  is a formal or scalar parameter, and for all  $X, Y$  in a Banach algebra and all scalars  $s$ ,*

$$e^{sX} Y e^{-sX} = e^{s \operatorname{ad}_X} Y = \sum_{n \geq 0} \frac{s^n}{n!} \operatorname{ad}_X^n Y. \quad (5.1)$$

*For invertible elements write  $\operatorname{Ad}_A Y = A Y A^{-1}$ ; then  $\operatorname{Ad}_{e^X} = e^{\operatorname{ad}_X}$ . The Banach-algebra series converges absolutely for every  $X, Y, s$ , with  $\sum_n |s|^n \|\operatorname{ad}_X^n Y\| / n! \leq e^{2|s|\|X\|} \|Y\|$ . An entirely associative version is*

$$\operatorname{ad}_X^n Y = \sum_{j=0}^n (-1)^j \binom{n}{j} X^{n-j} Y X^j. \quad (5.2)$$

*Proof.* Let  $L_X$  and  $R_X$  denote left and right multiplication by  $X$ . They commute, since  $L_X R_X A = X A X = R_X L_X A$ , and  $\operatorname{ad}_X = L_X - R_X$ . The binomial theorem for commuting operators gives  $\operatorname{ad}_X^n = \sum_j (-1)^j \binom{n}{j} L_X^{n-j} R_X^j$ , which applied to  $Y$  is (5.2).

Set  $F(s) = e^{sX} Y e^{-sX}$ . In a Banach algebra  $F$  is differentiable with  $F'(s) = X e^{sX} Y e^{-sX} - e^{sX} Y e^{-sX} X = \operatorname{ad}_X F(s)$  and  $F(0) = Y$ , by the product rule and  $\frac{d}{ds} e^{sX} = X e^{sX} = e^{sX} X$ . The map  $\operatorname{ad}_X$  is a bounded linear operator with  $\|\operatorname{ad}_X\| \leq 2\|X\|$ , so the linear differential equation  $F' = \operatorname{ad}_X F$  has the unique solution  $F(s) = e^{s \operatorname{ad}_X} Y = \sum_n s^n \operatorname{ad}_X^n Y / n!$ ; uniqueness holds because a difference  $G$  of two solutions satisfies  $\|G(s)\| \leq 2\|X\| \int_0^s \|G\|$ , hence vanishes by Grönwall’s inequality. The norm bound follows from  $\|\operatorname{ad}_X^n Y\| \leq (2\|X\|)^n \|Y\|$ . Formally, both sides of (5.1) are power series in  $s$  with coefficients in  $\hat{T}$ ; using (5.2), the coefficient of  $s^n$  on the left is  $\sum_j X^{n-j} Y (-X)^j / ((n-j)!j!) = \frac{1}{n!} \sum_j \binom{n}{j} (-1)^j X^{n-j} Y X^j = \operatorname{ad}_X^n Y / n!$ , which is the coefficient on the right. (This direct comparison also gives a second, purely algebraic proof in the Banach case.) Setting  $s = 1$  gives  $\operatorname{Ad}_{e^X} = e^{\operatorname{ad}_X}$ .  $\square$

Conjugation  $A \mapsto e^X A e^{-X}$  is an algebra automorphism, continuous in the Banach case and degree-preserving in the formal case, so it commutes with the exponential series. Therefore

$$e^X e^Y e^{-X} = \exp(e^X Y e^{-X}) = \exp(e^{\operatorname{ad}_X} Y), \quad (5.3)$$

$$e^X e^Y = \exp\left(\sum_{n \geq 0} \frac{\operatorname{ad}_X^n Y}{n!}\right) e^X. \quad (5.4)$$

The second follows from the first by multiplying on the right by  $e^X$ . These are the general braiding identities of the source page, including the iterated-commutator notation  $[X, Y]_n = \operatorname{ad}_X^n Y$  there. Unlike a general BCH series, the adjoint exponential in (5.4) is an entire series for bounded operators and requires no smallness.

## 5.2 The differential of the exponential

**Theorem 5.2** (Duhamel differential). *Let  $A(t)$  be a differentiable curve in a Banach algebra, with  $H = A'(t)$ , or a formal power series in  $t$  with coefficients in  $F^1 \hat{T}$ . Then*

$$(\operatorname{d} \exp)_A(H) := \frac{d}{dt} e^{A(t)} = \sum_{p,q \geq 0} \frac{A^p H A^q}{(p+q+1)!} = \int_0^1 e^{(1-s)A} H e^{sA} ds, \quad (5.5)$$

$$e^{-A} \frac{d}{dt} e^A = \int_0^1 e^{-s \operatorname{ad}_A} H ds = \sum_{n \geq 0} \frac{(-1)^n}{(n+1)!} \operatorname{ad}_A^n H = \phi(\operatorname{ad}_A) H, \quad (5.6)$$

$$\left(\frac{d}{dt} e^A\right) e^{-A} = \int_0^1 e^{s \operatorname{ad}_A} H ds = \sum_{n \geq 0} \frac{1}{(n+1)!} \operatorname{ad}_A^n H = \phi(-\operatorname{ad}_A) H. \quad (5.7)$$

All series converge absolutely for arbitrary bounded  $A, H$ .

*Proof.* The product rule, applied without moving any factor, gives  $\frac{d}{dt}A^n = \sum_{j=0}^{n-1} A^j H A^{n-1-j}$ ; in the Banach case the exponential series may be differentiated term by term because  $\sum_n \left\| \frac{d}{dt}A^n \right\| / n! \leq \sum_n n \|A\|^{n-1} \|H\| / n!$  converges uniformly on compact parameter sets, and in the formal case the statement is coefficientwise. Collecting terms, the coefficient of  $A^p H A^q$  in  $\frac{d}{dt}e^A$  is  $1/(p+q+1)!$ , which is the first equality in (5.5). Expanding the two exponentials in the integral, the coefficient of  $A^p H A^q$  is

$$\frac{1}{p!q!} \int_0^1 (1-s)^p s^q ds = \frac{1}{(p+q+1)!}.$$

The beta integral is evaluated by induction on  $p$ : for  $p=0$  it is  $1/(q+1)$ , and integration by parts gives  $\int_0^1 (1-s)^p s^q ds = \frac{p}{q+1} \int_0^1 (1-s)^{p-1} s^{q+1} ds$ , so  $\int_0^1 (1-s)^p s^q ds = \frac{p!}{(q+1)(q+2)\cdots(q+p)} \cdot \frac{1}{q+p+1} = \frac{p!q!}{(p+q+1)!}$ . This proves (5.5); interchanging sum and integral is justified by the absolutely convergent majorant  $e^{(1-s)\|A\|} \|H\| e^{s\|A\|}$ .

Multiplying (5.5) on the left by  $e^{-A}$  gives  $\int_0^1 e^{-sA} H e^{sA} ds$ , and theorem 5.1 with  $X=A$  and parameter  $-s$  turns the integrand into  $e^{-s\text{ad}_A} H$ . Integrating the absolutely convergent series  $\sum_n (-s)^n \text{ad}_A^n H / n!$  term by term gives  $\sum_n (-1)^n \text{ad}_A^n H / (n+1)!$ , which is (5.6). Multiplying (5.5) on the right by  $e^{-A}$  gives  $\int_0^1 e^{(1-s)A} H e^{-(1-s)A} ds = \int_0^1 e^{u\text{ad}_A} H du$ , and the same integration gives (5.7).  $\square$

The quotient in  $\phi(z) = (1 - e^{-z})/z$  is notation for an entire function with  $\phi(0) = 1$ ; (5.6) does not require  $\text{ad}_A$  to be invertible. The same statements hold intrinsically for finite-dimensional Lie groups, where multiplication by  $e^{-A}$  means left or right translation of tangent vectors; an intrinsic derivation is given in section 8. In particular  $(d\exp)_0 = \text{id}$ .

### 5.3 Bernoulli coefficients, including the convention at degree one

The formal reciprocal of  $\phi$  is

$$\beta(z) = \frac{z}{1 - e^{-z}} = \sum_{n \geq 0} b_n^+ z^n, \quad b_n^+ = \frac{B_n^+}{n!}, \quad (5.8)$$

which defines the Bernoulli numbers  $B_n^+$  in the convention  $B_1^+ = +\frac{1}{2}$  used in the note of the source page. The other common convention is generated by  $\gamma(z) = z/(e^z - 1) = \beta(-z) = \sum b_n^- z^n$ , with  $b_n^- = (-1)^n b_n^+$  and  $B_1^- = -\frac{1}{2}$ .

**Proposition 5.3** (Complete Bernoulli coefficients).  $b_0^+ = 1$ , and for  $n \geq 1$  the coefficients are determined by the recursion

$$b_n^+ = - \sum_{j=1}^n \frac{(-1)^j}{(j+1)!} b_{n-j}^+, \quad (5.9)$$

and by the finite nonrecursive formula

$$b_n^+ = \sum_{k=1}^n (-1)^{n+k} \sum_{\substack{j_1+\cdots+j_k=n \\ j_i \geq 1}} \prod_{i=1}^k \frac{1}{(j_i+1)!}. \quad (5.10)$$

Moreover

$$\beta(z) = \frac{z}{2} \left( 1 + \coth \frac{z}{2} \right) = 1 + \frac{z}{2} + \frac{z^2}{12} - \frac{z^4}{720} + \frac{z^6}{30240} - \frac{z^8}{1209600} + \cdots, \quad (5.11)$$

so  $b_{2m+1}^+ = 0$  for  $m \geq 1$ ; the even coefficients are

$$b_{2m}^+ = \frac{2(-1)^{m-1}\zeta(2m)}{(2\pi)^{2m}} \quad (m \geq 1), \quad (5.12)$$

where  $\zeta(2m) = \sum_{k \geq 1} k^{-2m}$ ; and the Taylor series of  $\beta$  has radius of convergence exactly  $2\pi$ .

*Proof.* Write  $\phi(z) = \sum_{j \geq 0} \phi_j z^j$  with  $\phi_j = (-1)^j/(j+1)!$ , so  $\phi_0 = 1$ . Comparing the coefficient of  $z^n$ ,  $n \geq 1$ , in  $\phi(z)\beta(z) = 1$  gives  $\sum_{j=0}^n \phi_j b_{n-j}^+ = 0$ , which is (5.9). For (5.10), write  $\phi = 1 + A$  with  $A(z) = \sum_{j \geq 1} \phi_j z^j$  of positive order and expand  $\beta = (1 + A)^{-1} = \sum_{k \geq 0} (-A)^k$ , a formally convergent geometric series. The coefficient of  $z^n$  in  $(-A)^k$  is  $(-1)^k \sum_{j_1 + \dots + j_k = n} \prod_i \phi_{j_i} = (-1)^k (-1)^n \sum \prod_i 1/(j_i + 1)!$ , and summing over  $k$  from 1 to  $n$  (no  $k > n$  contributes because each  $j_i \geq 1$ ) gives (5.10).

Multiplying numerator and denominator of  $\beta$  by  $e^{z/2}$ ,

$$\beta(z) = \frac{ze^{z/2}}{e^{z/2} - e^{-z/2}} = \frac{z}{2} \cdot \frac{e^{z/2} + e^{-z/2} + (e^{z/2} - e^{-z/2})}{e^{z/2} - e^{-z/2}} = \frac{z}{2} \left( \coth \frac{z}{2} + 1 \right).$$

The function  $\frac{z}{2} \coth \frac{z}{2} = \beta(z) - \frac{z}{2}$  is even in  $z$  and regular at zero, so all its odd Taylor coefficients vanish; this proves  $b_{2m+1}^+ = 0$  for  $m \geq 1$ . The displayed numerical coefficients follow from (5.9):  $b_1^+ = \frac{1}{2}$ ,  $b_2^+ = \frac{1}{12}$ ,  $b_4^+ = -\frac{1}{720}$ ,  $b_6^+ = \frac{1}{30240}$ ,  $b_8^+ = -\frac{1}{1209600}$ ; for instance  $b_1^+ = -\phi_1 b_0^+ = \frac{1}{2}$  and  $b_2^+ = -(\phi_1 b_1^+ + \phi_2 b_0^+) = -(-\frac{1}{2} \cdot \frac{1}{2} + \frac{1}{6}) = \frac{1}{12}$ .

For the even coefficients we derive the partial fraction expansion of  $\coth$  from an elementary Fourier series. Fix a real  $z > 0$  and let  $g$  be the 1-periodic extension of  $u \mapsto \cosh(z(u - \frac{1}{2}))$  from  $[0, 1]$ ; it is continuous, since  $g(0) = g(1) = \cosh(z/2)$ , and piecewise smooth. Its Fourier coefficients are

$$\begin{aligned} c_k &= \int_0^1 \cosh(z(u - \tfrac{1}{2})) e^{-2\pi i k u} du = \frac{1}{2} e^{-z/2} \int_0^1 e^{(z-2\pi i k)u} du + \frac{1}{2} e^{z/2} \int_0^1 e^{-(z+2\pi i k)u} du \\ &= \frac{e^{z/2} - e^{-z/2}}{2(z - 2\pi i k)} + \frac{e^{z/2} - e^{-z/2}}{2(z + 2\pi i k)} = \frac{2z \sinh(z/2)}{z^2 + 4\pi^2 k^2}, \end{aligned}$$

using  $e^{\pm 2\pi i k} = 1$ . Since  $c_k = O(k^{-2})$ , the Fourier series  $\sum_{k \in \mathbb{Z}} c_k e^{2\pi i k u}$  converges absolutely and uniformly to a continuous function with the same Fourier coefficients as  $g$ ; by uniqueness of Fourier coefficients for continuous functions, its sum is  $g$ . Evaluating at  $u = 0$ ,

$$\cosh \frac{z}{2} = \frac{2 \sinh(z/2)}{z} + \sum_{k \geq 1} \frac{4z \sinh(z/2)}{z^2 + 4\pi^2 k^2}, \quad \text{so} \quad \coth \frac{z}{2} = \frac{2}{z} + 4z \sum_{k \geq 1} \frac{1}{z^2 + 4\pi^2 k^2}.$$

Both sides are meromorphic functions of  $z \in \mathbb{C}$ ; the right side converges normally on compact sets avoiding the points  $\pm 2\pi i k$ , so the identity proved for real  $z > 0$  extends to all  $z$  away from the poles by the identity theorem. Substituting into (5.11),

$$\beta(z) = 1 + \frac{z}{2} + 2z^2 \sum_{k \geq 1} \frac{1}{z^2 + 4\pi^2 k^2}. \quad (5.13)$$

For  $|z| < 2\pi$  expand each term geometrically,  $\frac{1}{z^2 + 4\pi^2 k^2} = \sum_{m \geq 0} \frac{(-1)^m z^{2m}}{(2\pi k)^{2m+2}}$ ; the double series converges absolutely because  $\sum_k \sum_m |z|^{2m}/(2\pi k)^{2m+2} = \sum_k 1/(4\pi^2 k^2 - |z|^2) < \infty$ , so the order of summation may be interchanged, giving  $\beta(z) = 1 + \frac{z}{2} + \sum_{m \geq 0} \frac{2(-1)^m \zeta(2m+2)}{(2\pi)^{2m+2}} z^{2m+2}$ , which is (5.12). Finally,  $1 - e^{-z}$  vanishes exactly at  $2\pi i \mathbb{Z}$ ; at  $z = 0$  the zero of the denominator is cancelled by the numerator, while at  $\pm 2\pi i$  the numerator is nonzero, so  $\beta$  has genuine poles there and nowhere closer to the origin. The Taylor radius is therefore exactly  $2\pi$ . (Equivalently, (5.12) gives  $|b_{2m}^+|^{1/2m} \rightarrow 1/(2\pi)$  because  $\zeta(2m) \rightarrow 1$ .)  $\square$

Functional notation such as  $\beta(\text{ad}_X)$  does not require  $\text{ad}_X$  to be invertible. It means the formal series, or the analytic functional calculus in a domain avoiding the poles; in a finite-dimensional space the latter is defined whenever  $\text{spec}(\text{ad}_X)$  avoids the nonzero points  $2\pi i\ell$  (section 6.5).

## 5.4 The function $\psi$ and its complete expansion

The other scalar function on the source page is

$$\psi(x) = \frac{x \log x}{x-1} = 1 - \sum_{n \geq 1} \frac{(1-x)^n}{n(n+1)}, \quad |1-x| < 1, \quad \psi(1) = 1, \quad (5.14)$$

and, with the logarithm branch satisfying  $\log(e^z) = z$  near  $z = 0$ ,

$$\psi(e^z) = \beta(z) = \sum_{n \geq 0} B_n^+ \frac{z^n}{n!}. \quad (5.15)$$

*Proof of (5.14) and (5.15).* Put  $u = 1 - x$  and use  $\log(1 - u) = -\sum_{j \geq 1} u^j/j$  for  $|u| < 1$ . Then

$$\psi(1 - u) = \frac{(1 - u) \log(1 - u)}{-u} = (1 - u) \sum_{j \geq 1} \frac{u^{j-1}}{j} = \sum_{j \geq 1} \frac{u^{j-1}}{j} - \sum_{j \geq 1} \frac{u^j}{j} = 1 + \sum_{n \geq 1} \left( \frac{1}{n+1} - \frac{1}{n} \right) u^n,$$

and  $\frac{1}{n+1} - \frac{1}{n} = -\frac{1}{n(n+1)}$ . The same computation is valid as an identity of formal power series in  $u$ . Substituting  $x = e^z$  into  $x \log x / (x - 1)$  gives  $ze^z / (e^z - 1) = z / (1 - e^{-z}) = \beta(z)$ ; as a formal identity, substitute the positive-order series  $u = 1 - e^z$  into (5.14).  $\square$

Equation (5.15) must not be read as a branch-independent identity for arbitrary complex  $z$ ; it is an identity of power series at  $z = 0$ , and analytically it holds wherever the logarithm branch used in  $\psi$  satisfies  $\log(e^z) = z$ .

## 6 The integral formula and every order in either variable

### 6.1 The differential equation and the Poincaré integral

**Theorem 6.1** (Logarithmic differential equation). *In the degree completion, and analytically for  $X, Y$  in a sufficiently small neighborhood of  $(0, 0)$  in a Banach algebra or a finite-dimensional Lie algebra, the function  $Z(t) = \text{BCH}(X, tY) = \log(e^X e^{tY})$  is characterized by*

$$Z'(t) = \beta(\text{ad}_{Z(t)})Y, \quad Z(0) = X. \quad (6.1)$$

*Proof.* Let  $G(t) = e^X e^{tY}$ . Its left logarithmic derivative is  $G^{-1}G' = e^{-tY} e^{-X} e^X e^{tY} Y = Y$ . Since  $G(t) = e^{Z(t)}$ , (5.6) gives  $G^{-1}G' = \phi(\text{ad}_Z)Z'$ , hence  $\phi(\text{ad}_Z)Z' = Y$ . Formally,  $\text{ad}_Z$  raises total degree in  $X, Y$ , so  $\beta(\text{ad}_Z) = \sum_n b_n^+ \text{ad}_Z^n$  is a well-defined operator on  $\hat{T}$  and  $\beta(\text{ad}_Z)\phi(\text{ad}_Z) = \text{id}$  because  $\beta\phi = 1$  as scalar power series. Applying  $\beta(\text{ad}_Z)$  to both sides gives (6.1);  $Z(0) = \log e^X = X$ . Analytically, for  $Z$  small  $\|\text{ad}_Z\| \leq 2\|Z\| < 2\pi$  and  $\phi(\text{ad}_Z)$  is invertible with inverse  $\beta(\text{ad}_Z)$  given by its convergent Taylor series, so the same manipulation is valid; in a finite-dimensional Lie algebra the intrinsic version of (5.6) proved in section 8 is used.

Conversely, if  $Z(t)$  solves (6.1), then  $e^{Z(t)}$  has left logarithmic derivative  $\phi(\text{ad}_Z)Z' = \phi(\text{ad}_Z)\beta(\text{ad}_Z)Y = Y$  and initial value  $e^X$ . The equation  $G' = GY$ ,  $G(0) = e^X$ , has a unique solution: formally, comparing coefficients of  $t^n$  determines  $G$  recursively; analytically, a difference  $\Delta G$  of two solutions satisfies  $\Delta G' = \Delta G Y$  and  $\Delta G(0) = 0$ , so  $\Delta G(t) = \Delta G(0)e^{tY} = 0$  (differentiate  $\Delta G(t)e^{-tY}$  to see it is constant). Hence  $e^{Z(t)} = e^X e^{tY}$  and  $Z(t) = \text{BCH}(X, tY)$  by uniqueness of the formal, or of the local analytic, logarithm.  $\square$

This differential equation is a useful independent mechanism for constructing the BCH series: its right-hand side involves only iterated Lie brackets, and no appeal to the Hopf algebra structure is needed for the recursions derived from it below.

**Theorem 6.2** (Poincaré integral formula). *With the same interpretations,*

$$\text{BCH}(X, Y) = X + \int_0^1 \psi(e^{\text{ad}_X} e^{t \text{ad}_Y}) Y \, dt, \quad \psi(x) = \frac{x \log x}{x - 1}, \quad (6.2)$$

where  $\psi$  of an operator  $E$  near the identity is defined by the series (5.14) in  $I - E$ . In the formal setting the integral means termwise integration of the polynomial coefficients in  $t$ .

*Proof.* By theorem 5.1 and the multiplicativity of  $\text{Ad}$ ,

$$e^{\text{ad}_{Z(t)}} = \text{Ad}_{e^{Z(t)}} = \text{Ad}_{e^X e^{tY}} = \text{Ad}_{e^X} \text{Ad}_{e^{tY}} = e^{\text{ad}_X} e^{t \text{ad}_Y}.$$

Formally,  $\text{ad}_{Z(t)}$  has positive degree, so substituting it into the scalar identity  $\psi(e^z) = \beta(z)$  of (5.15) gives  $\psi(e^{\text{ad}_{Z(t)}}) = \beta(\text{ad}_{Z(t)})$  as operators on  $\widehat{T}$ ; here  $\psi(e^{\text{ad}_Z})$  is computed by substituting the positive-degree operator  $I - e^{\text{ad}_Z}$  into (5.14). Analytically, for small  $X, Y$  the operator  $E(t) = e^{\text{ad}_X} e^{t \text{ad}_Y}$  satisfies  $\|E(t) - I\| < 1$ , its logarithm series converges to  $\text{ad}_{Z(t)}$  (the unique small logarithm), and again  $\psi(E(t)) = \beta(\text{ad}_{Z(t)})$ . Hence (6.1) reads  $Z'(t) = \psi(e^{\text{ad}_X} e^{t \text{ad}_Y}) Y$ , and integrating from 0 to 1 with  $Z(0) = X$  and  $Z(1) = \text{BCH}(X, Y)$  gives (6.2). In the formal setting each fixed-degree coefficient of the integrand is a polynomial in  $t$ , so integration is termwise and no interchange of limits is involved.  $\square$

No invertibility of  $\text{ad}_Z$  is needed in this argument: it uses an identity of endomorphisms, not recovery of  $Z$  from  $\text{ad}_Z$ . Likewise  $\psi(E)$  near  $E = I$  is defined by its power series in  $E - I$ , not by multiplication by a nonexistent inverse of  $E - I$ .

## 6.2 An explicit all-bidegree formula without an unevaluated integral

Put  $A = \text{ad}_X$  and  $B = \text{ad}_Y$  and recall that  $Z_{p,q}$  denotes the component of  $Z$  with  $p$  letters  $X$  and  $q$  letters  $Y$ .

**Theorem 6.3** (Operator-block expansion of every bidegree). *The full BCH series is*

$$Z = X + Y + \sum_{k \geq 1} \frac{(-1)^{k+1}}{k(k+1)} \sum_{\substack{r_i, s_i \geq 0 \\ r_i + s_i > 0 \ (1 \leq i \leq k)}} \frac{A^{r_1} B^{s_1} \cdots A^{r_k} B^{s_k} Y}{(1 + \sum_i s_i) \prod_i r_i! s_i!}. \quad (6.3)$$

Consequently  $Z_{1,0} = X$ ,  $Z_{0,1} = Y$ ,  $Z_{p,0} = Z_{0,p} = 0$  for  $p \geq 2$ , and for  $q \geq 1$ ,  $p + q \geq 2$ ,

$$Z_{p,q} = \frac{1}{q} \sum_{k=1}^{p+q-1} \frac{(-1)^{k+1}}{k(k+1)} \sum_{\substack{r_i, s_i \geq 0, \ r_i + s_i > 0 \\ \sum_i r_i = p, \ \sum_i s_i = q-1}} \frac{A^{r_1} B^{s_1} \cdots A^{r_k} B^{s_k} Y}{\prod_i r_i! s_i!}. \quad (6.4)$$

At fixed total degree all sums are finite; every summand with  $s_k > 0$  vanishes, and so does every summand with  $p = 0$  and  $k \geq 1$ .

*Proof.* Set  $E(t) = e^A e^{tB} - I$ . Expanding the two exponentials,

$$E(t) = \sum_{\substack{r, s \geq 0 \\ r+s > 0}} \frac{t^s}{r! s!} A^r B^s,$$

so  $E(t)^k$  is the sum over  $k$ -tuples of nonempty blocks of  $\frac{t^{\sum s_i}}{\prod r_i! s_i!} A^{r_1} B^{s_1} \cdots A^{r_k} B^{s_k}$ . By (5.14) with  $1 - x$  replaced by  $-E(t)$ ,

$$\psi(I + E(t)) = I - \sum_{k \geq 1} \frac{(-E(t))^k}{k(k+1)} = I + \sum_{k \geq 1} \frac{(-1)^{k+1}}{k(k+1)} E(t)^k.$$

Insert this into (6.2). The term  $I$  integrates to  $Y$ . A block term carries  $t^{\sum s_i}$ , whose integral over  $[0, 1]$  is  $1/(1 + \sum s_i)$ ; since the operator is applied to a final letter  $Y$ , the number of letters  $Y$  in the result is  $q = 1 + \sum s_i$  and the number of letters  $X$  is  $p = \sum r_i$ . This proves (6.3) and, on collecting the terms of bidegree  $(p, q)$ , (6.4); the upper limit  $k \leq p + q - 1$  holds because every block has positive degree and the total degree of the blocks is  $p + q - 1$ . A summand with  $s_k > 0$  ends with  $BY = [Y, Y] = 0$  and vanishes; if  $p = 0$  every block has  $r_i = 0$ , so  $s_k > 0$  and the summand vanishes, which gives  $Z_{0,q} = 0$  for  $q \geq 2$ . Finally  $Z_{p,0} = 0$  for  $p \geq 2$  because  $\text{BCH}(X, 0) = X$  by (4.8).  $\square$

For example, for  $(p, q) = (1, 2)$  the only block tuples with  $\sum r_i = 1$ ,  $\sum s_i = 1$ , and  $s_k = 0$  are  $(B)(A)$  with  $k = 2$ , giving  $\frac{1}{2} \cdot \frac{-1}{6} BAY = -\frac{1}{12} [Y, [X, Y]]$ , in agreement with (4.11). Unlike a finite display with an unspecified remainder, (6.4) gives every term with three, four, or arbitrarily many occurrences of  $Y$ .

### 6.3 The part linear in $Y$ and a triangular recursion

**Corollary 6.4** (All single- $Y$  terms). *The complete part of  $Z(X, Y)$  that is linear in  $Y$  is*

$$\begin{aligned} \beta(\text{ad}_X)Y &= \frac{\text{ad}_X}{1 - e^{-\text{ad}_X}} Y = \frac{\text{ad}_X}{2} \left( 1 + \coth \frac{\text{ad}_X}{2} \right) Y \\ &= Y + \frac{1}{2} [X, Y] + \frac{1}{12} [X, [X, Y]] - \frac{1}{720} \text{ad}_X^4 Y + \frac{1}{30240} \text{ad}_X^6 Y - \cdots, \end{aligned} \quad (6.5)$$

so that  $Z(X, Y) = X + \beta(\text{ad}_X)Y + O(Y^2)$ , and the coefficient of  $\text{ad}_X^n Y$  is  $B_n^+/n!$ .

*Proof.* Replace  $Y$  by  $\varepsilon Y$  in (6.1):  $\frac{d}{dt} Z(X, t\varepsilon Y) = \varepsilon \beta(\text{ad}_Z)Y$ . The coefficient of  $\varepsilon^1$  in  $Z(X, \varepsilon Y)$  is obtained by integrating the coefficient of  $\varepsilon^1$  on the right from  $t = 0$  to 1; since  $Z = X + O(\varepsilon)$ , that coefficient is  $\beta(\text{ad}_X)Y$ . Alternatively, the terms with  $q = 1$  in (6.4) are those with all  $s_i = 0$ , that is  $Z_{p,1} = \sum_k \frac{(-1)^{k+1}}{k(k+1)} \sum_{r_1+\cdots+r_k=p, r_i \geq 1} \frac{A^p Y}{\prod r_i!}$ , which is the coefficient of  $z^p$  in  $\psi(e^z) = \beta(z)$  expanded by (5.14). The hyperbolic form is (5.11).  $\square$

This proves both versions of the source page's linear-in- $Y$  formula. The quotients and the product with  $\coth$  denote the regular function  $\beta$ ; neither expression calls for an inverse of  $\text{ad}_X$  at zero. The remainder  $O(Y^2)$  means membership in the subspace of terms with at least two occurrences of  $Y$ , not an unspecified bound on  $\|X\|$ . The linear formula by itself is not an exact BCH formula unless a further relation annihilates all higher- $Y$  terms, as in theorem 9.4.

The remainder can be resolved at every order. Write

$$Z(X, \varepsilon Y) = \sum_{q \geq 0} \varepsilon^q H_q(X; Y), \quad H_0 = X, \quad H_1 = \beta(A)Y, \quad (6.6)$$

where  $H_q$  is a formal series in  $X$  homogeneous of degree  $q$  in  $Y$ .

**Proposition 6.5** (Triangular recursion in the number of  $Y$ 's). *For every  $q \geq 1$ ,*

$$\boxed{qH_q = \sum_{m \geq 0} b_m^+ \sum_{\substack{j_1, \dots, j_m \geq 0 \\ j_1 + \cdots + j_m = q-1}} \text{ad}_{H_{j_1}} \cdots \text{ad}_{H_{j_m}} Y,} \quad (6.7)$$

where for  $m = 0$  the inner term is  $Y$  if  $q = 1$  and zero otherwise. Every index  $j_i$  is less than  $q$ , and in every fixed total degree the sum is finite. In particular the complete quadratic part is

$$H_2 = \frac{1}{2} \sum_{m \geq 1} b_m^+ \sum_{r=0}^{m-1} A^r [H_1, A^{m-1-r} Y]. \quad (6.8)$$

The nonrecursive alternative is  $H_q = \sum_{p \geq 0} Z_{p,q}$  with (6.4).

*Proof.* Differentiate (6.6) with respect to  $\varepsilon$ . By theorem 6.1 applied to  $Z(X, \varepsilon Y)$  as a function of  $\varepsilon$ ,  $\partial_\varepsilon Z = \beta(\text{ad}_Z)Y$ . Expand  $\text{ad}_Z = A + \sum_{k \geq 1} \varepsilon^k \text{ad}_{H_k}$  and each power  $\text{ad}_Z^m = \sum_{j_1, \dots, j_m \geq 0} \varepsilon^{j_1 + \dots + j_m} \text{ad}_{H_{j_1}} \cdots \text{ad}_{H_{j_m}}$  without changing the order of the factors. The coefficient of  $\varepsilon^{q-1}$  on the right is the double sum in (6.7), and on the left it is  $qH_q$ . An index  $j_i = q - 1$  forces all others to be zero, so only  $H_j$  with  $j \leq q - 1$  occur. Although there are arbitrarily many factors  $\text{ad}_{H_0} = A$ , each raises the total degree, so at fixed total degree only finitely many  $m$  contribute. For  $q = 2$  exactly one index equals 1 and the rest are 0; summing over its position  $r$  gives (6.8). The last assertion is theorem 6.3 summed over  $p$ .  $\square$

## 6.4 A bivariate generating function for the complete quadratic term

**Theorem 6.6** (Quadratic kernel). *Let*

$$K(u, v) = \frac{\beta(u)}{2u} (\beta(u + v) - \beta(v)) = \sum_{p, q \geq 0} k_{pq} u^p v^q, \quad (6.9)$$

where the quotient is the regular divided difference  $(\beta(u + v) - \beta(v))/u = \sum_m b_m^+ \sum_{r=0}^{m-1} (u + v)^r v^{m-1-r}$ . Then

$$H_2 = \sum_{p, q \geq 0} k_{pq} [A^p Y, A^q Y], \quad k_{pq} = \frac{1}{2} \sum_{a=0}^p b_a^+ b_{p+q+1-a}^+ \binom{p+q+1-a}{p+1-a}. \quad (6.10)$$

Thus every coefficient of the quadratic part of  $Z$  is an explicit finite sum of products of two Bernoulli coefficients.

*Proof.* Define a linear map  $\Phi$  from formal power series in two commuting variables  $u, v$  to  $\hat{T}$  by  $\Phi(u^a v^b) = [A^a Y, A^b Y]$ . By the derivation rule (1.3),  $A[A^a Y, A^b Y] = [A^{a+1} Y, A^b Y] + [A^a Y, A^{b+1} Y]$ , that is,  $A\Phi(P) = \Phi((u + v)P)$  for every series  $P$ , and therefore  $A^r \Phi(P) = \Phi((u + v)^r P)$ . Since  $H_1 = \beta(A)Y = \sum_a b_a^+ A^a Y$ ,

$$[H_1, A^{m-1-r} Y] = \sum_a b_a^+ [A^a Y, A^{m-1-r} Y] = \Phi(\beta(u) v^{m-1-r}).$$

Hence (6.8) becomes

$$H_2 = \frac{1}{2} \sum_{m \geq 1} b_m^+ \sum_{r=0}^{m-1} \Phi((u + v)^r \beta(u) v^{m-1-r}) = \frac{1}{2} \Phi\left(\beta(u) \sum_{m \geq 1} b_m^+ \frac{(u + v)^m - v^m}{u}\right),$$

using the finite geometric identity  $\sum_{r=0}^{m-1} (u + v)^r v^{m-1-r} = ((u + v)^m - v^m)/u$ . The sum over  $m \geq 1$  equals  $(\beta(u + v) - \beta(v))/u$ , since the  $m = 0$  term vanishes. This proves  $H_2 = \Phi(K)$ , which is the first part of (6.10). For the coefficients, expand  $(u + v)^m - v^m = \sum_{j \geq 1} \binom{m}{j} u^j v^{m-j}$ , so

$$K(u, v) = \frac{1}{2} \sum_{a \geq 0} b_a^+ u^a \sum_{m \geq 1} b_m^+ \sum_{j=1}^m \binom{m}{j} u^{j-1} v^{m-j}.$$

The coefficient of  $u^p v^q$  collects the terms with  $a + j - 1 = p$  and  $m - j = q$ , that is  $j = p + 1 - a$  and  $m = p + q + 1 - a$ , with  $0 \leq a \leq p$ ; this is the stated finite sum.  $\square$

Because brackets are antisymmetric,  $K$  may be replaced by its antisymmetrization  $\frac{1}{2}(K(u, v) - K(v, u))$  without changing  $\Phi(K)$ . For orientation, the first terms grouped by the number of  $X$ 's are

$$H_2 = -\frac{1}{12}[Y, AY] - \frac{1}{24}[Y, A^2Y] - \frac{1}{180}[Y, A^3Y] - \frac{1}{120}[AY, A^2Y] + O_X(4), \quad (6.11)$$

where  $O_X(4)$  means at least four  $X$ 's and exactly two  $Y$ 's. For example,  $k_{01} = \frac{1}{2}b_0^+b_2^+(\frac{2}{1}) = \frac{1}{12}$  and  $k_{10} = \frac{1}{2}(b_0^+b_2^+(\frac{2}{2}) + b_1^+b_1^+(\frac{1}{1})) = \frac{1}{2}(\frac{1}{12} + \frac{1}{4}) = \frac{1}{6}$ , so the coefficient of  $[Y, AY]$  is  $k_{01} - k_{10} = -\frac{1}{12}$ , in agreement with the term  $\frac{1}{12}[Y, [Y, X]]$  of (4.11). The kernel (6.9), not this short display, is the full expansion.

## 6.5 Analytic meaning when $X$ is not close to zero

The recursion (6.7) is a formal identity in the total-degree completion. When  $X$  is a fixed element of a finite-dimensional Lie algebra  $\mathfrak{g}$  that is not small, the Bernoulli series  $\beta(\text{ad}_X)$  may diverge even though the local expansion of  $\log(e^Xe^Y)$  around  $\varepsilon = 0$  exists. The following finite-at-each-order replacement gives the actual local analytic expansion.

Let  $A = \text{ad}_X$ , complexified if  $\mathfrak{g}$  is real, and assume the *regularity condition*

$$\text{spec}(A) \cap (2\pi i\mathbb{Z} \setminus \{0\}) = \emptyset. \quad (6.12)$$

By triangularization the eigenvalues of  $\phi(A)$  are the values  $\phi(\lambda)$  at the eigenvalues  $\lambda$  of  $A$ , and  $\phi$  vanishes exactly at  $2\pi i\mathbb{Z} \setminus \{0\}$ ; so (6.12) holds if and only if  $\phi(A)$  is invertible (see theorem 12.1 for the detailed argument). Then  $\beta(A)$  means the inverse  $\phi(A)^{-1}$ , which by holomorphic functional calculus equals  $\frac{1}{2\pi i} \oint_{\Gamma} \beta(z)(zI - A)^{-1}dz$  for any finite union  $\Gamma$  of positively oriented contours enclosing  $\text{spec}(A)$  and avoiding the poles of  $\beta$ ; it is not necessarily the Taylor series at zero. Since  $(\text{dexp})_X(H) = e^X\phi(A)H$  by (5.6) (in a Lie group, by the intrinsic version in section 8), the differential of the exponential at  $X$  is invertible, and the inverse function theorem provides an exponential chart centered at  $X$ ; thus  $\log(e^Xe^{\varepsilon Y})$  has a unique analytic branch  $Z(\varepsilon)$  with  $Z(0) = X$  for  $|\varepsilon|$  small, and it satisfies (6.1) in the form  $Z'(\varepsilon) = \phi(\text{ad}_{Z(\varepsilon)})^{-1}Y$ .

For operators  $E_1, \dots, E_m$  on  $\mathfrak{g}_{\mathbb{C}}$  define

$$T_m(A)[E_1, \dots, E_m] = \frac{1}{2\pi i} \oint_{\Gamma} \beta(z) R(z) E_1 R(z) E_2 \cdots E_m R(z) dz, \quad R(z) = (zI - A)^{-1}. \quad (6.13)$$

The symmetrization

$$D^m \beta(A)[E_1, \dots, E_m] = \sum_{\pi \in S_m} T_m(A)[E_{\pi(1)}, \dots, E_{\pi(m)}]$$

is the  $m$ th Fréchet derivative of the holomorphic matrix function  $\beta$ , but the unsymmetrized form is what the recursion needs.

**Theorem 6.7** (Local expansion at a regular base point). *Under (6.12),  $Z(\varepsilon) = \sum_{q \geq 0} \varepsilon^q H_q$  converges for  $|\varepsilon|$  small, with  $H_0 = X$ ,  $H_1 = \beta(A)Y$ , and for  $q \geq 2$*

$$qH_q = \sum_{m=1}^{q-1} \sum_{\substack{j_1, \dots, j_m \geq 1 \\ j_1 + \dots + j_m = q-1}} T_m(A)[\text{ad}_{H_{j_1}}, \dots, \text{ad}_{H_{j_m}}]Y. \quad (6.14)$$

*Both sums are finite, and only earlier  $H_j$  occur. In particular  $2H_2 = T_1(A)[\text{ad}_{\beta(A)Y}]Y$ . For a real Lie algebra all  $H_q$  are real.*

*Proof.* For an operator  $E$  with  $\|E\| \sup_{z \in \Gamma} \|R(z)\| < 1$ , the resolvent of  $A + E$  exists on  $\Gamma$  and is given by the Neumann series

$$(zI - A - E)^{-1} = (R(z)^{-1}(I - R(z)E))^{-1} = \sum_{k \geq 0} (R(z)E)^k R(z),$$

converging uniformly on  $\Gamma$ . For such  $E$  the spectrum of  $A + E$  lies inside  $\Gamma$  (a point  $z \in \Gamma$  or outside is in the resolvent set by the same Neumann series, and the interior of  $\Gamma$  contains  $\text{spec}(A)$  together with a neighborhood on which the estimate persists), so by the holomorphic functional calculus

$$\beta(A + E) = \frac{1}{2\pi i} \oint_{\Gamma} \beta(z)(zI - A - E)^{-1} dz = \sum_{k \geq 0} \frac{1}{2\pi i} \oint_{\Gamma} \beta(z) R(z) (ER(z))^k dz,$$

and  $\beta(A + E) = \phi(A + E)^{-1}$  because  $\phi\beta = 1$  and functional calculus is multiplicative. Now take  $E = E(\varepsilon) = \sum_{j \geq 1} \varepsilon^j \text{ad}_{H_j}$ , where the  $H_j$  are the Taylor coefficients of the analytic branch  $Z(\varepsilon)$ ; for  $|\varepsilon|$  small the norm condition holds. Substituting into the uniformly convergent double series and collecting the coefficient of  $\varepsilon^{q-1}$  gives

$$[\varepsilon^{q-1}] \beta(\text{ad}_{Z(\varepsilon)}) = \sum_{m \geq 0} \sum_{\substack{j_1, \dots, j_m \geq 1 \\ j_1 + \dots + j_m = q-1}} T_m(A) [\text{ad}_{H_{j_1}}, \dots, \text{ad}_{H_{j_m}}],$$

where the  $m = 0$  term is  $\beta(A)$  when  $q = 1$  and zero otherwise, and  $m \leq q - 1$  because each  $j_i \geq 1$ . The differential equation  $Z'(\varepsilon) = \beta(\text{ad}_{Z(\varepsilon)})Y$  holds for the analytic branch by the argument of [theorem 6.1](#) (the left logarithmic derivative of  $e^X e^{\varepsilon Y}$  is  $Y$ , and  $\phi(\text{ad}_{Z(\varepsilon)})$  is invertible for small  $\varepsilon$  by continuity of the spectrum), so comparing Taylor coefficients gives  $qH_q = [\varepsilon^{q-1}] \beta(\text{ad}_{Z(\varepsilon)})Y$ , which is (6.14). Convergence of the Taylor series of  $Z(\varepsilon)$  follows from analyticity of the local inverse of the exponential map. For a real Lie algebra, complex conjugation on  $\mathfrak{g}_{\mathbb{C}}$  commutes with  $\beta(\text{ad}_Z)$  for real  $Z$  (the contour may be chosen symmetric and  $\beta$  has real Taylor coefficients), so the uniquely determined coefficients  $H_q$  are real.  $\square$

This separates the all-orders local expansion around a regular base point  $X$  from the total-degree expansion around  $(0, 0)$ : in the former one must not evaluate the infinite Bernoulli sum in (6.7) literally when that series diverges.

## 6.6 Two finite all-order recursions by total degree

There is also a recursive proof of Lie-series existence that uses neither the primitive-element theorem nor the Poincaré integral. Let  $G(t) = e^{tX} e^{tY}$ ,  $Z(t) = \log G(t) = \sum_{n \geq 1} t^n Z_n$ , and

$$V(t) = X + e^{t \text{ad}_X} Y = \sum_{j \geq 0} t^j V_j, \quad V_0 = X + Y, \quad V_j = \frac{\text{ad}_X^j Y}{j!} \quad (j \geq 1). \quad (6.15)$$

**Theorem 6.8** (Total-degree Bernoulli recursion). *For  $n \geq 1$ ,*

$$nZ_n = \sum_{m=0}^{n-1} b_m^- \sum_{\substack{k_0 \geq 0, k_1, \dots, k_m \geq 1 \\ k_0 + k_1 + \dots + k_m = n-1}} \text{ad}_{Z_{k_1}} \cdots \text{ad}_{Z_{k_m}} V_{k_0}, \quad (6.16)$$

where for  $m = 0$  the inner expression means  $V_{n-1}$ . Every  $Z_{k_i}$  on the right has index less than  $n$ ; hence  $Z_1 = X + Y$  and, by induction, every  $Z_n$  is a Lie polynomial with rational coefficients. Equivalently, with  $S = X + Y$  and  $D = X - Y$ ,

$$Z'(t) = S + \frac{1}{2}[D, Z] + \sum_{p \geq 1} b_{2p}^+ \text{ad}_Z^{2p} S, \quad (6.17)$$

so that for  $n \geq 2$

$$nZ_n = \frac{1}{2}[D, Z_{n-1}] + \sum_{p=1}^{\lfloor (n-1)/2 \rfloor} b_{2p}^+ \sum_{\substack{i_1 + \dots + i_{2p} = n-1 \\ i_j \geq 1}} \text{ad}_{Z_{i_1}} \cdots \text{ad}_{Z_{i_{2p}}} S. \quad (6.18)$$

*Proof.* The right logarithmic derivative of  $G$  is  $G'G^{-1} = Xe^{tX}e^{tY}e^{-tY}e^{-tX} + e^{tX}Ye^{tY}e^{-tY}e^{-tX} = X + e^{tX}Ye^{-tX} = X + e^{t\text{ad}_X}Y = V(t)$  by theorem 5.1. On the other hand (5.7) gives  $G'G^{-1} = \phi(-\text{ad}_Z)Z'$ . Formally,  $\text{ad}_{Z(t)}$  has positive degree in  $t$ , so  $\gamma(\text{ad}_Z) = \beta(-\text{ad}_Z) = \sum_m b_m^- \text{ad}_Z^m$  is a well-defined operator inverting  $\phi(-\text{ad}_Z)$ , and

$$Z'(t) = \gamma(\text{ad}_{Z(t)})V(t). \quad (6.19)$$

Expand  $\text{ad}_Z^m = \sum t^{k_1 + \dots + k_m} \text{ad}_{Z_{k_1}} \cdots \text{ad}_{Z_{k_m}}$  over  $k_i \geq 1$ , multiply by  $V(t)$ , and compare the coefficients of  $t^{n-1}$ : the left side gives  $nZ_n$  and the right side gives (6.16). All indices  $k_i$  are at least 1 and sum to at most  $n-1$ , so each is less than  $n$ . Starting from  $Z_1 = V_0 = X + Y$ , induction shows that every  $Z_n$  is a rational Lie polynomial. The series so constructed satisfies (6.19), hence  $e^{Z(t)}$  has right logarithmic derivative  $V(t)$  and value 1 at  $t = 0$ , so it equals  $G(t)$  by uniqueness for  $G' = VG$  (compare coefficients of  $t$ ); thus it coincides with the logarithmic construction (2.2) at  $t = 1$ .

For the compact form, note that  $e^{\text{ad}_Z} = \text{Ad}_{e^Z} = \text{Ad}_{e^{tX}} \text{Ad}_{e^{tY}} = e^{t\text{ad}_X} e^{t\text{ad}_Y}$  and  $e^{t\text{ad}_Y}Y = Y$ , so  $e^{\text{ad}_Z}Y = e^{t\text{ad}_X}Y$  and  $V = X + e^{\text{ad}_Z}Y$ . Since  $\gamma(z)e^z = ze^z/(e^z - 1) = \beta(z)$ , (6.19) becomes  $Z' = \gamma(\text{ad}_Z)X + \beta(\text{ad}_Z)Y$ . The two scalar functions satisfy  $\gamma(z) + \beta(z) = 2\sum_{p \geq 0} b_{2p}^+ z^{2p}$  (their odd parts cancel because  $\gamma(z) = \beta(-z)$ ) and  $\beta(z) - \gamma(z) = z$  (their odd part is  $b_1^+ z = z/2$  each, with opposite signs, and  $b_{2p+1}^+ = 0$  for  $p \geq 1$ ). Writing  $X = \frac{1}{2}(S + D)$  and  $Y = \frac{1}{2}(S - D)$ ,

$$Z' = \frac{1}{2}(\gamma + \beta)(\text{ad}_Z)S + \frac{1}{2}(\gamma - \beta)(\text{ad}_Z)D = \sum_{p \geq 0} b_{2p}^+ \text{ad}_Z^{2p} S - \frac{1}{2} \text{ad}_Z D,$$

which is (6.17). Comparing coefficients of  $t^{n-1}$  gives (6.18).  $\square$

The recursion (6.16) gives  $2Z_2 = V_1 + b_1^- [Z_1, V_0] = [X, Y] - \frac{1}{2}[X + Y, X + Y] = [X, Y]$  and

$$\begin{aligned} 3Z_3 &= V_2 + b_1^- ([Z_1, V_1] + [Z_2, V_0]) + b_2^- [Z_1, [Z_1, V_0]] \\ &= \frac{1}{2}[X, [X, Y]] - \frac{1}{2}[X + Y, [X, Y]] + \frac{1}{4}[X + Y, [X, Y]] = \frac{1}{4}[X, [X, Y]] - \frac{1}{4}[Y, [X, Y]], \end{aligned}$$

which is (4.11). The compact form (6.18) with  $C = [X, Y]$  gives  $2Z_2 = \frac{1}{2}[D, S] = C$ ,  $3Z_3 = \frac{1}{4}[D, C]$  (since  $\text{ad}_S^2 S = 0$ ), and

$$4Z_4 = \frac{1}{2}[D, Z_3] + b_2^+ [S, [Z_2, S]] = \frac{1}{24}[D, [D, C]] - \frac{1}{24}[S, [S, C]] = -\frac{1}{6}[Y, [X, C]],$$

using  $[X, [Y, C]] = [Y, [X, C]]$ ; this reproduces (4.12). Eichler's associativity-based argument [3] proves the same structural theorem by a different route; no claim that his recursion coincides with (6.16) is needed here.

## 7 Convergence, remainder estimates, and logarithm obstructions

The formal theorem is unconditional, but its numerical evaluation is not. Four different questions must be kept apart: convergence of the Mercator series  $\sum_k (-1)^{k-1} (e^X e^Y - I)^k / k$ ; absolute convergence of its expansion into individual associative words or into the individual commutator summands of Dynkin's formula; convergence of the homogeneous Lie series  $\sum_n Z_n$ ; and existence of some logarithm of  $e^X e^Y$ . These are different summation procedures, and outside a common domain of absolute convergence they can behave differently; examples and further discussion appear in [7]. This section proves elementary quantitative statements sufficient for every convergence claim on the source page, without claiming optimal constants.

## 7.1 Absolute convergence in a Banach algebra

Let  $\mathcal{A}$  be a real or complex unital Banach algebra with a submultiplicative norm,  $\|AB\| \leq \|A\| \|B\|$ . Unless said otherwise we do not assume  $\|I\| = 1$ : every estimate below involves only products of positive-length words, so the norm of the identity never enters. In particular the Hilbert–Schmidt norm on  $d \times d$  matrices is admissible: by Cauchy–Schwarz,

$$\|AB\|_{\text{HS}}^2 = \sum_{i,j} \left| \sum_k A_{ik} B_{kj} \right|^2 \leq \sum_{i,j} \left( \sum_k |A_{ik}|^2 \right) \left( \sum_k |B_{kj}|^2 \right) = \|A\|_{\text{HS}}^2 \|B\|_{\text{HS}}^2,$$

although  $\|I\|_{\text{HS}} = \sqrt{d}$ . Throughout,  $s = \|X\| + \|Y\|$ .

**Theorem 7.1** (Two convergence domains and explicit remainders). *(i) If  $s < \log 2$ , the associative block expansion (2.3) is absolutely convergent, the homogeneous BCH series  $\sum_n Z_n(X, Y)$  converges absolutely, and its sum  $Z$  satisfies  $e^Z = e^X e^Y$ . More precisely, for every  $r > 0$  with  $rs < \log 2$ ,*

$$\sum_{k \geq 1} \frac{1}{k} \sum_{r_i + s_i > 0} \frac{\|X^{r_1} Y^{s_1} \dots X^{r_k} Y^{s_k}\|}{\prod_i r_i! s_i!} r^{\sum (r_i + s_i)} \leq M_s(r) := -\log(2 - e^{rs}),$$

$$\sum_{n \geq 1} r^n \|Z_n(X, Y)\| \leq M_s(r). \quad (7.1)$$

*(ii) If  $1 < r < \log 2/s$  and  $Z^{[N]} = \sum_{n=1}^N Z_n$ , then*

$$\|Z - Z^{[N]}\| \leq r^{-(N+1)} M_s(r), \quad (7.2)$$

$$\|e^{Z^{[N]}} - e^X e^Y\| \leq c_{\mathcal{A}} e^{M_s(1)} r^{-(N+1)} M_s(r), \quad (7.3)$$

where  $c_{\mathcal{A}} = \|I\|^2$ ; thus  $c_{\mathcal{A}} = 1$  for a norm with  $\|I\| = 1$ .

*(iii) If  $s < \frac{1}{2} \log 2$ , the individual right-nested commutator summands of Dynkin's formula (4.1) are absolutely summable, with*

$$\sum_{\text{all Dynkin summands}} \|\text{summand}\| \leq M_D(s) := -\frac{1}{2} \log(2 - e^{2s}), \quad (7.4)$$

and if  $Rs < \frac{1}{2} \log 2$  with  $R > 1$  the tail after degree  $N$  of this absolute sum is at most  $R^{-(N+1)} M_D(Rs)$ .

*(iv) For a finite-dimensional normed Lie algebra  $\mathfrak{g}$  with  $\|[u, v]\| \leq \kappa \|u\| \|v\|$ , Dynkin's Lie series converges absolutely whenever  $\kappa(\|X\| + \|Y\|) < \log 2$ , with absolute sum at most  $\kappa^{-1}[-\log(2 - e^{\kappa s})]$ .*

*Proof.* (i) Expanding into nonempty words,

$$\|e^X e^Y - I\| \leq \sum_{r+s > 0} \frac{\|X\|^r \|Y\|^s}{r! s!} = e^s - 1,$$

where a block with  $r = 0$  or  $s = 0$  is a pure power and contributes no factor  $\|I\|$ . Replacing each term of the block expansion of  $(e^X e^Y - I)^k$  by the product of the norms of its blocks bounds the sum of the norms of all terms of the  $k$ th power by  $(e^s - 1)^k$ . If  $s < \log 2$  then  $e^s - 1 < 1$ , and

$$\sum_{k \geq 1} \frac{(e^s - 1)^k}{k} = -\log(1 - (e^s - 1)) = -\log(2 - e^s).$$

Replacing  $X, Y$  by  $rX, rY$  multiplies a term of total degree  $n$  by  $r^n$  and gives the first inequality in (7.1). Absolute convergence of the expanded series permits regrouping by total degree; the group of degree  $n$  is  $Z_n(X, Y)$  by theorem 2.2 (the formal identity is a statement about which word coefficients occur), so  $\sum_n r^n \|Z_n\| \leq M_s(r)$ , and with  $r = 1$  the homogeneous series converges absolutely. Its sum is the Mercator logarithm  $\log(I + U)$ ,  $U = e^X e^Y - I$ ,  $\|U\| < 1$ . The scalar identity  $\exp(\log(1 + u)) = 1 + u$  holds for  $|u| < 1$  as an identity of absolutely convergent series, and substituting  $u \mapsto U$  is legitimate in the closed commutative subalgebra generated by  $U$ , exactly as in theorem 2.1. Hence  $e^Z = I + U = e^X e^Y$ .

(ii) For  $n > N$ ,  $r^n \geq r^{N+1}$ , so  $\sum_{n>N} \|Z_n\| \leq r^{-(N+1)} \sum_{n>N} r^n \|Z_n\| \leq r^{-(N+1)} M_s(r)$ , which is (7.2). For the exponentials, differentiating  $u \mapsto e^{(1-u)A} e^{uB}$  gives

$$e^B - e^A = \int_0^1 e^{(1-u)A} (B - A) e^{uB} du, \quad (7.5)$$

since the derivative of the integrand's antiderivative  $e^{(1-u)A} e^{uB}$  is  $e^{(1-u)A} (-A + B) e^{uB}$  (the factor  $A$  commutes with  $e^{(1-u)A}$  and  $B$  with  $e^{uB}$ ). Submultiplicativity gives  $\|I\| = \|I^2\| \leq \|I\|^2$ , so  $\|I\| \geq 1$ , and therefore  $\|e^C\| \leq \|I\| + \sum_{n \geq 1} \|C\|^n / n! \leq \|I\| e^{\|C\|}$ . Consequently

$$\|e^B - e^A\| \leq \|I\|^2 \int_0^1 e^{(1-u)\|A\|} e^{u\|B\|} du \|B - A\| \leq \|I\|^2 e^{\max(\|A\|, \|B\|)} \|B - A\|.$$

Apply this with  $A = Z^{[N]}$  and  $B = Z$ : both have norm at most  $\sum_n \|Z_n\| \leq M_s(1)$ , and  $\|Z - Z^{[N]}\|$  is bounded by (7.2). Since  $e^Z = e^X e^Y$  by (i), this proves (7.3).

(iii) For a right-nested bracket of length  $n$ , iterating  $\|[U, V]\| \leq 2\|U\| \|V\|$  gives  $\|R(a_1 \cdots a_n)\| \leq 2^{n-1} \prod_j \|a_j\|$ . In (4.1) discard the factor  $1/d \leq 1$ . The sum of the norms of the degree- $n$  summands is then at most  $2^{n-1}$  times the sum of the norms of the corresponding associative block terms with  $X, Y$  replaced by  $2X, 2Y$ , divided by  $2^n$ ; hence the total is at most  $\frac{1}{2} \sum_{k \geq 1} (e^{2s} - 1)^k / k = -\frac{1}{2} \log(2 - e^{2s})$ , finite when  $2s < \log 2$ . The tail bound follows exactly as in (ii) by applying the estimate to  $RX, RY$ .

(iv) Replace  $2^{n-1}$  by  $\kappa^{n-1}$  in (iii); the norm bound on a right-nested bracket of length  $n$  becomes  $\kappa^{n-1} \prod_j \|a_j\|$ , and the same computation gives the bound  $\kappa^{-1} [-\log(2 - e^{\kappa s})]$  under  $\kappa s < \log 2$ . Every finite-dimensional Lie algebra admits such a  $\kappa$ , because a bilinear map is bounded on the product of the unit spheres, so this proves local convergence without a matrix representation. If the bracket is identically zero, BCH terminates at  $X + Y$  and no positive  $\kappa$  is needed.  $\square$

The two radii  $\log 2$  and  $\frac{1}{2} \log 2$  concern different ways of summing expressions. The larger applies to the series of already-collected homogeneous Lie polynomials and to the associative word expansion; the smaller controls all the unreduced nested-commutator terms of Dynkin's multiple sum separately, and it is the bound stated in the source page's convergence discussion. Neither is claimed to be optimal.

## 7.2 The Mercator tail and the role of $\|Z\| < \log 2$

For a direct truncation of the Mercator series, put  $q = e^s - 1 < 1$ . Bounding  $1/k \leq 1/(K+1)$  in the tail gives

$$\left\| \log(e^X e^Y) - \sum_{k=1}^K \frac{(-1)^{k-1}}{k} (e^X e^Y - I)^k \right\| \leq \sum_{k>K} \frac{q^k}{k} \leq \frac{q^{K+1}}{(K+1)(1-q)}. \quad (7.6)$$

The source page displays the additional condition  $\|Z\| < \log 2$  next to its associative expansion. That condition is *not* needed to define  $Z = \log(e^X e^Y)$  by the convergent Mercator series: the

bound actually proved is  $\|Z\| \leq M_s(1)$ , and one should not silently infer a further norm bound on this constructed  $Z$ . The condition is relevant in the converse direction. Suppose  $Z_0$  is *given* with  $e^{Z_0} = e^X e^Y$  and  $\|Z_0\| < \log 2$ . Then for  $|t| \|Z_0\| < \log 2$  we have  $\|e^{tZ_0} - I\| \leq e^{|t|\|Z_0\|} - 1 < 1$ , so the Mercator logarithm  $L(t) = \log(e^{tZ_0})$  is a convergent power series in  $t$  on that disk. Near  $t = 0$  it equals  $tZ_0$  by the formal identity  $\log(e^u) = u$  applied in the commutative closed subalgebra generated by  $Z_0$ ; both sides being analytic in  $t$  on the disk, they agree throughout it, and at  $t = 1$  we get  $\log(e^{Z_0}) = Z_0$ . Combined with  $s < \log 2$ , this shows that the page’s two restrictions do suffice to conclude that its given  $Z_0$  coincides with the Mercator logarithm of  $e^X e^Y$ , and hence with the BCH sum. Without a restriction on  $Z_0$  the conclusion fails: other logarithms exist, as the trace discussion next shows.

### 7.3 The trace identity and its branch restriction

**Proposition 7.2.** *For the formal BCH series, and for finite complex or real matrices whenever the homogeneous series converges,*

$$\operatorname{tr} Z(X, Y) = \operatorname{tr} X + \operatorname{tr} Y. \quad (7.7)$$

*For an arbitrary complex matrix  $L$  with  $e^L = e^X e^Y$ , the branch-independent conclusion is only*

$$\operatorname{tr} L - \operatorname{tr} X - \operatorname{tr} Y \in 2\pi i\mathbb{Z}. \quad (7.8)$$

*Proof.* The identity  $\operatorname{tr}(AB) = \operatorname{tr}(BA)$  follows by interchanging the two summation indices in  $\sum_{i,j} A_{ij} B_{ji}$ . Hence every commutator has trace zero, and every nested bracket of length at least two, being a commutator, has trace zero. By theorem 3.6 each  $Z_n$  with  $n \geq 2$  is a sum of such brackets, so  $\operatorname{tr} Z_n = 0$  for  $n \geq 2$  and  $\operatorname{tr} Z_1 = \operatorname{tr} X + \operatorname{tr} Y$ . This proves (7.7) degree by degree; when the series converges, continuity of the trace (a linear functional on a finite-dimensional space) gives it for the sum.

For (7.8) we first show  $\det e^A = e^{\operatorname{tr} A}$  for every complex square matrix  $A$ . Every complex matrix is triangularizable: choose an eigenvector  $v_1$ , complete it to a basis, and observe that in this basis  $A$  is block upper triangular with a  $1 \times 1$  block; induct on the dimension of the quotient. In an upper-triangular basis with diagonal  $\lambda_1, \dots, \lambda_d$ , every power  $A^k$  is upper triangular with diagonal  $\lambda_j^k$ , so  $e^A$  is upper triangular with diagonal  $e^{\lambda_j}$ , and  $\det e^A = \prod_j e^{\lambda_j} = e^{\sum \lambda_j} = e^{\operatorname{tr} A}$ . Applying this to  $e^L = e^X e^Y$  and using multiplicativity of the determinant gives  $e^{\operatorname{tr} L} = e^{\operatorname{tr} X} e^{\operatorname{tr} Y}$ , that is  $e^{\operatorname{tr} L - \operatorname{tr} X - \operatorname{tr} Y} = 1$ , and the scalar exponential takes the value 1 exactly on  $2\pi i\mathbb{Z}$ .  $\square$

A scalar counterexample to unrestricted use of the principal logarithm: take  $X = Y = 3\pi i/4$  in dimension one. The BCH logarithm is  $X + Y = 3\pi i/2$ , whereas the principal logarithm of the product  $e^X e^Y = e^{3\pi i/2} = -i$  is  $-\pi i/2$ ; they differ by  $2\pi i$ . Thus the word “logarithm” in (7.7) cannot be left branch-independent. When  $L, X, Y$  are all real matrices the difference in (7.8) is real, hence zero.

### 7.4 The $\mathfrak{sl}_2$ example, completely resolved

**Theorem 7.3.** *Let*

$$X = \begin{pmatrix} 0 & i\pi \\ i\pi & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}. \quad (7.9)$$

*Then*

$$e^X e^Y = \begin{pmatrix} -1 & -1 \\ 0 & -1 \end{pmatrix} = -I - Y. \quad (7.10)$$

The complete set of complex  $2 \times 2$  logarithms of this matrix is

$$L_k = (2k + 1)\pi i I + Y, \quad k \in \mathbb{Z}. \quad (7.11)$$

None of them is traceless, so none lies in  $\mathfrak{sl}_2(\mathbb{C})$ , and the homogeneous BCH series  $\sum_n Z_n(X, Y)$  does not converge, even conditionally, although the product has infinitely many logarithms in  $\mathfrak{gl}_2(\mathbb{C})$ .

*Proof.* Since  $X^2 = -\pi^2 I$ , the even and odd parts of the exponential series give  $e^X = \sum_m \frac{(-\pi^2)^m}{(2m)!} I + \sum_m \frac{(-\pi^2)^m}{(2m+1)!} X = \cos \pi I + \frac{\sin \pi}{\pi} X = -I$ . Since  $Y^2 = 0$ ,  $e^Y = I + Y$ . Hence  $e^X e^Y = -I - Y$ , which is (7.10).

Let  $L$  be any logarithm,  $e^L = -I - Y$ . A matrix commutes with every power of itself, hence with its exponential; so  $L$  commutes with  $-I - Y$  and therefore with  $Y$ . Writing  $L = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ , the equation  $LY = YL$  reads  $\begin{pmatrix} 0 & a \\ 0 & c \end{pmatrix} = \begin{pmatrix} c & d \\ 0 & 0 \end{pmatrix}$ , so  $c = 0$  and  $a = d$ : the centralizer of  $Y$  consists of the matrices  $aI + bY$ . For such a matrix,  $aI$  and  $bY$  commute and  $(bY)^2 = 0$ , so  $e^{aI+bY} = e^a(I + bY)$ . Equality with  $-I - Y$  forces  $e^a = -1$  and  $e^a b = -1$ , that is  $a = (2k + 1)\pi i$  and  $b = 1$ . This proves (7.11). The trace of  $L_k$  is  $2(2k + 1)\pi i \neq 0$ .

Suppose, to reach a contradiction, that  $\sum_{n \geq 1} Z_n(X, Y)$  converged to some matrix  $S$ . Then  $Z_n \rightarrow 0$ , so the coefficients are bounded and the matrix power series  $Z(t) = \sum_{n \geq 1} t^n Z_n(X, Y)$  converges absolutely, entry by entry, for  $|t| < 1$ , defining an analytic matrix function there. For  $|t|(\|X\| + \|Y\|) < \log 2$  we have  $Z(t) = \text{BCH}(tX, tY)$  by homogeneity, and theorem 7.1(i) gives  $e^{Z(t)} = e^{tX} e^{tY}$ . Both sides are analytic in  $t$  on the unit disk (the exponential of an analytic matrix function is analytic, being an absolutely and locally uniformly convergent series of analytic functions), so by the identity theorem  $e^{Z(t)} = e^{tX} e^{tY}$  for all  $|t| < 1$ .

Abel's limit theorem now gives  $Z(t) \rightarrow S$  as  $t \uparrow 1$  along the real axis. For completeness: let  $S_m = \sum_{n=1}^m Z_n$  and  $S_0 = 0$ . For  $0 < t < 1$ , summation by parts gives

$$Z(t) = \sum_{n \geq 1} t^n (S_n - S_{n-1}) = \sum_{n \geq 1} t^n S_n - \sum_{n \geq 1} t^{n+1} S_n = (1 - t) \sum_{n \geq 1} t^n S_n,$$

all series converging absolutely because the  $S_n$  are bounded. Since  $(1 - t) \sum_{n \geq 1} t^n = t$ , we obtain  $Z(t) - tS = (1 - t) \sum_{n \geq 1} t^n (S_n - S)$ . Given  $\varepsilon > 0$  choose  $N$  with  $\|S_n - S\| < \varepsilon$  for all  $n > N$ . The finite part  $(1 - t) \sum_{n \leq N} t^n (S_n - S)$  tends to zero as  $t \uparrow 1$ , and the tail is bounded by  $\varepsilon(1 - t) \sum_{n > N} t^n \leq \varepsilon$ . Hence  $\limsup_{t \uparrow 1} \|Z(t) - tS\| \leq \varepsilon$  for every  $\varepsilon > 0$ , so  $Z(t) - tS \rightarrow 0$  and  $Z(t) \rightarrow S$ . Continuity of the exponential then gives  $e^S = \lim e^{Z(t)} = \lim e^{tX} e^{tY} = e^X e^Y = -I - Y$ . But every  $Z_n$  is traceless by theorem 7.2, so  $\text{tr } S = \lim \text{tr } S_m = 0$ , contradicting the classification (7.11). Hence the series does not converge.  $\square$

This example simultaneously distinguishes existence of a logarithm, existence of a logarithm in the specified Lie algebra, and convergence of the BCH series. They are genuinely different questions. The obstruction is not that the invertible matrix  $-I - Y$  lacks a logarithm; it has infinitely many. The obstruction is that it has no logarithm in the traceless Lie algebra generated by  $X$  and  $Y$ .

## 8 Lie groups: exponential coordinates, the local–global distinction, and nilpotent group laws

### 8.1 Matrix groups

For a matrix Lie group  $G \subseteq \text{GL}_d$ , its Lie algebra  $\mathfrak{g}$  is the tangent space at the identity. The left-invariant vector field associated with  $X \in \mathfrak{g}$  is  $X^L(g) = gX$ . Regarding these as vector fields

on the open subset  $\mathrm{GL}_d$  of the vector space of matrices, the bracket of two vector fields  $U, V$  is  $[U, V](g) = DV(g)[U(g)] - DU(g)[V(g)]$ , and since  $DX^L(g)[H] = HX$ ,

$$[X^L, Y^L](g) = (gX)Y - (gY)X = g(XY - YX) = [X, Y]^L(g).$$

Thus the Lie bracket of  $\mathfrak{g}$ , defined through left-invariant fields, is the matrix commutator. The integral curve of  $X^L$  through  $I$  solves  $g' = gX$ ,  $g(0) = I$ ; the matrix exponential  $e^{tX}$  solves this equation, and the solution is unique (differentiate  $g(t)e^{-tX}$ ), so the Lie-group exponential is the ordinary matrix exponential. This proves the matrix statements of the source page.

## 8.2 The Maurer–Cartan form and an intrinsic differential of the exponential

None of the assertions on the page requires a matrix embedding. For an arbitrary finite-dimensional real or complex Lie group  $G$  with Lie algebra  $\mathfrak{g}$ , define the left Maurer–Cartan form by  $\theta_g = (L_{g^{-1}})_* : T_g G \rightarrow T_e G = \mathfrak{g}$ ; in a matrix group it is  $g^{-1}dg$ . On a left-invariant field it takes the constant value  $\theta(U^L) = U$ .

**Proposition 8.1** (Maurer–Cartan equation). *The  $\mathfrak{g}$ -valued one-form  $\theta$  satisfies*

$$d\theta + \frac{1}{2}[\theta, \theta] = 0, \quad \frac{1}{2}[\theta, \theta](U, V) := [\theta(U), \theta(V)]. \quad (8.1)$$

Consequently, for a smooth two-parameter map  $g(s, t)$  into  $G$ , the left-trivialized partial derivatives  $\alpha = \theta(\partial_s g)$  and  $\beta = \theta(\partial_t g)$  satisfy the mixed-variation identity

$$\partial_s \beta - \partial_t \alpha + [\alpha, \beta] = 0. \quad (8.2)$$

*Proof.* For a one-form  $\omega$  and vector fields  $U, V$  one has  $d\omega(U, V) = U(\omega(V)) - V(\omega(U)) - \omega([U, V])$ . Evaluate on left-invariant fields  $U^L, V^L$ : the functions  $\theta(V^L) = V$  and  $\theta(U^L) = U$  are constant, so the first two terms vanish, and  $[U^L, V^L] = [U, V]^L$  by definition of the Lie algebra bracket; hence  $d\theta(U^L, V^L) = -\theta([U, V]^L) = -[U, V]$ . On the other hand  $\frac{1}{2}[\theta, \theta](U^L, V^L) = [\theta(U^L), \theta(V^L)] = [U, V]$ . Both sides of (8.1) are tensorial (bilinear over functions) and left-invariant fields span every tangent space, so (8.1) holds identically. In a matrix group the same result is the direct computation  $d(g^{-1}) = -g^{-1}(dg)g^{-1}$ , whence  $d(g^{-1}dg) = -g^{-1}dg \wedge g^{-1}dg$ .

For (8.2), pull  $\theta$  back by  $g$ :  $g^*\theta = \alpha ds + \beta dt$ . Pullback commutes with  $d$  and with the bracket of forms, so (8.1) gives  $d(g^*\theta) = -[\alpha, \beta] ds \wedge dt$ , while directly  $d(\alpha ds + \beta dt) = (\partial_s \beta - \partial_t \alpha) ds \wedge dt$ . Comparing coefficients gives (8.2). (In a matrix group:  $\partial_s(g^{-1}\partial_t g) = -\alpha\beta + g^{-1}\partial_s\partial_t g$  and  $\partial_t(g^{-1}\partial_s g) = -\beta\alpha + g^{-1}\partial_t\partial_s g$ ; subtracting gives  $-\alpha\beta$ .)  $\square$

**Proposition 8.2** (Intrinsic differential of the exponential). *For  $X, H \in \mathfrak{g}$ ,*

$$(L_{\exp(-X)})_*(d\exp)_X(H) = \phi(\mathrm{ad}_X)H = \int_0^1 e^{-u\mathrm{ad}_X} H du, \quad (8.3)$$

and similarly  $(R_{\exp(-X)})_*(d\exp)_X(H) = \phi(-\mathrm{ad}_X)H$ . In particular  $(d\exp)_0 = \mathrm{id}_{\mathfrak{g}}$ , and  $\mathrm{Ad}_{\exp X} = e^{\mathrm{ad}_X}$  intrinsically.

*Proof.* Put  $g(s, t) = \exp(s(X + tH))$ . For fixed  $t$ ,  $s \mapsto g(s, t)$  is the one-parameter subgroup with velocity  $X + tH$ , so its left-trivialized velocity is  $\alpha(s, t) = \theta(\partial_s g) = X + tH$ . Let  $\beta(s, t) = \theta(\partial_t g)$ ; then  $\beta(0, t) = 0$  because  $g(0, t) = e$  for all  $t$ . By (8.2),  $\partial_s \beta = \partial_t \alpha - [\alpha, \beta] = H - [X + tH, \beta]$ . At  $t = 0$  this is the linear equation  $\partial_s \beta = H - \mathrm{ad}_X \beta$  with  $\beta(0) = 0$ , whose unique solution is  $\beta(s) = \int_0^s e^{-(s-u)\mathrm{ad}_X} H du$  (check by differentiation:  $\partial_s \beta = H - \mathrm{ad}_X \beta$ ). At  $s = 1$ , substituting  $v = 1 - u$ ,  $\beta(1, 0) = \int_0^1 e^{-v\mathrm{ad}_X} H dv = \phi(\mathrm{ad}_X)H$  by the termwise integration of theorem 5.2. Since  $\beta(1, 0) = \theta(\partial_t g(1, 0)) = (L_{\exp(-X)})_*(d\exp)_X(H)$ , this is (8.3). The right-trivialized version follows by applying the same argument to the right Maurer–Cartan form, or from

$(R_{\exp(-X)})_* = \text{Ad}_{\exp X} \circ (L_{\exp(-X)})_*$  together with  $\text{Ad}_{\exp X} \phi(\text{ad}_X) = e^{\text{ad}_X} \phi(\text{ad}_X) = \phi(-\text{ad}_X)$ . At  $X = 0$  the formula gives  $(d\exp)_0(H) = H$ . Finally, differentiating the one-parameter group of automorphisms  $t \mapsto \text{Ad}_{\exp(tX)}$  gives  $\frac{d}{dt} \text{Ad}_{\exp(tX)} = \text{ad}_X \text{Ad}_{\exp(tX)}$  with  $\text{Ad}_{\exp(0)} = I$ , whose unique solution is  $e^{t\text{ad}_X}$ ; this is the intrinsic Campbell identity.  $\square$

### 8.3 Local exponential coordinates and the local BCH law

**Theorem 8.3** (Local BCH multiplication). *Let  $G$  be a finite-dimensional real or complex Lie group. There is a neighborhood  $U$  of 0 in  $\mathfrak{g}$  on which  $\exp$  is a diffeomorphism onto a neighborhood of the identity. After shrinking  $U$ , for all  $X, Y \in U$ ,*

$$\exp X \exp Y = \exp(\text{BCH}(X, Y)), \quad (8.4)$$

where the universal series converges absolutely in  $\mathfrak{g}$ . Hence the local multiplication law of  $G$  is determined entirely by the Lie bracket.

*Proof.* By theorem 8.2,  $(d\exp)_0 = \text{id}$ , so the inverse function theorem provides the chart. Choose a norm on  $\mathfrak{g}$  and  $\kappa$  with  $\|[u, v]\| \leq \kappa \|u\| \|v\|$ . By theorem 7.1(iv), for  $\kappa(\|X\| + \|Y\|) < \log 2$  the series  $Z(t) = \text{BCH}(X, tY) = \sum_n Z_n(X, tY)$  converges absolutely, uniformly for  $t \in [0, 1]$ , and so does the series with  $X, Y$  replaced by  $rX, rY$  for some  $r > 1$ , provided  $X, Y$  are small enough. Consequently  $Z(t)$  is given by a power series in  $t$  with radius greater than 1, and it may be differentiated termwise in  $t$ . The differential equation (6.1),  $Z'(t) = \beta(\text{ad}_{Z(t)})Y$ , holds in the formal setting coefficient by coefficient in  $t$  (each coefficient being a Lie polynomial identity in  $X, Y$ , valid in  $\mathfrak{g}$  by universality, section 3.3); for  $\|Z(t)\|$  small enough that  $\|\text{ad}_{Z(t)}\| < 2\pi$ , the right side is a convergent series in  $t$  with the same coefficients, so the equation holds analytically.

Now let  $L(t) = \log(\exp X \exp(tY))$  in the exponential chart, defined for small  $X, Y$  and  $t \in [0, 1]$  after shrinking  $U$  so that the relevant products stay in the chart. Its left-trivialized derivative is  $\theta(\frac{d}{dt} \exp X \exp(tY)) = Y$  (the curve  $t \mapsto \exp X \exp(tY)$  is the left translate by  $\exp X$  of a one-parameter subgroup). By theorem 8.2 the same left-trivialized derivative equals  $\phi(\text{ad}_{L(t)})L'(t)$ . For  $L$  small,  $\phi(\text{ad}_L)$  is invertible with inverse  $\beta(\text{ad}_L)$ , so  $L' = \beta(\text{ad}_L)Y$  with  $L(0) = X$ . The map  $L \mapsto \beta(\text{ad}_L)Y$  is analytic on a neighborhood of 0, so the initial value problem has a unique solution by the Picard–Lindelöf theorem. Both  $L(t)$  and  $Z(t)$  solve it; hence  $L(1) = Z(1) = \text{BCH}(X, Y)$ , which is (8.4) since  $\exp$  is injective on  $U$ .  $\square$

The local assertion is not a global classification. The additive group  $\mathbb{R}$  and the circle group  $S^1 = \{e^{it}\}$  have the same one-dimensional abelian Lie algebra, but one is noncompact and the other compact, so they are not isomorphic as Lie groups; they differ in topology, not in local structure. The identity map of their Lie algebras integrates from  $\mathbb{R}$  to  $S^1$  as  $t \mapsto e^{it}$ , but not from  $S^1$  to  $\mathbb{R}$ : a continuous homomorphism from the compact group  $S^1$  to  $\mathbb{R}$  has compact image, which is a compact subgroup of  $\mathbb{R}$ , hence  $\{0\}$ , so its differential is zero.

### 8.4 Homomorphisms, representations, and integration

**Theorem 8.4** (Local Lie correspondence). *Let  $f : \mathfrak{g} \rightarrow \mathfrak{h}$  be a homomorphism of Lie algebras of Lie groups  $G, H$ . Then*

$$F(\exp_G X) = \exp_H(fX) \quad (8.5)$$

defines a smooth map on an identity neighborhood of  $G$  which is a local homomorphism:  $F(gh) = F(g)F(h)$  whenever  $g, h, gh$  lie in a suitable smaller neighborhood. Conversely, the differential at the identity of a smooth local homomorphism preserves Lie brackets, and a local homomorphism is determined near the identity by that differential.

*Proof.* On the exponential neighborhood of [theorem 8.3](#), (8.5) defines a smooth map, being the composition  $\exp_H \circ f \circ \log_G$ . By [theorem 4.2](#),  $f$  preserves every  $Z_n$ , and by continuity it preserves the convergent sum:  $f(\text{BCH}(X, Y)) = \text{BCH}(fX, fY)$ . Hence, for  $X, Y$  small enough that (8.4) applies in both groups,

$$\begin{aligned} F(\exp X \exp Y) &= F(\exp \text{BCH}(X, Y)) = \exp_H(f \text{BCH}(X, Y)) = \exp_H(\text{BCH}(fX, fY)) \\ &= \exp_H(fX) \exp_H(fY) = F(\exp X) F(\exp Y). \end{aligned}$$

For the converse, let  $F$  be a smooth local homomorphism with differential  $A = (dF)_e$ . Differentiating  $F(g \exp(tX)) = F(g)F(\exp(tX))$  at  $t = 0$  shows that  $F$  relates the left-invariant field  $X^L$  on  $G$  to the field  $(AX)^L$  on  $H$ , wherever both are defined. Related vector fields have related brackets (a standard consequence of the chain rule applied to  $F_*$ ), so  $[X^L, Y^L] = [X, Y]^L$  is related to  $[(AX)^L, (AY)^L] = [AX, AY]^L$ ; evaluating at  $e$  gives  $A[X, Y] = [AX, AY]$ . Finally, since  $F$  maps the one-parameter subgroup  $\exp(tX)$  to a local one-parameter subgroup of  $H$  with initial velocity  $AX$ , uniqueness of integral curves gives  $F(\exp(tX)) = \exp_H(tAX)$  for small  $t$ ; as  $\exp$  is onto an identity neighborhood,  $F$  is determined near  $e$  by  $A$ .  $\square$

**Theorem 8.5** (Integration from a simply connected group). *If  $G$  is connected and simply connected and  $H$  is a Lie group, every Lie algebra homomorphism  $f : \mathfrak{g} \rightarrow \mathfrak{h}$  integrates to a unique smooth homomorphism  $F : G \rightarrow H$  with  $(dF)_e = f$ . In particular every finite-dimensional representation of  $\mathfrak{g}$  integrates to a representation of  $G$ .*

*Proof.* Let  $F_0$  be the local homomorphism of [theorem 8.4](#), defined and multiplicative on an open identity neighborhood  $W$  in the sense that  $F_0(uv) = F_0(u)F_0(v)$  whenever  $u, v, uv \in W$ . Choose a smaller symmetric identity neighborhood  $W_1$  with  $W_1 W_1 \subseteq W$ .

*Step 1: paths.* Let  $\gamma : [0, 1] \rightarrow G$  be a continuous path with  $\gamma(0) = e$  and  $\gamma(1) = g$ . By uniform continuity there is a partition  $0 = t_0 < t_1 < \dots < t_m = 1$  such that every increment  $u_j = \gamma(t_{j-1})^{-1}\gamma(t_j)$  lies in  $W_1$ . Define

$$F_\gamma(g) = F_0(u_1)F_0(u_2) \cdots F_0(u_m).$$

This does not depend on the partition. Inserting one additional point  $t'$  between  $t_{j-1}$  and  $t_j$  replaces  $u_j$  by the product  $u'u''$  of two increments; if the partition is fine enough that  $u', u'' \in W_1$ , then  $u'u'' = u_j \in W$  and  $F_0(u_j) = F_0(u')F_0(u'')$  by local multiplicativity, so the product is unchanged. Any two sufficiently fine partitions have a common refinement obtained by inserting points one at a time, so  $F_\gamma(g)$  is well defined.

*Step 2: homotopy invariance.* Let  $\Gamma : [0, 1]^2 \rightarrow G$  be a continuous homotopy with  $\Gamma(\cdot, 0) = \gamma_0$ ,  $\Gamma(\cdot, 1) = \gamma_1$ ,  $\Gamma(0, \cdot) = e$ ,  $\Gamma(1, \cdot) = g$ . By uniform continuity there is a rectangular grid so fine that for any two grid points  $p, q$  of the same closed cell,  $\Gamma(p)^{-1}\Gamma(q) \in W_1$ . Around a single cell with corners  $a, b, c, d$  (in cyclic order), the four increments  $u = \Gamma(a)^{-1}\Gamma(b)$ ,  $v = \Gamma(b)^{-1}\Gamma(c)$ ,  $u' = \Gamma(a)^{-1}\Gamma(d)$ ,  $v' = \Gamma(d)^{-1}\Gamma(c)$  satisfy  $uv = u'v' = \Gamma(a)^{-1}\Gamma(c) \in W_1$ , so local multiplicativity gives  $F_0(u)F_0(v) = F_0(uv) = F_0(u')F_0(v')$ . Hence the products of  $F_0$  along the two edge paths of the cell from  $a$  to  $c$  coincide. Sweeping across the grid one cell at a time, replacing the lower-right edge pair of a cell by its upper-left pair, converts the grid path along the bottom and right boundary into the path along the left and top boundary without changing the product. The left and right boundary edges are constant paths (at  $e$  and at  $g$ ) and contribute factors  $F_0(e) = e$ . The bottom and top boundary paths are  $\gamma_0$  and  $\gamma_1$  evaluated at the grid points, whose increments form fine partitions in the sense of Step 1. Therefore  $F_{\gamma_0}(g) = F_{\gamma_1}(g)$ .

*Step 3: definition of  $F$ .* Since  $G$  is simply connected, any two paths from  $e$  to  $g$  are homotopic with fixed endpoints, and since  $G$  is connected such paths exist. Define  $F(g) = F_\gamma(g)$  for any such path.

*Step 4: multiplicativity, smoothness, differential.* Given paths  $\gamma$  from  $e$  to  $g$  and  $\delta$  from  $e$  to  $h$ , the concatenation of  $\gamma$  with the left translate  $g\delta$  is a path from  $e$  to  $gh$ . The increments of  $g\delta$

are  $(g\delta(t_{j-1}))^{-1}g\delta(t_j) = \delta(t_{j-1})^{-1}\delta(t_j)$ , the original increments of  $\delta$ . Hence  $F(gh) = F(g)F(h)$ . For  $u \in W_1$ , appending to a path to  $g$  the short path  $t \mapsto g \exp(t \log u)$  from  $g$  to  $gu$  gives  $F(gu) = F(g)F_0(u)$ ; thus  $F$  agrees near  $g$  with the smooth map  $u \mapsto F(g)F_0(u)$ , so  $F$  is smooth, and at  $g = e$  it agrees with  $F_0$ , so  $(dF)_e = f$ .

*Step 5: uniqueness.* Any homomorphism  $F'$  with differential  $f$  satisfies  $F'(\exp X) = \exp_H(fX)$  (it maps one-parameter subgroups to one-parameter subgroups with the given initial velocity), so  $F' = F_0 = F$  on the exponential neighborhood. The set where  $F' = F$  is a subgroup containing an identity neighborhood, hence an open subgroup; its cosets are open, so it is also closed; connectedness makes it all of  $G$ .

Taking  $H = \mathrm{GL}(E)$  for a finite-dimensional vector space  $E$  gives the statement about representations.  $\square$

Without simple connectedness the path construction can yield a nontrivial value around a loop; this is the precise global obstruction absent from the local BCH law. Descent from a simply connected covering group to a specified quotient requires the covering kernel to act trivially.

## 8.5 Nilpotent Lie algebras and the polynomial group law

Write  $\mathfrak{g}^1 = \mathfrak{g}$  and  $\mathfrak{g}^{r+1} = [\mathfrak{g}, \mathfrak{g}^r]$  for the lower central series. The algebra is nilpotent of class at most  $c$  if  $\mathfrak{g}^{c+1} = 0$ .

**Theorem 8.6** (Termination and the nilpotent group law). *In a nilpotent Lie algebra of class at most  $c$ , every  $Z_n$  with  $n > c$  vanishes identically, and*

$$X * Y = \mathrm{BCH}(X, Y) = \sum_{n=1}^c Z_n(X, Y) \quad (8.6)$$

*is a polynomial group law on the vector space  $\mathfrak{g}$  with identity 0 and inverse  $-X$ . For finite-dimensional real  $\mathfrak{g}$  this makes  $\mathfrak{g}$  a connected, simply connected Lie group whose Lie algebra is  $\mathfrak{g}$  with its original bracket and whose exponential map is the identity of  $\mathfrak{g}$ . Any connected simply connected Lie group with Lie algebra  $\mathfrak{g}$  is isomorphic to it, and in every Lie group with Lie algebra  $\mathfrak{g}$  the identity  $\exp X \exp Y = \exp \mathrm{BCH}(X, Y)$  holds for all  $X, Y \in \mathfrak{g}$ .*

*Proof.* By the Jacobi identity in the form (1.3), induction on  $r$  gives  $[\mathfrak{g}^r, \mathfrak{g}^s] \subseteq \mathfrak{g}^{r+s}$ : for  $r = 1$  this is the definition, and if it holds for  $r$  then for  $x \in \mathfrak{g}, y \in \mathfrak{g}^r, z \in \mathfrak{g}^s$ ,  $[[x, y], z] = [x, [y, z]] - [y, [x, z]] \in [\mathfrak{g}, \mathfrak{g}^{r+s}] + [\mathfrak{g}^r, \mathfrak{g}^{s+1}] \subseteq \mathfrak{g}^{r+s+1}$ . Hence every bracket monomial of degree  $n$  in elements of  $\mathfrak{g}$  lies in  $\mathfrak{g}^n$ , and vanishes for  $n > c$ . Every  $Z_n$  is such a combination, so  $Z_n = 0$  for  $n > c$ .

The formal identities (4.7) and (4.8) are identities between Lie series in three free variables. Evaluating them in  $\mathfrak{g}$  turns both sides into finite polynomial identities, because all Lie monomials of degree greater than  $c$  vanish, including those produced by substituting one polynomial into another. Thus  $(X * Y) * T = X * (Y * T)$ ,  $X * 0 = 0 * X = X$ , and  $X * (-X) = 0$  hold for all  $X, Y, T \in \mathfrak{g}$ , not merely near zero. So  $(\mathfrak{g}, *)$  is a group, with polynomial multiplication and inversion; it is a Lie group whose underlying manifold is a vector space, hence connected and simply connected.

Its Lie algebra is computed from the group commutator. Write  $a = sX$ ,  $b = tY$ , and let  $O(3)$  denote terms of total degree at least three in  $s, t$ . Using  $u * v = u + v + \frac{1}{2}[u, v] + O(3)$  three times,

$$\begin{aligned} (a * b) * (-a) &= (a + b + \tfrac{1}{2}[a, b]) - a + \tfrac{1}{2}[a + b, -a] + O(3) = b + [a, b] + O(3), \\ ((a * b) * (-a)) * (-b) &= (b + [a, b]) - b + \tfrac{1}{2}[b, -b] + O(3) = [a, b] + O(3) = st[X, Y] + O(3). \end{aligned}$$

The mixed second derivative  $\partial_s \partial_t$  at  $s = t = 0$  of the group commutator  $g_s h_t g_s^{-1} h_t^{-1}$  of two one-parameter subgroups is the Lie bracket of their initial velocities in the Lie algebra of any Lie

group, so the bracket of  $(\mathfrak{g}, *)$  is the original bracket of  $\mathfrak{g}$ . (Alternatively, the left-invariant field of  $X$  is  $g \mapsto \frac{d}{dt}g * (tX)|_{t=0}$ , and computing brackets of these polynomial fields gives the same result.) Since  $Z(sX, tX) = (s + t)X$  (all brackets of multiples of  $X$  vanish), the one-parameter subgroups are  $t \mapsto tX$ , so the exponential map is the identity.

If  $G$  is a connected simply connected Lie group with Lie algebra  $\mathfrak{g}$ , [theorem 8.5](#) integrates the identity map of  $\mathfrak{g}$  in both directions, giving homomorphisms  $(\mathfrak{g}, *) \rightarrow G$  and  $G \rightarrow (\mathfrak{g}, *)$  whose composites have differential the identity, hence are the identity by the uniqueness part; so they are inverse isomorphisms. In particular the exponential map of  $G$  is a global diffeomorphism. For an arbitrary Lie group  $G'$  with Lie algebra  $\mathfrak{g}$ , the homomorphism  $(\mathfrak{g}, *) \rightarrow G'$  integrating the identity map sends  $X$  to  $\exp_{G'} X$  (it maps one-parameter subgroups to one-parameter subgroups), and multiplicativity reads  $\exp_{G'}(X * Y) = \exp_{G'} X \exp_{G'} Y$ .  $\square$

Termination is a statement about Lie brackets, not about matrix powers: a nilpotent Lie algebra may be represented by matrices that are not themselves nilpotent.

For a basis  $P, Q, C$  with  $[P, Q] = C$  and all other brackets zero (the Heisenberg algebra, of class two), (8.6) reads

$$(x, y, z) * (x', y', z') = (x + x', y + y', z + z' + \tfrac{1}{2}(xy' - yx')) \quad (8.7)$$

in coordinates  $xP + yQ + zC$ , since  $\frac{1}{2}[xP + yQ, x'P + y'Q] = \frac{1}{2}(xy' - yx')C$ . A concrete matrix instance is  $P = E_{12}$ ,  $Q = E_{23}$ ,  $C = E_{13}$  in the strictly upper triangular  $3 \times 3$  matrices, where both sides of  $e^X e^Y = e^{X+Y+[X,Y]/2}$  are finite matrix polynomials. This group law underlies the Weyl and displacement identities of [section 13](#).

## 9 Exactly summable commutator relations

The universal series is infinite, but algebraic relations can remove almost all of it. The proofs below establish *global* exponential identities, not merely small-neighborhood Taylor expansions. For Banach algebras all exponentials exist; for finite-dimensional Lie groups the same proofs use one-parameter subgroups and their right-trivialized logarithmic derivatives ([theorem 8.2](#)), together with uniqueness of solutions of the ordinary differential equation  $g' = V(t)^R(g)$  on the group. Complex scalar parameters are understood in a complex Banach algebra or complex Lie group; in a real Lie group all parameters and linear combinations must be real.

### 9.1 Commuting elements and central commutators

**Proposition 9.1** (Commuting case). *If  $[X, Y] = 0$ , then  $Z(X, Y) = X + Y$  formally, and  $e^X e^Y = e^{X+Y}$  for all bounded  $X, Y$  without a smallness hypothesis.*

*Proof.* For commuting elements the binomial theorem holds, and the Cauchy product of the two absolutely convergent series gives  $e^X e^Y = \sum_{r,s} X^r Y^s / (r!s!) = \sum_n \frac{1}{n!} \sum_{r+s=n} \binom{n}{r} X^r Y^s = \sum_n (X + Y)^n / n! = e^{X+Y}$ . Formally, the same computation is valid coefficientwise, and then  $Z(X, Y) = \log e^{X+Y} = X + Y$  by [theorem 2.1](#); this is also the content of [theorem 3.6](#), since every  $Z_n$  with  $n \geq 2$  lies in the ideal generated by  $[X, Y]$ .  $\square$

**Theorem 9.2** (Central-commutator identity). *Suppose  $C = [X, Y]$  commutes with both  $X$  and  $Y$ . Then, for every scalar parameter  $t$ ,*

$$e^{tX} Y e^{-tX} = Y + tC, \quad (9.1)$$

$$e^{tX} e^{tY} = \exp(t(X + Y) + \tfrac{1}{2}t^2 C). \quad (9.2)$$

In particular,

$$e^X e^Y = e^{X+Y+C/2}, \quad e^{X+Y} = e^X e^Y e^{-C/2}, \quad e^{X/2} e^Y e^{X/2} = e^{X+Y}, \quad e^X e^Y e^{-X} e^{-Y} = e^C. \quad (9.3)$$

These are formal identities, global identities for bounded elements of a Banach algebra, and global identities for elements of a finite-dimensional Lie group whose Lie algebra elements satisfy the stated bracket relations.

*Proof.* In the Campbell series (5.1),  $\text{ad}_X^2 Y = [X, C] = 0$ , so all terms after the first commutator vanish, giving (9.1). Let  $g(t) = e^{tX} e^{tY}$ . Its right logarithmic derivative is

$$g'(t)g(t)^{-1} = X + e^{tX} Y e^{-tX} = X + Y + tC,$$

so  $g' = (X + Y + tC)g$  with  $g(0) = I$ . Let  $K(t) = t(X + Y) + \frac{1}{2}t^2C$ . Since  $C$  commutes with  $X + Y$ ,  $K(t)$  commutes with  $K'(t) = X + Y + tC$ , and therefore  $\text{ad}_K K' = 0$  and (5.7) gives  $\frac{d}{dt} e^{K(t)} = \phi(-\text{ad}_K) K' e^K = K' e^K = (X + Y + tC) e^{K(t)}$ , with  $e^{K(0)} = I$ . Both  $g$  and  $e^K$  solve  $h' = (X + Y + tC)h$ ,  $h(0) = I$ ; the difference  $\Delta$  of two solutions satisfies  $\Delta' = (X + Y + tC)\Delta$ ,  $\Delta(0) = 0$ , hence  $\Delta \equiv 0$  by uniqueness for linear equations with continuous bounded coefficients (Grönwall), or by comparing coefficients of  $t$  formally. This proves (9.2). In a Lie group the same argument uses the right-trivialized derivative and uniqueness of integral curves of the time-dependent right-invariant field  $t \mapsto (X + Y + tC)^R$ .

Setting  $t = 1$  gives the first identity of (9.3); since  $C/2$  commutes with  $X + Y + C/2$ , theorem 9.1 gives  $e^{X+Y+C/2} e^{-C/2} = e^{X+Y}$ , the second identity. For the symmetric identity apply the first identity to the pair  $(X/2, Y)$ , whose commutator is  $C/2$ , and then to the pair  $(X/2 + Y + C/4, X/2)$ , whose commutator is  $\frac{1}{2}[Y, X] = -C/2$  (all other brackets vanish and  $C$  is central):

$$e^{X/2} e^Y e^{X/2} = e^{X/2+Y+C/4} e^{X/2} = e^{X/2+Y+C/4+X/2-C/4} = e^{X+Y}.$$

Finally, by (9.1) and multiplicativity of conjugation,  $e^X e^Y e^{-X} = e^{Y+C}$ , and since  $Y$  and  $C$  commute,  $e^{Y+C} e^{-Y} = e^C$ .  $\square$

Centrality is needed only relative to the Lie subalgebra generated by  $X$  and  $Y$ , not relative to a larger ambient algebra. This proof is global and does not infer convergence from the formal truncation alone; for unbounded operators a common invariant domain and integrability hypotheses are still necessary, as section 13 shows.

**Corollary 9.3** (Several factors with central pairwise commutators). *If all commutators  $[X_i, X_j]$  commute with every  $X_k$ , then*

$$\prod_{j=1}^m e^{X_j} = \exp\left(\sum_{j=1}^m X_j + \frac{1}{2} \sum_{1 \leq i < j \leq m} [X_i, X_j]\right), \quad (9.4)$$

the product being ordered by increasing  $j$ .

*Proof.* Induct on  $m$ ; the case  $m = 1$  is trivial. Let  $E_{m-1} = \sum_{j < m} X_j + \frac{1}{2} \sum_{i < j < m} [X_i, X_j]$ , so that  $\prod_{j < m} e^{X_j} = e^{E_{m-1}}$  by the induction hypothesis. The commutator  $[E_{m-1}, X_m] = \sum_{j < m} [X_j, X_m]$ , because the central part of  $E_{m-1}$  commutes with  $X_m$ ; it is a sum of central elements, hence commutes with  $E_{m-1}$  and  $X_m$ . Theorem 9.2 applied to  $(E_{m-1}, X_m)$  gives  $e^{E_{m-1}} e^{X_m} = \exp(E_{m-1} + X_m + \frac{1}{2} \sum_{j < m} [X_j, X_m])$ , which is (9.4) for  $m$  factors.  $\square$

This is the all-factors form behind the Weyl and displacement multiplication laws of section 13. In the Heisenberg algebra all higher BCH terms vanish because every Lie monomial of degree at least three contains a bracket with the central element  $C$ , in agreement with theorem 8.6.

## 9.2 An eigenvector of the adjoint action

**Theorem 9.4** (The relation  $[X, Y] = sY$ ). *Suppose  $[X, Y] = sY$  with  $s$  a scalar. Put*

$$q_s(t) = \int_0^t e^{-su} du = \begin{cases} (1 - e^{-st})/s, & s \neq 0, \\ t, & s = 0, \end{cases} \quad q_s(1) = \phi(s).$$

Then for every scalar  $c$  and every  $t$ ,

$$e^{t(X+cY)} = e^{tX} e^{cq_s(t)Y}, \quad \text{in particular} \quad e^{X+cY} = e^X e^{c\phi(s)Y}. \quad (9.5)$$

Consequently, if  $s \notin 2\pi i\mathbb{Z} \setminus \{0\}$ ,

$$\boxed{e^X e^Y = \exp\left(X + \frac{s}{1 - e^{-s}} Y\right) = \exp(X + \beta(s)Y)}, \quad (9.6)$$

with the coefficient interpreted as  $\beta(0) = 1$  at  $s = 0$ . For every  $s$ , including the resonant values,

$$e^X e^Y e^{-X} = e^{e^s Y}, \quad e^X e^Y = e^{e^s Y} e^X. \quad (9.7)$$

The formal BCH series is

$$Z(X, Y) = X + \sum_{n \geq 0} b_n^+ s^n Y = X + \beta(s)Y, \quad (9.8)$$

where the scalar series converges for  $|s| < 2\pi$  and diverges for  $|s| \geq 2\pi$  if  $Y \neq 0$ . All exponential identities are global in the same settings as [theorem 9.2](#).

*Proof.* The relation gives  $\text{ad}_X^n Y = s^n Y$  by induction, so the Campbell series (5.1) yields  $e^{tX} Y e^{-tX} = e^{st} Y$ ; conjugating the exponential series of  $Y$  gives  $e^{tX} e^Y e^{-tX} = e^{e^{st} Y}$ , and at  $t = 1$  this is (9.7), whose second form follows by multiplying on the right by  $e^X$ . No division has occurred, so these hold for every  $s$ .

For (9.5), let  $F(t) = e^{tX} e^{cq_s(t)Y}$ . Its right logarithmic derivative is

$$F'F^{-1} = X + e^{tX} (cq'_s(t)Y) e^{-tX} = X + ce^{-st} e^{st} Y = X + cY,$$

using  $q'_s(t) = e^{-st}$  and the fact that  $cq_s(t)Y$  commutes with its own derivative, so that (5.7) for the second factor reduces to plain differentiation. Hence  $F' = (X + cY)F$  with  $F(0) = I$ . The function  $e^{t(X+cY)}$  solves the same initial value problem, and uniqueness (as in the proof of [theorem 9.2](#)) gives (9.5). At  $t = 1$ ,  $q_s(1) = \phi(s)$ , with the removable value 1 at  $s = 0$ .

If  $s \notin 2\pi i\mathbb{Z} \setminus \{0\}$ , then  $\phi(s) \neq 0$  (the zeros of  $\phi$  are exactly the nonzero points of  $2\pi i\mathbb{Z}$ ), and choosing  $c = 1/\phi(s) = \beta(s)$  in (9.5) gives (9.6).

For the formal assertion (9.8), we show that every Lie monomial in  $X, Y$  containing at least two occurrences of  $Y$  vanishes under the relation. Consider a bracket tree evaluating such a monomial. A branch containing no  $Y$  is a bracket of copies of  $X$  alone, which is  $X$  if it has length one and 0 otherwise. A branch containing exactly one  $Y$  is, by induction on its size, a scalar multiple of  $Y$ : at each node one combines either  $X$  with a multiple of  $Y$  (giving a multiple of  $sY$ ) or a zero branch with something (giving 0). At the first node where two branches each containing at least one  $Y$  meet, both are multiples of  $Y$  and their bracket is zero. Hence only the terms linear in  $Y$  survive, and by [theorem 6.4](#) these are  $\beta(\text{ad}_X)Y = \sum_n b_n^+ \text{ad}_X^n Y = \sum_n b_n^+ s^n Y$ . The scalar series is the Taylor series of  $\beta$  at  $s$ , with radius exactly  $2\pi$  by [theorem 5.3](#); by (5.12) its even terms have magnitude  $2\zeta(2m)(|s|/2\pi)^{2m} \geq 2(|s|/2\pi)^{2m}$ , which does not tend to zero when  $|s| \geq 2\pi$ , so the series diverges there.  $\square$

The differential-equation proof of (9.6) establishes the exponential identity even where the Taylor series (9.8) diverges: resummation and Taylor convergence are distinct assertions. In the same vein,

$$\text{BCH}(tX, tY) = tX + t\beta(ts)Y \quad (9.9)$$

as a Taylor series in  $t$  at 0, whose radius is exactly  $2\pi/|s|$  when  $s \neq 0$  and  $Y \neq 0$ ; real  $s > 2\pi$  therefore makes this series divergent at  $t = 1$  although the resummed identity remains valid. The braiding law (9.7), being an entire exponential identity, has no resonance exclusion at all.

**Example 9.5** (Resonance). Let

$$X = \begin{pmatrix} s & 0 \\ 0 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad [X, Y] = sY. \quad (9.10)$$

Direct exponentiation gives

$$e^X = \begin{pmatrix} e^s & 0 \\ 0 & 1 \end{pmatrix}, \quad e^X e^Y = \begin{pmatrix} e^s & e^s \\ 0 & 1 \end{pmatrix}, \quad e^{X+cY} = \begin{pmatrix} e^s & c(e^s - 1)/s \\ 0 & 1 \end{pmatrix},$$

the last by the triangular structure ( $X + cY$  has eigenvalues  $s, 0$  and its exponential's off-diagonal entry is  $c$  times the divided difference of  $e^z$  at  $s, 0$ ), with the continuous value  $c$  at  $s = 0$ . If  $s = 3\pi$ , the choice  $c = \beta(3\pi)$  works although the BCH series (9.8) diverges. If  $s = 2\pi i k \neq 0$ , then  $e^s = 1$ , every  $e^{X+cY}$  equals  $I$ , while  $e^X e^Y = I + Y \neq I$ ; thus no logarithm of the form  $X + cY$  exists at a resonance, although the logarithm  $Y$  does exist, and the braiding identity  $e^X e^Y = e^{e^s Y} e^X = e^Y e^X$  remains true. What fails at resonance is the shape  $X + cY$  of the logarithm, not the existence of a logarithm.

## 10 Zassenhaus factorization: every exponent, three recursions, a tree formula, and convergence

BCH combines a product of exponentials into one exponential. Zassenhaus factorization performs the ordered operation in the opposite direction: it writes the exponential of a sum as an ordered product of exponentials of homogeneous Lie corrections. The order of the factors is part of the statement. Recursive constructions are central in the computational literature [6]; explicit reorganizations are considered in [8]. We prove existence and uniqueness, three recursive constructions, a finite nonrecursive formula for every exponent, a globally convergent ordered-integral expansion, and a quantitative convergence theorem for the product.

### 10.1 Existence, uniqueness, and the residual algorithm

**Theorem 10.1** (Formal Zassenhaus factorization). *There is a unique sequence of homogeneous Lie polynomials  $C_n(X, Y)$  of degree  $n$ ,  $n \geq 2$ , such that*

$$e^{t(X+Y)} = e^{tX} e^{tY} \prod_{n=2}^{\infty} e^{t^n C_n(X, Y)} = e^{tX} e^{tY} e^{t^2 C_2} e^{t^3 C_3} e^{t^4 C_4} \dots \quad (10.1)$$

*The infinite product is ordered by increasing  $n$  from left to right and is well defined coefficientwise in  $t$ : modulo  $t^{N+1}$  only the factors through  $C_N$  matter. Define*

$$R_1(t) = e^{-tY} e^{-tX} e^{t(X+Y)}, \quad (10.2)$$

*and recursively, for  $n \geq 2$ ,*

$$C_n = [t^n] R_{n-1}(t) = [t^n] \log R_{n-1}(t), \quad R_n(t) = e^{-t^n C_n} R_{n-1}(t). \quad (10.3)$$

*Then  $R_n(t) = I + O(t^{n+1})$  for every  $n \geq 1$ , and (10.3) computes all the exponents.*

*Proof.* Work in  $\widehat{T}[[t]]$ , or equivalently in  $\widehat{T}$  with  $t$  tracking total degree. The coefficient of  $t^1$  in  $R_1$  is  $-Y - X + (X + Y) = 0$ , so  $R_1 = I + O(t^2)$ . Suppose inductively that  $R_{n-1} = I + t^n C_n + O(t^{n+1})$ . Then  $\log R_{n-1} = t^n C_n + O(t^{n+1})$ , which gives the second expression for  $C_n$  in (10.3), and  $R_n = e^{-t^n C_n} R_{n-1} = (I - t^n C_n + O(t^{2n}))(I + t^n C_n + O(t^{n+1})) = I + O(t^{n+1})$ . Thus the recursion removes one degree at a time.

Each residual is group-like in the completed Hopf algebra of section 3 (extended by the central scalar  $t$ ):  $R_1$  is a product of exponentials of the primitive elements  $-tY$ ,  $-tX$ ,  $t(X + Y)$ , hence group-like by theorem 3.3; and if  $R_{n-1}$  is group-like and  $C_n$  is a Lie polynomial, then  $e^{-t^n C_n}$  is group-like and so is the product  $R_n$ . To see that  $C_n$  is a Lie polynomial, compare the coefficients of  $t^n$  in  $\Delta R_{n-1} = R_{n-1} \otimes R_{n-1}$ : on the left it is  $\Delta C_n$ ; on the right, since  $R_{n-1} = I + t^n C_n + O(t^{n+1})$  has no terms of  $t$ -degree between 1 and  $n - 1$ , it is  $C_n \otimes 1 + 1 \otimes C_n$ . Hence  $C_n$  is primitive, and by theorem 3.5 it is a Lie polynomial; it is homogeneous of degree  $n$  in the letters because every input has matching degree in  $t$  and in  $X, Y$ . This completes the induction.

Undoing the residuals,  $R_1 = e^{t^2 C_2} e^{t^3 C_3} \dots e^{t^N C_N} R_N$  with  $R_N = I + O(t^{N+1})$ , so modulo  $t^{N+1}$  the product of the first  $N - 1$  factors equals  $R_1$ ; multiplying by  $e^{tX} e^{tY}$  on the left proves (10.1) degree by degree. For uniqueness, suppose  $e^{t(X+Y)} = e^{tX} e^{tY} \prod_n e^{t^n C'_n}$  with homogeneous  $C'_n$  of degree  $n$ . Then  $\prod_{n \geq 2} e^{t^n C'_n} = R_1$ . If  $C'_j = C_j$  for  $j < n$ , then multiplying on the left by the inverses of the first factors gives  $\prod_{j \geq n} e^{t^j C'_j} = R_{n-1}$ , and comparing coefficients of  $t^n$  gives  $C'_n = C_n$ . Induction proves uniqueness.  $\square$

When the identity is written as  $e^{-tX} e^{t(X+Y)} = \prod_{n \geq 1} e^{t^n C_n}$ , as on the source page, the convention *must* be  $C_1 = Y$ ; the notation would otherwise omit a necessary first-degree factor. Extracting  $t^n$  from  $R_{n-1} = e^{-t^{n-1} C_{n-1}} \dots e^{-t^2 C_2} R_1$  gives the finite coefficient recurrence

$$C_n = \sum_{\substack{k_2, \dots, k_{n-1}, a, b, c \geq 0 \\ \sum_{j=2}^{n-1} j k_j + a + b + c = n}} \left( \prod_{j=n-1}^2 \frac{(-C_j)^{k_j}}{k_j!} \right) \frac{(-Y)^a}{a!} \frac{(-X)^b}{b!} \frac{(X+Y)^c}{c!}, \quad (10.4)$$

in which the product over  $j$  is taken in *descending* order of  $j$  from left to right. It involves associative products, but the result is a Lie polynomial by the theorem, and (3.6) converts it into nested commutators.

## 10.2 A recursion consisting entirely of commutators

Define the right logarithmic derivatives  $F_n(t) = R'_n(t) R_n(t)^{-1}$ .

**Proposition 10.2** (Differential Zassenhaus algorithm).

$$F_1(t) = e^{-t \operatorname{ad}_Y} (e^{-t \operatorname{ad}_X} - I) Y = \sum_{k \geq 1} t^k f_{1,k}, \quad f_{1,k} = (-1)^k \sum_{j=1}^k \frac{\operatorname{ad}_Y^{k-j} \operatorname{ad}_X^j Y}{(k-j)! j!}, \quad (10.5)$$

$$C_n = \frac{1}{n} [t^{n-1}] F_{n-1}(t), \quad F_n(t) = e^{-t^n \operatorname{ad}_{C_n}} (F_{n-1}(t) - n t^{n-1} C_n) \quad (n \geq 2). \quad (10.6)$$

Writing  $F_n = \sum_{k \geq 0} t^k f_{n,k}$ , with  $f_{n,k} = 0$  for  $k < 0$ , the coefficients satisfy the fully finite update

$$\boxed{f_{n,k} = \sum_{j=0}^{\lfloor k/n \rfloor} \frac{(-1)^j}{j!} \operatorname{ad}_{C_n}^j f_{n-1, k-nj} - n C_n \delta_{k, n-1}, \quad C_n = \frac{f_{n-1, n-1}}{n}.} \quad (10.7)$$

Only previously computed homogeneous Lie polynomials occur; this is a Lie-only algorithm at every degree.

*Proof.* For a product  $ABC$  of differentiable invertible curves,  $(ABC)'(ABC)^{-1} = A'A^{-1} + A(B'B^{-1})A^{-1} + AB(C'C^{-1})B^{-1}A^{-1}$ . With  $A = e^{-tY}$ ,  $B = e^{-tX}$ ,  $C = e^{t(X+Y)}$  we have  $A'A^{-1} = -Y$ ,  $B'B^{-1} = -X$ ,  $C'C^{-1} = X + Y$ , and conjugations are adjoint exponentials by theorem 5.1:

$$\begin{aligned} F_1 &= -Y - e^{-t\text{ad}_Y} X + e^{-t\text{ad}_Y} e^{-t\text{ad}_X} (X + Y) \\ &= -Y - e^{-t\text{ad}_Y} X + e^{-t\text{ad}_Y} X + e^{-t\text{ad}_Y} e^{-t\text{ad}_X} Y = e^{-t\text{ad}_Y} (e^{-t\text{ad}_X} - I)Y, \end{aligned}$$

using  $e^{-t\text{ad}_X} X = X$  and  $e^{-t\text{ad}_Y} Y = Y$ . Expanding both adjoint exponentials gives the double series  $\sum_{i \geq 0, j \geq 1} (-1)^{i+j} t^{i+j} \text{ad}_Y^i \text{ad}_X^j Y / (i!j!)$ , and collecting  $t^k$  gives  $f_{1,k}$ .

Since  $R_{n-1} = I + t^n C_n + O(t^{n+1})$ , we have  $R'_{n-1} = nt^{n-1} C_n + O(t^n)$  and  $R_{n-1}^{-1} = I + O(t^n)$ , so  $F_{n-1} = nt^{n-1} C_n + O(t^n)$ ; in particular  $f_{n-1,k} = 0$  for  $k < n-1$  and  $f_{n-1,n-1} = nC_n$ , the first part of (10.6). Differentiating  $R_n = e^{-t^n C_n} R_{n-1}$ ,

$$F_n = \left( \frac{d}{dt} e^{-t^n C_n} \right) e^{t^n C_n} + e^{-t^n C_n} F_{n-1} e^{t^n C_n} = -nt^{n-1} C_n + e^{-t^n \text{ad}_{C_n}} F_{n-1},$$

because  $t^n C_n$  commutes with its derivative and conjugation by  $e^{-t^n C_n}$  is  $e^{-t^n \text{ad}_{C_n}}$ . As  $e^{-t^n \text{ad}_{C_n}} C_n = C_n$ , the correction may be moved inside the operator exponential, giving the second part of (10.6). Expanding  $e^{-t^n \text{ad}_{C_n}} = \sum_j (-1)^j t^{nj} \text{ad}_{C_n}^j / j!$  and extracting the coefficient of  $t^k$  gives (10.7); the term  $-nt^{n-1} C_n$  is annihilated by  $\text{ad}_{C_n}^j$  for  $j \geq 1$  and contributes only through  $j = 0$ .  $\square$

### 10.3 The exponents displayed on the source page

**Proposition 10.3.**

$$C_2 = -\frac{1}{2}[X, Y], \tag{10.8}$$

$$C_3 = \frac{1}{6}[X, [X, Y]] + \frac{1}{3}[Y, [X, Y]] = \frac{1}{6}[X, [X, Y]] - \frac{1}{3}[[X, Y], Y], \tag{10.9}$$

$$C_4 = -\frac{1}{24}[X, [X, [X, Y]]] - \frac{1}{8}[Y, [X, [X, Y]]] - \frac{1}{8}[Y, [Y, [X, Y]]] \tag{10.10}$$

$$= -\frac{1}{24}([[[X, Y], X], X] + 3[[[X, Y], X], Y] + 3[[[X, Y], Y], Y]). \tag{10.11}$$

*Proof.* From (10.5),

$$\begin{aligned} f_{1,1} &= -[X, Y], \\ f_{1,2} &= \text{ad}_Y \text{ad}_X Y + \frac{1}{2} \text{ad}_X^2 Y = [Y, [X, Y]] + \frac{1}{2}[X, [X, Y]], \\ f_{1,3} &= -\frac{1}{2} \text{ad}_Y^2 \text{ad}_X Y - \frac{1}{2} \text{ad}_Y \text{ad}_X^2 Y - \frac{1}{6} \text{ad}_X^3 Y. \end{aligned}$$

By (10.7),  $C_2 = f_{1,1}/2 = -\frac{1}{2}[X, Y]$ . Next,  $f_{2,2} = f_{1,2} - \text{ad}_{C_2} f_{1,0} = f_{1,2}$  since  $f_{1,0} = 0$ , so  $C_3 = f_{2,2}/3$ , which is (10.9); the second form uses  $[Y, [X, Y]] = -[[X, Y], Y]$ . Then  $f_{2,3} = f_{1,3} - \text{ad}_{C_2} f_{1,1} = f_{1,3} + \frac{1}{2}[[X, Y], [X, Y]] = f_{1,3}$ , and  $f_{3,3} = f_{2,3} - \text{ad}_{C_3} f_{2,0} = f_{2,3}$ , so  $C_4 = f_{3,3}/4$ , which is (10.10). For (10.11), antisymmetry gives  $[[X, Y], X] = -[X, [X, Y]]$ , hence  $[[[X, Y], X], X] = [X, [X, [X, Y]]]$ ,  $[[[X, Y], X], Y] = [Y, [X, [X, Y]]]$ , and  $[[[X, Y], Y], Y] = [Y, [Y, [X, Y]]]$ . Substituting these into (10.10) gives (10.11), exactly the left-nested form on the source page. Since the corrections at each step were computed exactly, no higher-order approximation has been left implicit.  $\square$

The complete fifth and sixth exponents are printed in section B (table 4), and all exponents through degree twelve accompany the article as data. There is also a left-oriented companion factorization: replacing  $t$  by  $-t$  in (10.1) and inverting both sides reverses the factor order and gives

$$e^{t(X+Y)} = \dots e^{t^4 \hat{C}_4} e^{t^3 \hat{C}_3} e^{t^2 \hat{C}_2} e^{tY} e^{tX}, \quad \hat{C}_n = (-1)^{n+1} C_n, \tag{10.12}$$

since  $(e^{(-t)^n C_n})^{-1} = e^{(-1)^{n+1} t^n C_n}$ .

## 10.4 A finite nonrecursive formula using weighted trees

The previous algorithms determine the exponents recursively. For a finite nonrecursive expression at every fixed order, first define the explicit associative polynomials

$$A_n = [t^n] R_1(t) = \sum_{a+b+c=n} \frac{(-1)^{a+b}}{a! b! c!} Y^a X^b (X + Y)^c \quad (n \geq 0), \quad (10.13)$$

where  $(X + Y)^c$  is the sum of all  $2^c$  words of length  $c$ ;  $A_0 = I$  and  $A_1 = 0$ . Coefficient extraction from the ordered product  $R_1 = \prod_{j \geq 2} e^{t^j C_j}$  gives the forward partition recurrence

$$C_n = A_n - \sum_{\substack{k_2, \dots, k_{n-1} \geq 0 \\ \sum_{j=2}^{n-1} j k_j = n}} \frac{C_2^{k_2} C_3^{k_3} \cdots C_{n-1}^{k_{n-1}}}{k_2! k_3! \cdots k_{n-1}!}, \quad (10.14)$$

with the factors in *increasing* order of degree. Indeed, choosing from each factor  $e^{t^j C_j}$  the term  $t^{j k_j} C_j^{k_j} / k_j!$  with  $\sum j k_j = n$  gives all contributions to  $t^n$ ; the unique choice with  $k_n = 1$  contributes  $C_n$ , and every other choice has  $k_n = 0$  and at least two factors. The sum is empty for  $n = 2, 3$ .

**Definition 10.4.** Let  $\mathcal{T}_n$  be the set of finite plane rooted trees with a positive integer weight at each vertex, subject to: the root has weight  $n$ ; every weight is at least 2; and each internal vertex  $v$  of weight  $n_v$  has  $r \geq 2$  ordered children whose weights  $2 \leq j_1 \leq j_2 \leq \cdots \leq j_r$  are weakly increasing from left to right and sum to  $n_v$ . Leaves carry no further restriction. Let  $I(T)$  be the number of internal vertices, let  $m_{v,j}$  be the number of children of weight  $j$  at the internal vertex  $v$ , and let  $\ell_1, \dots, \ell_q$  be the weights of the leaves read from left to right in planar order. Define

$$\omega(T) = \frac{(-1)^{I(T)}}{\prod_{v \text{ internal}} \prod_{j \geq 2} m_{v,j}!}, \quad A(T) = A_{\ell_1} A_{\ell_2} \cdots A_{\ell_q}. \quad (10.15)$$

Children of equal weight are distinct ordered slots whose subtrees may differ; interchanging two unequal subtrees changes  $A(T)$  in general, because the  $A_j$  do not commute.

**Theorem 10.5** (All-terms Zassenhaus tree formula). *For every  $n \geq 2$ ,  $\mathcal{T}_n$  is finite and*

$$\boxed{C_n = \sum_{T \in \mathcal{T}_n} \omega(T) A(T)}. \quad (10.16)$$

*This is a finite formula involving only factorials, sums, and products of  $X$  and  $Y$ . An explicitly nested-commutator version is*

$$C_n = \frac{1}{n} \sum_{T \in \mathcal{T}_n} \omega(T) \sum_{w \in \{X, Y\}^n} [w] A(T) R(w). \quad (10.17)$$

*Proof.* Finiteness: the leaf weights are at least 2 and sum to  $n$  (every internal vertex's weight is the sum of its children's weights, so by induction the root weight is the sum of the leaf weights), so there are at most  $\lfloor n/2 \rfloor$  leaves; since every internal vertex has at least two children, a plane tree with  $q$  leaves has fewer than  $q$  internal vertices; and each vertex weight is an integer in  $[2, n]$ . Hence only finitely many shapes and weight assignments occur.

We prove (10.16) by strong induction on  $n$ . For  $n = 2, 3$  the sum in (10.14) is empty,  $C_n = A_n$ , and  $\mathcal{T}_n$  consists of the single-vertex tree (a root of weight  $n$  with no children, since no weakly increasing list of at least two integers  $\geq 2$  sums to 2 or 3), for which  $\omega = 1$  and  $A(T) = A_n$ . For  $n \geq 4$ , assume the formula for all degrees below  $n$ . In (10.14), a choice

of exponents  $(k_2, \dots, k_{n-1})$  with  $\sum j k_j = n$  corresponds bijectively to the weakly increasing list of child weights of a root of weight  $n$ , namely  $k_2$  copies of 2, then  $k_3$  copies of 3, and so on; the product  $C_2^{k_2} \dots C_{n-1}^{k_{n-1}}$  is the ordered product of  $C_{j_i}$  over this list, and the coefficient  $-1/\prod_j k_j!$  is exactly the root's contribution  $(-1)\prod_j 1/m_{\text{root},j}!$  to  $\omega$ . Substituting the induction hypothesis  $C_{j_i} = \sum_{T_i \in \mathcal{T}_{j_i}} \omega(T_i) A(T_i)$  for each factor and expanding the product gives a sum over all tuples  $(T_1, \dots, T_r)$  of subtrees, one in each child slot, of  $(-1)\prod_j \frac{1}{m_{\text{root},j}!} \prod_i \omega(T_i) \prod_i A(T_i)$ . Attaching  $T_1, \dots, T_r$  in order to the root produces exactly one tree  $T \in \mathcal{T}_n$  with at least one internal vertex; conversely every such tree arises exactly once (remove the root). For that tree,  $\omega(T) = (-1)\prod_j \frac{1}{m_{\text{root},j}!} \prod_i \omega(T_i)$  (the internal vertices of  $T$  are the root and those of the  $T_i$ ), and  $A(T) = \prod_i A(T_i)$  because the leaves of  $T$  in planar order are the leaves of  $T_1$ , then of  $T_2$ , and so on. The remaining term  $A_n$  is the contribution of the single-vertex tree. This proves (10.16). Finally,  $C_n$  is a homogeneous Lie polynomial of degree  $n$  by [theorem 10.1](#), hence primitive, so [theorem 3.4](#) gives  $C_n = \frac{1}{n} D(C_n) = \frac{1}{n} \sum_w [w] C_n R(w)$ ; substituting the word expansion of (10.16) gives (10.17).  $\square$

For orientation, the first instances are

$$\begin{aligned} C_2 &= A_2, & C_3 &= A_3, & C_4 &= A_4 - \frac{1}{2} A_2^2, & C_5 &= A_5 - A_2 A_3, \\ C_6 &= A_6 - A_2 A_4 - \frac{1}{2} A_3^2 + \frac{1}{3} A_2^3, & C_7 &= A_7 - A_2 A_5 - A_3 A_4 + \frac{1}{2} A_2^2 A_3 + \frac{1}{2} A_3 A_2^2. \end{aligned} \quad (10.18)$$

The coefficient  $\frac{1}{3}$  of  $A_2^3$  in  $C_6$  is  $\frac{1}{2} - \frac{1}{6}$ : the tree with root children  $(2, 4)$  and the child of weight 4 split into  $(2, 2)$  contributes  $(-1)^2 \cdot \frac{1}{2!} = \frac{1}{2}$ , and the tree with root children  $(2, 2, 2)$  contributes  $-\frac{1}{3!}$ . In  $C_7$  the two terms  $A_2^2 A_3$  and  $A_3 A_2^2$  are different polynomials: the first collects the tree with root list  $(2, 5)$  and  $5 \rightarrow (2, 3)$  (weight  $+1$ ) and the tree with root list  $(2, 2, 3)$  (weight  $-\frac{1}{2!}$ ), while the second comes from the root list  $(3, 4)$  with  $4 \rightarrow (2, 2)$  (weight  $+\frac{1}{2!}$ ). This illustrates why an unordered product would lose information. The data file `zassenhaus_tree_polynomials.json` lists these polynomials in the  $A_j$  through degree ten, and the verification program of [section 14](#) confirms that (10.16) agrees with the two recursive constructions through that degree.

## 10.5 A globally convergent ordered-simplex expansion

There is another useful all-terms expansion, a sum of ordered products rather than a product of exponentials; its convergence is therefore a separate and easier question than Zassenhaus convergence.

**Theorem 10.6** (Ordered-simplex expansion). *For bounded  $X, Y$  in a unital Banach algebra and every scalar  $t$ ,*

$$\begin{aligned} e^{t(X+Y)} &= e^{tX} \left[ I + \sum_{k \geq 1} \sum_{n_1, \dots, n_k \geq 0} \frac{(-1)^{n_1 + \dots + n_k} t^{k+n_1+\dots+n_k}}{\prod_{j=1}^k n_j! \prod_{j=1}^k (n_j + n_{j+1} + \dots + n_k + k - j + 1)} \right. \\ &\quad \left. \cdot (\text{ad}_X^{n_1} Y) (\text{ad}_X^{n_2} Y) \dots (\text{ad}_X^{n_k} Y) \right], \end{aligned} \quad (10.19)$$

and the multiple series is absolutely convergent, with the sum of the norms of its terms bounded by  $\exp(|t| \|Y\| e^{2|t| \|X\|})$ .

*Proof.* Put  $H(t) = e^{-tX} e^{t(X+Y)}$ . Differentiating,  $H'(t) = -X e^{-tX} e^{t(X+Y)} + e^{-tX} (X + Y) e^{t(X+Y)} = e^{-tX} Y e^{t(X+Y)} = (e^{-tX} Y e^{tX}) H(t) = B(t) H(t)$  with  $B(t) = e^{-t \text{ad}_X} Y$  by [theorem 5.1](#), and  $H(0) = I$ . First take real  $t \geq 0$ . Integrating,  $H(t) = I + \int_0^t B(s) H(s) ds$ , and

substituting this expression for  $H(s)$  repeatedly  $N$  times gives

$$\begin{aligned} H(t) &= I + \sum_{k=1}^N \int_{0 < t_k < \dots < t_1 < t} B(t_1) \cdots B(t_k) dt_k \cdots dt_1 \\ &\quad + \int_{0 < t_{N+1} < \dots < t_1 < t} B(t_1) \cdots B(t_{N+1}) H(t_{N+1}) dt_{N+1} \cdots dt_1. \end{aligned}$$

With  $b = \sup_{0 \leq s \leq t} \|B(s)\| \leq \|Y\| e^{2t\|X\|}$  and  $\|H(s)\| \leq \|I\| e^{2s\|X\| + s\|X+Y\|}$  bounded on  $[0, t]$ , the remainder has norm at most  $b^{N+1}t^{N+1}/(N+1)!$  times that bound, since the ordered simplex has volume  $t^{N+1}/(N+1)!$ . Hence the remainder tends to zero and  $H(t) = I + \sum_{k \geq 1} \int_{\text{simplex}} B(t_1) \cdots B(t_k)$ . Now expand  $B(s) = \sum_{n \geq 0} (-s)^n \text{ad}_X^n Y / n!$  in each factor. The resulting multiple series of monomials is absolutely convergent: the sum of the norms of the terms of  $B(s)$  is at most  $\sum_n s^n (2\|X\|)^n \|Y\| / n! = \|Y\| e^{2s\|X\|} \leq b$ , so the sum of the norms of all monomials in the  $k$ th ordered integral is at most  $b^k t^k / k!$ , and summing over  $k$  gives the bound  $e^{bt}$ . Absolute convergence justifies integrating term by term. The iterated monomial integral is evaluated from the inside out: integrating  $t_k^{n_k}$  from 0 to  $t_{k-1}$  gives  $t_{k-1}^{n_k+1} / (n_k + 1)$ ; then  $t_{k-1}^{n_{k-1}+n_k+1}$  integrates to  $t_{k-2}^{n_{k-1}+n_k+2} / (n_{k-1} + n_k + 2)$ ; continuing,

$$\int_{0 < t_k < \dots < t_1 < t} \prod_{j=1}^k t_j^{n_j} dt_k \cdots dt_1 = \frac{t^{k+n_1+\dots+n_k}}{\prod_{j=1}^k (n_j + n_{j+1} + \dots + n_k + k - j + 1)}.$$

This proves the coefficients in (10.19) for real  $t \geq 0$ . For complex  $t$ , parametrize the segment from 0 to  $t$  by  $s = tu$ ,  $0 \leq u \leq 1$ ; the same Picard iteration and the same integrals, now with the factor  $t^{k+\sum n_j}$  pulled out, give the identical formula and the bound with  $|t|$ . Multiplying by  $e^{tX}$  gives (10.19).  $\square$

## 10.6 An elementary sufficient convergence condition for the product

Formal stabilization of a product does not automatically imply analytic convergence. The following bound supplies a concrete domain; it is deliberately elementary and is not asserted to be optimal.

**Theorem 10.7** (Convergence of the ordered Zassenhaus product). *Let  $X, Y$  be elements of a unital Banach algebra with  $\|I\| = 1$  and put  $a = \|X\| + \|Y\|$ . Let  $r > 0$  satisfy*

$$h(r) := e^{2ar} - 1 - 2ar < 2 \log 2 - 1. \quad (10.20)$$

*Then  $\sum_{n \geq 2} r^n \|C_n\| < \infty$ , the ordered product in (10.1) converges in norm uniformly for  $|t| \leq r$ , and the identity (10.1) holds there. Condition (10.20) is satisfied whenever  $ar < u_*/2 = 0.38349\dots$ , where  $u_* = 0.76699\dots$  is the positive solution of  $e^u - 1 - u = 2 \log 2 - 1$ ; a particularly simple sufficient condition is  $a|t| \leq \frac{1}{4}$ .*

*Proof. Majorant.* Expanding the three exponentials in (10.2), the sum of the norms of all terms of degree  $n$  in  $t$  is at most the coefficient of  $t^n$  in  $e^{t\|Y\|} e^{t\|X\|} e^{t(\|X\|+\|Y\|)} = e^{2at}$ , so  $\|A_n\| \leq (2a)^n / n!$  for  $n \geq 2$ , and  $h(z) = \sum_{n \geq 2} (2a)^n z^n / n!$  is a nonnegative-coefficient majorant of  $R_1 - I$ . Define a scalar formal series  $c(z) = \sum_{n \geq 2} c_n z^n$  by

$$c = h + (e^c - 1 - c). \quad (10.21)$$

Since  $c$  has order 2, the expression  $e^c - 1 - c = \sum_{m \geq 2} c^m / m!$  has its degree- $n$  coefficient determined by  $c_2, \dots, c_{n-2}$ ; so (10.21) determines the  $c_n$  recursively, and they are nonnegative because all ingredients are. In the commutative scalar algebra,  $\prod_{j \geq 2} e^{c_j z^j} = e^{c(z)}$ , whose coefficient of  $z^n$

is  $\sum_{j k_j = n} \prod_j c_j^{k_j} / k_j!$ ; the choice  $k_n = 1$  contributes  $c_n$ , and the remaining choices, all with at least two factors, contribute exactly  $[z^n](e^c - 1 - c)$ . Comparing with (10.14) and using the triangle inequality and submultiplicativity,

$$\|C_n\| \leq \|A_n\| + \sum_{\substack{j k_j = n \\ k_n = 0}} \prod_j \frac{\|C_j\|^{k_j}}{k_j!} \leq h_n + \sum_{\substack{j k_j = n \\ k_n = 0}} \prod_j \frac{c_j^{k_j}}{k_j!} = c_n,$$

where the middle inequality uses the induction hypothesis  $\|C_j\| \leq c_j$  for  $j < n$  and monotonicity in the  $c_j \geq 0$ .

*Convergence of the majorant.* The function  $q(d) = 2d - e^d + 1$  is strictly increasing on  $[0, \log 2)$ , since  $q'(d) = 2 - e^d > 0$  there, from  $q(0) = 0$  to  $q(\log 2) = 2 \log 2 - 1$ . Since  $h$  is increasing and continuous and (10.20) is strict, choose  $R > r$  with  $h(R) < 2 \log 2 - 1$ , and then  $d \in [0, \log 2)$  with  $q(d) = h(R)$ . Let  $\mathcal{B}$  be the Banach space of functions holomorphic on  $|z| < R$  and continuous on  $|z| \leq R$  with the supremum norm, and let  $\mathcal{B}_d$  be its closed ball of radius  $d$ . For  $f \in \mathcal{B}_d$  define  $\mathcal{F}(f)(z) = h(z) + e^{f(z)} - 1 - f(z)$ . Since  $h$  has nonnegative coefficients,  $|h(z)| \leq h(R)$  on the closed disk, and  $|e^f - 1 - f| \leq \sum_{m \geq 2} |f|^m / m! \leq e^d - 1 - d$ ; hence  $\|\mathcal{F}(f)\| \leq h(R) + e^d - 1 - d = q(d) + e^d - 1 - d = d$ , so  $\mathcal{F}$  maps  $\mathcal{B}_d$  into itself. For  $f, g \in \mathcal{B}_d$ ,  $e^f - f - (e^g - g) = \int_0^1 (e^{g + \tau(f-g)} - 1) d\tau (f - g)$ , so  $\|\mathcal{F}(f) - \mathcal{F}(g)\| \leq (e^d - 1) \|f - g\|$  with  $e^d - 1 < 1$ . By the contraction mapping theorem  $\mathcal{F}$  has a unique fixed point  $f_* \in \mathcal{B}_d$ . Its Taylor coefficients satisfy (10.21) (coefficient extraction is continuous and  $e^{f_*} - 1 - f_*$  has the same coefficient recursion),  $f_*$  has order at least 2 because  $h$  does and  $e^f - 1 - f$  has order at least twice that of  $f$ , and the formal solution of (10.21) is unique; hence  $f_*(z) = \sum_n c_n z^n$  on  $|z| < R$ . As  $c_n \geq 0$  and  $r < R$ ,

$$\sum_{n \geq 2} r^n \|C_n\| \leq \sum_{n \geq 2} c_n r^n = f_*(r) \leq d < \log 2.$$

*Convergence of the product.* Let  $P_N(t) = \prod_{n=2}^N e^{t^n C_n}$ . For  $|t| \leq r$  and  $M > N$ ,  $P_M - P_N = P_N(\prod_{n=N+1}^M e^{t^n C_n} - I)$ . Expanding each factor as  $I + (e^{t^n C_n} - I)$  and multiplying out, every term other than  $I$  contains at least one factor  $e^{t^n C_n} - I$  of norm at most  $e^{r^n \|C_n\|} - 1$ , so  $\|\prod_{n=N+1}^M e^{t^n C_n} - I\| \leq \exp(\sum_{n > N} r^n \|C_n\|) - 1$ , which tends to zero as  $N \rightarrow \infty$ ; and  $\|P_N\| \leq \exp(\sum_n r^n \|C_n\|) \leq e^d$ . Hence  $(P_N)$  is uniformly Cauchy on  $|t| \leq r$  and converges uniformly to a limit  $P(t)$ , continuous on the closed disk and analytic on  $|t| < r$  as a uniform limit of entire functions. For  $n \leq N$  the coefficient of  $t^n$  in  $P_N$  does not depend on  $N$  (the omitted factors contribute only degrees above  $N$ ), and by the Cauchy coefficient formula uniform convergence gives  $[t^n]P = [t^n]P_N$ . The formal identity of theorem 10.1 says that  $[t^n](e^{tX} e^{tY} P_N(t)) = [t^n]e^{t(X+Y)}$  for  $n \leq N$ . Thus the analytic functions  $e^{tX} e^{tY} P(t)$  and  $e^{t(X+Y)}$  have the same Taylor coefficients at 0, so they agree on  $|t| < r$ , and by continuity on  $|t| \leq r$ .

*The numerical conditions.* The function  $u \mapsto e^u - 1 - u$  is strictly increasing on  $[0, \infty)$  from 0, so  $e^u - 1 - u = 2 \log 2 - 1 = 0.386294\dots$  has a unique positive solution  $u_*$ ; evaluating,  $e^{0.767} - 1 - 0.767 = 0.38629\dots$  and  $e^{0.766} - 1 - 0.766 = 0.38514\dots$ , so  $u_* = 0.7669980\dots$  and (10.20) holds exactly when  $2ar < u_*$ . In particular  $ar \leq \frac{1}{4}$  gives  $h(r) \leq e^{1/2} - \frac{3}{2} < 1.6488 - 1.5 = 0.1488 < 0.386$ .  $\square$

The assumption  $\|I\| = 1$  was used only in the bound  $\|P_N\| \leq e^d$ ; for a general submultiplicative norm a fixed factor  $\|I\|$  appears and nothing else changes.

## 10.7 Entire resummation in the eigenvector case

When  $[X, Y] = sY$ , the factorization (9.5) with  $c = 1$  gives  $e^{t(X+Y)} = e^{tX} e^{q_s(t)Y}$ , hence

$$R_1(t) = e^{-tY} e^{-tX} e^{tX} e^{q_s(t)Y} = e^{(q_s(t)-t)Y}, \quad q_s(t) - t = \sum_{n \geq 2} \frac{(-1)^{n-1} s^{n-1}}{n!} t^n.$$

All exponents are multiples of the same  $Y$  and commute, so by [theorem 2.1](#)

$$e^{(q_s(t)-t)Y} = \prod_{n \geq 2} \exp\left(\frac{(-1)^{n-1} s^{n-1}}{n!} t^n Y\right),$$

and uniqueness in [theorem 10.1](#) yields the all-orders closed form

$$C_n = \frac{(-1)^{n-1} s^{n-1}}{n!} Y \quad (n \geq 2). \quad (10.22)$$

This product converges for every finite  $s$  and  $t$ , a useful contrast with the Bernoulli BCH series (9.8), whose radius in  $s$  is only  $2\pi$ . If instead the commutator  $C = [X, Y]$  is central, (9.2) gives  $e^{t(X+Y)} = e^{tX} e^{tY} e^{-t^2 C/2}$ , so  $C_2 = -\frac{1}{2}[X, Y]$  and  $C_n = 0$  for every  $n \geq 3$ .

## 11 Trotter, symmetric, and higher-order product formulas with complete error series

The source page mentions the Suzuki–Trotter decomposition as a consequence of the Zassenhaus formula without displaying a particular version. We prove the basic product limit, the symmetric second-order formula, and a recursive construction of arbitrarily high even order, each with its complete logarithmic error series. All analytic statements in this section concern bounded elements  $X, Y$  of a unital Banach algebra with  $\|I\| = 1$ ; they do not assert a product theorem for unbounded semigroup generators, for which the classical sources [10] require additional hypotheses. Throughout  $a = \|X\| + \|Y\|$  and  $S = X + Y$ .

We first record the multifactor version of [theorem 7.1\(i\)](#): for bounded  $A_1, \dots, A_m$  with  $\sum_i \|A_i\| = \sigma < \log 2$ , the Mercator logarithm of  $\prod_i e^{A_i}$  converges absolutely, equals the sum of the homogeneous parts of (2.7), exponentiates to the product, and the sum of the norms of its terms of degree at least  $N + 1$  is at most  $\rho^{-(N+1)} [-\log(2 - e^{\rho\sigma})]$  for any  $\rho > 1$  with  $\rho\sigma < \log 2$ . The proof is identical: the sum of the norms of the nonempty words of  $\prod e^{A_i} - I$  is at most  $e^\sigma - 1$ .

**Lemma 11.1.** *If  $\sigma < \frac{1}{2} \log 2$ , the terms of degree at least 2 in the logarithm of  $\prod_i e^{A_i}$  have total norm at most  $c_0 \sigma^2$ , and those of degree at least 3 have total norm at most  $c'_0 \sigma^3$ , with absolute constants  $c_0 = 4(\log 2)^{-2} \log \frac{1}{2-\sqrt{2}}$  and  $c'_0 = 8(\log 2)^{-3} \log \frac{1}{2-\sqrt{2}}$ .*

*Proof.* Take  $\rho = \log 2 / (2\sigma) > 1$ , so  $e^{\rho\sigma} = \sqrt{2}$  and  $-\log(2 - e^{\rho\sigma}) = \log \frac{1}{2-\sqrt{2}}$ . The tail bound with  $N = 1$  gives  $\rho^{-2} \log \frac{1}{2-\sqrt{2}} = \frac{4\sigma^2}{(\log 2)^2} \log \frac{1}{2-\sqrt{2}}$ , and with  $N = 2$  it gives the cubic bound.  $\square$

### 11.1 The Lie–Trotter product formula

**Theorem 11.2** (Lie–Trotter formula and its complete logarithmic error). *For every fixed scalar  $t$ ,*

$$\lim_{n \rightarrow \infty} (e^{tX/n} e^{tY/n})^n = e^{t(X+Y)} \quad (11.1)$$

in norm, with error  $O(n^{-1})$  uniformly for  $t$  in compact sets. For all  $n > |t|a/\log 2$  one has the exact convergent expansion

$$(e^{tX/n}e^{tY/n})^n = \exp\left(t(X+Y) + \sum_{k \geq 2} \frac{t^k}{n^{k-1}} Z_k(X, Y)\right), \quad (11.2)$$

so every term of the logarithmic error is explicitly available from (2.5) or Dynkin's formula.

*Proof.* For  $n > |t|a/\log 2$  the pair  $(tX/n, tY/n)$  satisfies the hypothesis of theorem 7.1(i), so  $L_n := \log(e^{tX/n}e^{tY/n}) = \sum_k Z_k(tX/n, tY/n) = \sum_k (t/n)^k Z_k(X, Y)$  converges absolutely, using homogeneity of  $Z_k$ , and  $e^{L_n} = e^{tX/n}e^{tY/n}$ . Powers of a single exponential satisfy  $(e^L)^n = e^{nL}$  by theorem 2.1 (in its analytic form), so the  $n$ th power equals  $\exp(nL_n) = \exp(tS + \sum_{k \geq 2} t^k n^{1-k} Z_k)$ , which is (11.2). Let  $E_n = \sum_{k \geq 2} t^k n^{1-k} Z_k = n \sum_{k \geq 2} Z_k(tX/n, tY/n)$ . For  $n \geq 2|t|a/\log 2$ , theorem 11.1 with  $\sigma = |t|a/n$  gives  $\|E_n\| \leq nc_0(|t|a/n)^2 = c_0 t^2 a^2/n$ . By the exponential difference estimate (7.5) with  $\|I\| = 1$ ,

$$\|(e^{tX/n}e^{tY/n})^n - e^{tS}\| = \|e^{tS+E_n} - e^{tS}\| \leq e^{|t|a+\|E_n\|} \|E_n\| \leq e^{|t|a+c_0 t^2 a^2/n} \frac{c_0 t^2 a^2}{n},$$

which is  $O(n^{-1})$  uniformly for  $|t|$  bounded.  $\square$

Thus the Zassenhaus viewpoint explains which correction factors are discarded, while BCH provides the most direct proof of the limit and of all corrections to its effective exponent.

## 11.2 Symmetry removes every even degree

Put

$$S_2(t) = e^{tX/2}e^{tY}e^{tX/2}. \quad (11.3)$$

**Proposition 11.3** (Symmetric splitting). *The formal logarithm of  $S_2(t)$  is odd in  $t$ :*

$$\log S_2(t) = t(X+Y) + \sum_{j \geq 1} t^{2j+1} E_{2j+1}, \quad E_3 = -\frac{1}{24}[X, [X, Y]] - \frac{1}{12}[Y, [X, Y]]. \quad (11.4)$$

Every  $E_{2j+1}$  is a Lie polynomial given by the finite multifactor word-cut formula (2.7) with  $(A_1, A_2, A_3) = (X/2, Y, X/2)$ , followed by the Dynkin projection. For bounded  $X, Y$  and  $n > |t|a/\log 2$ ,

$$S_2(t/n)^n = \exp\left(t(X+Y) + \sum_{j \geq 1} \frac{t^{2j+1}}{n^{2j}} E_{2j+1}\right) = e^{t(X+Y)} + O(n^{-2}), \quad (11.5)$$

uniformly for  $t$  in compact sets.

*Proof.* Since  $(e^A)^{-1} = e^{-A}$  and inversion reverses products,  $S_2(-t) = e^{-tX/2}e^{-tY}e^{-tX/2} = S_2(t)^{-1}$ . Let  $L(t) = \log S_2(t)$  be the formal logarithm, a series in  $\widehat{T}$  whose degree- $n$  part is  $t^n$  times a polynomial. Then  $e^{L(-t)} = S_2(-t) = S_2(t)^{-1} = e^{-L(t)}$ , and uniqueness of the formal logarithm gives  $L(-t) = -L(t)$ ; hence only odd powers of  $t$  occur. By theorem 2.4 and theorem 3.5 applied to the three-letter alphabet (the product of exponentials of primitives is group-like), each homogeneous part is a Lie polynomial given by the finite cut sum.

For  $E_3$  we compute by two applications of BCH through degree three. Put  $A = tX/2$  and  $B = tY$ , so that  $S_2 = e^A e^B e^A$ . By (4.9)–(4.11),

$$W := \text{BCH}(A, B) = A + B + \frac{1}{2}[A, B] + \frac{1}{12}[A, [A, B]] + \frac{1}{12}[B, [B, A]] + O(t^4).$$

Then  $\log S_2 = \text{BCH}(W, A) = W + A + \frac{1}{2}[W, A] + \frac{1}{12}[W, [W, A]] + \frac{1}{12}[A, [A, W]] + O(t^4)$ . Writing  $W = W_1 + W_2 + W_3 + O(t^4)$  with  $W_1 = A + B$ ,  $W_2 = \frac{1}{2}[A, B]$ ,  $W_3 = \frac{1}{12}[A, [A, B]] + \frac{1}{12}[B, [B, A]]$ , the degree-two part is  $W_2 + \frac{1}{2}[W_1, A] = \frac{1}{2}[A, B] + \frac{1}{2}[B, A] = 0$ , as oddness predicts. The degree-three part is

$$W_3 + \frac{1}{2}[W_2, A] + \frac{1}{12}[W_1, [W_1, A]] + \frac{1}{12}[A, [A, W_1]],$$

and the four pieces are, respectively,  $\frac{1}{12}[A, [A, B]] + \frac{1}{12}[B, [B, A]]$ ;  $\frac{1}{4}[[A, B], A] = -\frac{1}{4}[A, [A, B]]$ ;  $\frac{1}{12}[A + B, [B, A]] = -\frac{1}{12}[A, [A, B]] + \frac{1}{12}[B, [B, A]]$ ; and  $\frac{1}{12}[A, [A, B]]$ . Collecting, the coefficient of  $[A, [A, B]]$  is  $\frac{1}{12} - \frac{1}{4} - \frac{1}{12} + \frac{1}{12} = -\frac{1}{6}$  and that of  $[B, [B, A]]$  is  $\frac{1}{12} + \frac{1}{12} = \frac{1}{6}$ . Substituting  $A = tX/2$ ,  $B = tY$ :  $-\frac{1}{6} \cdot \frac{t^3}{4}[X, [X, Y]] + \frac{1}{6} \cdot \frac{t^3}{2}[Y, [Y, X]] = -\frac{t^3}{24}[X, [X, Y]] - \frac{t^3}{12}[Y, [X, Y]]$ , which is  $E_3$ .

For (11.5), apply the multifactor convergence statement to the three factors of  $S_2(t/n)$ , whose norms sum to  $|t|a/n$ ; the logarithm converges for  $n > |t|a/\log 2$ , and raising to the  $n$ th power multiplies it by  $n$ , giving the displayed exponent by homogeneity. The error exponent  $E'_n = n \sum_{j \geq 1} (t/n)^{2j+1} E_{2j+1}$  consists of the terms of degree at least 3 of  $n \log S_2(t/n)$ , so [theorem 11.1](#) gives  $\|E'_n\| \leq nc'_0(|t|a/n)^3 = O(n^{-2})$  for  $n \geq 2|t|a/\log 2$ , and the exponential difference estimate finishes the proof exactly as in [theorem 11.2](#).  $\square$

If  $[X, Y]$  is central, (9.3) shows that already  $S_2(t) = e^{t(X+Y)}$  exactly.

### 11.3 Arbitrarily high even order by a five-factor recursion

The recursive symmetric construction below is Suzuki's fractal decomposition [11, 12]; its order conditions follow directly from BCH.

**Theorem 11.4** (Symmetric product formulas of every even order). *For  $k \geq 2$  set*

$$p_k = \frac{1}{4 - 4^{1/(2k-1)}}, \quad (11.6)$$

and define recursively, starting from (11.3),

$$S_{2k}(t) = S_{2k-2}(p_k t)^2 S_{2k-2}((1 - 4p_k)t) S_{2k-2}(p_k t)^2. \quad (11.7)$$

Then  $S_{2k}(-t) = S_{2k}(t)^{-1}$  and

$$\log S_{2k}(t) = t(X + Y) + O(t^{2k+1}), \quad S_{2k}(t/n)^n = e^{t(X+Y)} + O(n^{-2k}) \quad (11.8)$$

for fixed  $k$  and  $t$ ; the first statement is formal and locally analytic, the second is a norm statement for bounded  $X, Y$ . Every term of the local logarithm of  $S_{2k}$  is a Lie polynomial with an explicit finite multifactor word-cut expansion.

*Proof.* We prove by induction on  $k \geq 1$  that  $S_{2k}(-t) = S_{2k}(t)^{-1}$  and  $\log S_{2k}(t) = tS + t^{2k+1}E^{(k)} + O(t^{2k+3})$  for some Lie polynomial  $E^{(k)}$ , where the logarithm is the formal one. The case  $k = 1$  is [theorem 11.3](#). Assume the statement for  $k - 1 \geq 1$  and abbreviate  $p = p_k$ ,  $c = (p, p, 1 - 4p, p, p)$ . Symmetry: since inversion reverses the order of a product and  $S_{2k-2}(-ct) = S_{2k-2}(ct)^{-1}$ ,

$$\begin{aligned} S_{2k}(-t) &= S_{2k-2}(pt)^{-2} S_{2k-2}((1 - 4p)t)^{-1} S_{2k-2}(pt)^{-2} \\ &= (S_{2k-2}(pt)^2 S_{2k-2}((1 - 4p)t) S_{2k-2}(pt)^2)^{-1} = S_{2k}(t)^{-1}, \end{aligned}$$

using that the list  $c$  is palindromic. Hence  $\log S_{2k}$  is odd in  $t$ , by the uniqueness argument of [theorem 11.3](#).

Order: the five factors are exponentials of  $W_i = c_i tS + c_i^{2k-1} t^{2k-1} E^{(k-1)} + O(t^{2k+1})$ ,  $i = 1, \dots, 5$ . By the formal BCH theorem for several factors ([theorem 2.4](#) together with [theorem 3.5](#)),

$\log \prod_i e^{W_i} = \sum_i W_i + (\text{Lie polynomials of degree at least two in the } W_i)$ . A bracket in which every argument is a multiple of  $tS$  vanishes, since  $[S, S] = 0$ ; a bracket containing at least one factor of order  $t^{2k-1}$  and at least one other factor has order at least  $t^{2k}$ . Therefore

$$\log S_{2k}(t) = tS \sum_i c_i + t^{2k-1} E^{(k-1)} \sum_i c_i^{2k-1} + O(t^{2k}).$$

Now  $\sum_i c_i = 4p + (1 - 4p) = 1$ . From (11.6),  $4 - 4^{1/(2k-1)} = 1/p$ , so  $1 - 4p = -4^{1/(2k-1)}p$  and  $(1 - 4p)^{2k-1} = -4p^{2k-1}$ ; hence  $\sum_i c_i^{2k-1} = 4p^{2k-1} + (1 - 4p)^{2k-1} = 0$ . Thus  $\log S_{2k}(t) = tS + O(t^{2k})$ , and oddness removes the degree- $2k$  term, leaving  $tS + t^{2k+1} E^{(k)} + O(t^{2k+3})$  with  $E^{(k)}$  a Lie polynomial. This completes the induction and proves the first part of (11.8); the finite word-cut description of every term follows from theorem 2.4 applied to the  $5^{k-1} \cdot 3$  exponential factors of  $S_{2k}$  with their scalar weights.

For the  $n$ -step estimate,  $S_{2k}(t/n)$  is a finite product of exponentials whose norms sum to  $\sigma_n = \lambda_k |t|a/n$ , where  $\lambda_k = \prod_{j=2}^k (4p_j + |1 - 4p_j|)$  accounts for the (possibly negative) substeps. For  $n$  large,  $\sigma_n < \frac{1}{2} \log 2$  and the multifactor logarithm converges;  $n \log S_{2k}(t/n) = tS + E_n''$  where  $E_n''$  consists of the terms of degree at least  $2k + 1$  of  $n \log S_{2k}(t/n)$ , hence has norm at most  $n\rho^{-(2k+1)} \log \frac{1}{2-\sqrt{2}}$  with  $\rho = \log 2/(2\sigma_n)$ , that is  $O(n\sigma_n^{2k+1}) = O(n^{-2k})$ . The exponential difference estimate then gives the second part of (11.8).  $\square$

The middle coefficient  $1 - 4p_k$  is negative (for  $k = 2$ ,  $p_2 = 1/(4 - 4^{1/3}) \approx 0.4145$  and  $1 - 4p_2 \approx -0.658$ ). Negative substeps are harmless for bounded operators and for unitary groups but can be a genuine obstruction for one-sided unbounded semigroups; the boundedness hypothesis of this section is therefore part of the result, not an optional technicality. The proof asserts nothing about convergence as  $k \rightarrow \infty$  at fixed  $t$ .

## 12 Exponential coordinates, coframes, and metrics to all orders

The differential identity (5.6) has an intrinsic geometric meaning. This section develops it into complete coordinate expansions and isolates the corrections needed in the geometric paragraph of the source page.

### 12.1 The infinitesimal formula and its full component expansion

Let  $G$  be a finite-dimensional Lie group with Lie algebra  $\mathfrak{g}$ , basis  $e_1, \dots, e_d$ , and structure constants

$$[e_i, e_j] = f_{ij}^k e_k, \quad X = X^i e_i, \quad (12.1)$$

with repeated upper-lower indices summed. For a differentiable  $\mathfrak{g}$ -valued function  $X$  on a manifold, applying (5.6) (intrinsically, theorem 8.2) to every tangent vector gives the complete infinitesimal identity

$$\begin{aligned} e^{-X} \mathrm{d}e^X &:= \exp(X)^* \theta = \phi(\mathrm{ad}_X) \mathrm{d}X = \sum_{n \geq 0} \frac{(-1)^n}{(n+1)!} \mathrm{ad}_X^n(\mathrm{d}X) \\ &= \mathrm{d}X - \frac{1}{2}[X, \mathrm{d}X] + \frac{1}{6}[X, [X, \mathrm{d}X]] - \dots, \end{aligned} \quad (12.2)$$

an equality of  $\mathfrak{g}$ -valued one-forms; in a matrix group the left side is literally  $g^{-1} \mathrm{d}g$  with  $g = e^X$ . Inserting (12.1) into the successive commutators gives the component terms displayed on the source page,

$$e^{-X} \mathrm{d}e^X = e_k \mathrm{d}X^k - \frac{1}{2} X^i \mathrm{d}X^j f_{ij}^k e_k + \frac{1}{6} X^i X^j \mathrm{d}X^k f_{jk}^\ell f_{i\ell}^m e_m - \dots, \quad (12.3)$$

whose indices and signs are fixed by the bracket convention rather than guessed. Define the matrix of  $\text{ad}_X$  and the coframe matrix by

$$M^a{}_b = X^i f_{ib}{}^a, \quad W(X) = \phi(M) = \sum_{n \geq 0} \frac{(-1)^n}{(n+1)!} M^n = \int_0^1 e^{-uM} du, \quad (12.4)$$

so that  $\text{ad}_X e_b = M^a{}_b e_a$ . Then every component at every order is specified by

$$e^{-X} d e^X = e_a W^a{}_b(X) dX^b, \quad (M^n)^a{}_b = \sum_{i_1, \dots, i_n} \sum_{c_1, \dots, c_{n-1}} X^{i_1} \dots X^{i_n} f_{i_1 c_{n-1}}{}^a f_{i_2 c_{n-2}}{}^{c_{n-1}} \dots f_{i_n b}{}^{c_1}, \quad (12.5)$$

which is ordinary matrix multiplication written out; for  $n = 1$  it reads  $X^i f_{ib}{}^a$ , and for  $n = 2$  it reproduces the third term of (12.3) after renaming dummy indices. The factorial series in (12.4) converges for every finite matrix  $M$ , and the integral form follows from (5.6).

## 12.2 Which matrix is the vielbein?

**Proposition 12.1** (The coframe and its regularity). *In exponential coordinates  $X^1, \dots, X^d$  near the identity, the one-forms  $\theta^a = W^a{}_b dX^b$  form a coframe (the components of the left Maurer–Cartan form), and the dual frame has coefficient matrix  $W^{-1}$ . The matrix  $M = \text{ad}_X$  is not this coframe: in positive dimension it is always singular, whereas  $W(0) = I$ . More precisely,*

$$W(X) \text{ is invertible} \iff \text{spec}(\text{ad}_X) \cap (2\pi i\mathbb{Z} \setminus \{0\}) = \emptyset, \quad (12.6)$$

*the spectrum being computed after complexification for a real Lie algebra, and on this set  $W^{-1} = \beta(M)$  by holomorphic functional calculus (not necessarily by a convergent Taylor series about  $M = 0$ ).*

*Proof.* Equation (12.5) identifies  $\theta^a$  with the components of the pullback of  $\theta$  under  $\exp$ , and since  $\exp$  is a local diffeomorphism at 0 (theorem 8.3) and  $W(0) = I$ , the  $\theta^a$  are linearly independent near 0: they form a coframe, and the dual frame is given by the inverse matrix. If  $X \neq 0$  then  $MX = [X, X] = 0$  with  $X \neq 0$ , so  $M$  has a nontrivial kernel; if  $X = 0$  then  $M = 0$ . Hence  $M$  is singular in every positive-dimensional Lie algebra and cannot be a frame or coframe matrix. Over  $\mathbb{C}$  triangularize  $M$ ; a power series in a triangular matrix is triangular with diagonal entries equal to the scalar function evaluated at the diagonal entries, so  $\det W = \prod_{\lambda \in \text{spec } M} \phi(\lambda)$  with multiplicities. The zeros of the entire function  $\phi(z) = (1 - e^{-z})/z$  are exactly the nonzero points of  $2\pi i\mathbb{Z}$  ( $1 - e^{-z} = 0$  iff  $z \in 2\pi i\mathbb{Z}$ , and at  $z = 0$  the value is 1). This proves (12.6); in particular the eigenvalue 0 forced by  $MX = 0$  is harmless for  $W$ . Where no zero is present, the analytic identity  $\phi\beta = 1$  and multiplicativity of functional calculus give  $W^{-1} = \beta(M)$ .  $\square$

Some authors reserve “vielbein” for the dual vector frame rather than the coframe; with that convention the frame coefficients are  $W^{-1}$ , still not  $M$ . The quotient  $(I - e^{-M})M^{-1}$  used on the source page is a shorthand for the entire function  $\phi(M)$ ; a literal inverse of  $M$  does not exist in this setting. This is a correction of a mathematical misidentification on the page, not an alternative sign convention. The defining differential relation of the coframe is the Maurer–Cartan equation of theorem 8.1 in components,

$$d\theta^a = -\frac{1}{2} f_{bc}{}^a \theta^b \wedge \theta^c, \quad (12.7)$$

obtained by writing  $\theta = e_a \theta^a$  and  $\frac{1}{2}[\theta, \theta] = \frac{1}{2} \theta^b \wedge \theta^c [e_b, e_c]$ .

### 12.3 Pullback tensors: the Jacobian and the nondegeneracy hypotheses

Let  $B$  be a symmetric bilinear form on  $\mathfrak{g}$  with matrix  $B_{ab} = B(e_a, e_b)$ . Left translation defines a left-invariant symmetric tensor  $g_G$  on  $G$  by  $g_G(u, v) = B(\theta u, \theta v)$ ; it is a pseudo-Riemannian metric exactly when  $B$  is nondegenerate, and Riemannian when  $B$  is positive definite. No invariance of  $B$  under the adjoint action is needed for left invariance.

**Theorem 12.2** (Pullback formula with its hypotheses). *Let  $F : N \rightarrow G$  be smooth. Near a point  $q_0 \in N$ , after a fixed left translation (which does not change the pullback of a left-invariant tensor), write  $F(q) = \exp X(q)$  with  $X(q) = X^b(q)e_b$  in coordinates  $q^\mu$  on  $N$ , and set*

$$E^a{}_\mu(q) = W^a{}_b(X(q)) \partial_\mu X^b(q). \quad (12.8)$$

Then

$$(F^*\theta)^a = E^a{}_\mu dq^\mu, \quad (F^*g_G)_{\mu\nu} = B_{ab} E^a{}_\mu E^b{}_\nu. \quad (12.9)$$

In exponential coordinates themselves,  $X^b = q^b$ , the Jacobian is the identity and

$$g_{ij} = B_{ab} W^a{}_i W^b{}_j, \quad g = W^\top B W. \quad (12.10)$$

If  $B$  is positive definite,  $F^*g_G$  is a Riemannian metric on  $N$  if and only if  $F$  is an immersion. If  $B$  is indefinite,  $F^*g_G$  is a pseudo-Riemannian metric if and only if  $F$  is an immersion and  $B$  restricted to  $\theta(dF(T_q N))$  is nondegenerate at every  $q$ .

*Proof.* The chain rule applied to (12.5) gives  $F^*\theta = (\exp \circ X)^*\theta = e_a W^a{}_b(X) dX^b = e_a W^a{}_b(X) \partial_\mu X^b dq^\mu$ , the first identity. By definition of the pullback of a bilinear form,  $(F^*g_G)(\partial_\mu, \partial_\nu) = B(\theta F_* \partial_\mu, \theta F_* \partial_\nu) = B(e_a E^a{}_\mu, e_b E^b{}_\nu) = B_{ab} E^a{}_\mu E^b{}_\nu$ . If  $B$  is positive definite, then  $(F^*g_G)(v, v) = B(\theta F_* v, \theta F_* v) \geq 0$  with equality iff  $F_* v = 0$ , so the pullback is positive definite iff  $dF$  is injective at every point. If  $B$  is indefinite, the pullback is the restriction of the nondegenerate form  $B$  to the subspace  $\theta(dF(T_q N))$  transported by the linear map  $\theta \circ dF$ , which is an isomorphism onto that subspace iff  $dF$  is injective; the transported form is nondegenerate iff the restriction is.  $\square$

Three distinct qualifications follow. An arbitrary smooth map does not induce a metric: a constant map has zero pullback. For an indefinite  $B$  even an immersion can have a degenerate pullback: the immersion  $q \mapsto (q, q)$  into  $(\mathbb{R}^2, dx^2 - dy^2)$  pulls the metric back to  $(1 - 1)dq^2 = 0$ . And an arbitrary previously specified metric on  $N$  cannot be asserted to be such a pullback without additional data. The Jacobian factors in (12.8) are indispensable unless the coordinates on  $N$  are the exponential coordinates  $X^i$  themselves.

The Killing form is

$$B_K(u, v) = \text{tr}(\text{ad}_u \text{ad}_v), \quad (B_K)_{ab} = f_{ac}{}^d f_{bd}{}^c, \quad (12.11)$$

the second expression being the trace of the product of the two adjoint matrices. It is symmetric by  $\text{tr}(AB) = \text{tr}(BA)$ , and it is invariant:

$$\begin{aligned} B_K([x, u], v) + B_K(u, [x, v]) &= \text{tr}([\text{ad}_x, \text{ad}_u] \text{ad}_v + \text{ad}_u [\text{ad}_x, \text{ad}_v]) \\ &= \text{tr}(\text{ad}_x \text{ad}_u \text{ad}_v - \text{ad}_u \text{ad}_x \text{ad}_v + \text{ad}_u \text{ad}_x \text{ad}_v - \text{ad}_u \text{ad}_v \text{ad}_x) \\ &= \text{tr}[\text{ad}_x, \text{ad}_u \text{ad}_v] = 0, \end{aligned}$$

using  $\text{ad}_{[x, u]} = [\text{ad}_x, \text{ad}_u]$  from (1.3). But the Killing form need not be nondegenerate: in an abelian Lie algebra every adjoint map is zero and  $B_K \equiv 0$ , and more generally it vanishes on the center of any Lie algebra. (By Cartan's criterion, which is not needed here,  $B_K$  is nondegenerate exactly for semisimple Lie algebras; and for compact semisimple algebras it is negative definite,

so even then it is  $-B_K$  that is Riemannian.) Consequently calling its pullback a “metric” without a nondegeneracy hypothesis is not a valid general assertion, and it is not a universal choice of Riemannian metric; an arbitrary positive definite inner product at the identity always gives a left-invariant Riemannian metric instead. When  $B_K$  is nondegenerate it may be used as  $B$  in [theorem 12.2](#), with its signature respected. Formula (12.10) is valid regardless of whether it defines a metric or only a possibly degenerate tensor.

## 12.4 All-orders metric coefficients, their even resummation, and the volume density

Without assuming invariance, substituting (12.4) into (12.10) gives the absolutely convergent double series

$$g_{ij} = \sum_{p,q \geq 0} \frac{(-1)^{p+q}}{(p+1)!(q+1)!} B_{ab} (M^p)^a{}_i (M^q)^b{}_j, \quad (12.12)$$

valid for every  $M$ . If  $B$  is invariant the answer is considerably simpler.

**Theorem 12.3** (Complete invariant-metric expansion). *Suppose  $B([X, u], v) + B(u, [X, v]) = 0$  for all  $u, v$  (for example  $B = B_K$ ). Then*

$$g = W^\top B W = B \phi(-M) \phi(M) = B \frac{e^M + e^{-M} - 2I}{M^2} = B \frac{2(\cosh M - I)}{M^2} = B \left( \frac{\sinh(M/2)}{M/2} \right)^2 \quad (12.13)$$

$$= B \sum_{n \geq 0} \frac{2M^{2n}}{(2n+2)!} = B \left( I + \frac{M^2}{12} + \frac{M^4}{360} + \frac{M^6}{20160} + \cdots \right). \quad (12.14)$$

All quotients denote entire functions with their removable values at zero; no inverse of  $M$  or  $M^2$  is assumed. If moreover  $B$  and  $W$  are invertible,  $g^{-1} = \beta(M) B^{-1} \beta(M)^\top$ .

*Proof.* In components, invariance reads  $B_{ab} M^a{}_c u^c v^b = -B_{ab} u^a M^b{}_c v^c$  for all  $u, v$ , that is  $M^\top B = -BM$ . Induction gives  $(M^\top)^n B = (-1)^n B M^n$ : if it holds for  $n$ , then  $(M^\top)^{n+1} B = M^\top (-1)^n B M^n = (-1)^{n+1} B M^{n+1}$ . Hence

$$W^\top B = \sum_n \frac{(-1)^n}{(n+1)!} (M^\top)^n B = B \sum_n \frac{M^n}{(n+1)!} = B \phi(-M),$$

and  $W^\top B W = B \phi(-M) \phi(M)$ . The scalar identity

$$\phi(-z) \phi(z) = \frac{(e^z - 1)(1 - e^{-z})}{z^2} = \frac{e^z + e^{-z} - 2}{z^2} = \frac{2(\cosh z - 1)}{z^2} = \frac{4 \sinh^2(z/2)}{z^2} = \sum_{n \geq 0} \frac{2z^{2n}}{(2n+2)!}$$

holds between entire functions (expand  $\cosh z - 1 = \sum_{m \geq 1} z^{2m}/(2m)!$ ), and since all the matrix functions involved are power series in the single matrix  $M$ , they commute and the scalar identity transfers to  $M$ . The coefficients are  $2/2! = 1$ ,  $2/4! = \frac{1}{12}$ ,  $2/6! = \frac{1}{360}$ ,  $2/8! = \frac{1}{20160}$ . The resulting matrix is symmetric because it was obtained as  $W^\top B W$ . Finally  $g^{-1} = W^{-1} B^{-1} (W^\top)^{-1} = \beta(M) B^{-1} \beta(M)^\top$  when the inverses exist.  $\square$

Note that all odd orders disappear only under the invariance hypothesis; no such cancellation occurs in (12.12).

When  $B$  is nondegenerate and the  $X^i$  are exponential coordinates, the metric matrix is  $W^\top B W$ , so its volume density is

$$\sqrt{|\det g|} = |\det W| \sqrt{|\det B|}, \quad \det W = \prod_{\lambda \in \text{spec } M} \phi(\lambda), \quad (12.15)$$

with the factor 1 at  $\lambda = 0$  and multiplicities included, by the triangularization used in [theorem 12.1](#). If the spectral radius of  $M$  is less than  $2\pi$ , then

$$\det W = \exp\left(-\frac{1}{2} \operatorname{tr} M + \sum_{m \geq 1} \frac{b_{2m}^+}{2m} \operatorname{tr}(M^{2m})\right) = \exp\left(-\frac{1}{2} \operatorname{tr} M + \frac{\operatorname{tr} M^2}{24} - \frac{\operatorname{tr} M^4}{2880} + \cdots\right). \quad (12.16)$$

*Proof of (12.16).* For  $|z| < 2\pi$ ,  $\phi(z) \neq 0$  and

$$\frac{d}{dz} \log \phi(z) = \frac{e^{-z}}{1 - e^{-z}} - \frac{1}{z} = \frac{1}{e^z - 1} - \frac{1}{z} = \frac{\gamma(z) - 1}{z} = \sum_{m \geq 1} b_m^- z^{m-1} = -\frac{1}{2} + \sum_{m \geq 1} b_{2m}^+ z^{2m-1},$$

using  $b_1^- = -\frac{1}{2}$  and  $b_{2m}^- = b_{2m}^+$ ,  $b_{2m+1}^- = 0$  for  $m \geq 1$ . Integrating with  $\log \phi(0) = 0$  gives  $\log \phi(z) = -\frac{z}{2} + \sum_{m \geq 1} \frac{b_{2m}^+}{2m} z^{2m}$ , a series with radius  $2\pi$  (the nearest zeros of  $\phi$ ). Hence  $\phi(\lambda_j) = \exp(-\frac{\lambda_j}{2} + \sum_m \frac{b_{2m}^+}{2m} \lambda_j^{2m})$  for every eigenvalue  $\lambda_j$  with  $|\lambda_j| < 2\pi$ ; multiplying over  $j$  and using  $\sum_j \lambda_j^k = \operatorname{tr} M^k$  (absolute convergence allows the interchange of the sums over  $j$  and  $m$ ) gives (12.16). The first coefficients are  $b_2^+/2 = \frac{1}{24}$  and  $b_4^+/4 = -\frac{1}{2880}$ .  $\square$

The determinant formula (12.15) itself has no radius restriction.

## 12.5 A concrete three-dimensional resummation

Identify the rotation Lie algebra  $\mathfrak{so}(3)$  with  $\mathbb{R}^3$  and the bracket with the cross product,  $[x, v] = x \times v$ . Then  $Mv = x \times v$  and, with  $r = |x|$ , the vector triple-product identity gives  $M^2v = x \times (x \times v) = x(x \cdot v) - r^2v$  and hence  $M^3v = x \times (x(x \cdot v) - r^2v) = -r^2x \times v$ , that is  $M^3 = -r^2M$ . Consequently  $M^{2k+1} = (-r^2)^k M$  and  $M^{2k+2} = (-r^2)^k M^2$  for  $k \geq 0$ , and summing the exponential series,

$$e^{-sM} = I + \sum_{k \geq 0} \frac{(-s)^{2k+1}(-r^2)^k}{(2k+1)!} M + \sum_{k \geq 0} \frac{(-s)^{2k+2}(-r^2)^k}{(2k+2)!} M^2 = I - \frac{\sin(sr)}{r} M + \frac{1 - \cos(sr)}{r^2} M^2,$$

which is Rodrigues' formula. Integrating over  $s \in [0, 1]$  by (12.4) gives the complete coframe

$$W = I - \frac{1 - \cos r}{r^2} M + \frac{r - \sin r}{r^3} M^2, \quad (12.17)$$

with the Taylor values  $\frac{1}{2}$  and  $\frac{1}{6}$  of the two coefficients at  $r = 0$ . The Euclidean dot product is invariant, since  $(x \times u) \cdot v + u \cdot (x \times v) = 0$ . For  $r > 0$  let  $P_{\parallel} = xx^{\top}/r^2$  and  $P_{\perp} = I - P_{\parallel}$ ; then  $M = 0$  on the range of  $P_{\parallel}$  and  $M^2 = -r^2I$  on the range of  $P_{\perp}$  (the plane orthogonal to  $x$ , where  $x \cdot v = 0$ ). Formula (12.13) with  $B = I$  therefore gives

$$W^{\top}W = P_{\parallel} + \frac{2(1 - \cos r)}{r^2} P_{\perp} = P_{\parallel} + \left(\frac{\sin(r/2)}{r/2}\right)^2 P_{\perp}, \quad \det W = \left(\frac{\sin(r/2)}{r/2}\right)^2, \quad (12.18)$$

the determinant following either from the eigenvalues  $0, ir, -ir$  of  $M$  and  $\phi(ir)\phi(-ir) = |1 - e^{-ir}|^2/r^2 = (2 - 2\cos r)/r^2$ , or directly from (12.17). Thus the coordinate coframe of the rotation group in exponential coordinates is singular exactly at the radii  $r = 2\pi, 4\pi, \dots$ , precisely as (12.6) predicts; at  $r = 0$  the whole expression has the unambiguous limit  $I$ .

## 13 Quantum-mechanical identities with explicit operator domains

The formal central-commutator calculation of [theorem 9.2](#) predicts the exponential formulas on the source page. For operators on a Hilbert space, however, unboundedness and domains cannot

be suppressed: we first prove the identities directly in the Schrödinger and Bargmann–Fock realizations, and then exhibit why a commutator identity on a dense domain alone is insufficient. This avoids the Stone–von Neumann classification theorem, which is mentioned but not stated on the page; background on these realizations may be found in [14, 13, 9]. Inner products are linear in the second argument, and  $|n\rangle, \langle n|$  denote basis vectors and functionals.

### 13.1 Position, momentum, and the Weyl relations

On  $L^2(\mathbb{R}, dx)$  fix  $\hbar > 0$  and define, on the Schwartz space  $\mathcal{S}(\mathbb{R})$ ,

$$(Qf)(x) = xf(x), \quad (Pf)(x) = -i\hbar f'(x). \quad (13.1)$$

Both map  $\mathcal{S}$  into itself, and direct differentiation gives  $(QPf)(x) - (PQf)(x) = -i\hbar xf'(x) + i\hbar(xf)'(x) = i\hbar f(x)$ , that is,

$$[Q, P]f = i\hbar f \quad (f \in \mathcal{S}(\mathbb{R})). \quad (13.2)$$

Both operators are symmetric on  $\mathcal{S}$  (integration by parts for  $P$ , with boundary terms vanishing by decay). Their self-adjoint realizations are the maximal multiplication operator  $\overline{Q}$  on  $\{f : xf \in L^2\}$  and  $\overline{P} = \mathcal{F}^{-1}(\hbar k)\mathcal{F}$ , the Fourier conjugate of multiplication by  $\hbar k$ , on  $\{f : k\hat{f} \in L^2\}$ ; these agree with (13.1) on  $\mathcal{S}$ , since the Fourier transform of  $-i\hbar f'$  is  $\hbar k\hat{f}$  for  $f \in \mathcal{S}$ . Schwartz space is a core for each. For multiplication by a real function  $m$  (here  $m(x) = x$  or, after Fourier transform,  $m(k) = \hbar k$ , and  $\mathcal{F}$  maps  $\mathcal{S}$  onto  $\mathcal{S}$ ) this follows from the basic criterion: a symmetric operator  $A$  is essentially self-adjoint iff  $(A \pm i)\text{Dom } A$  are dense; if  $g \perp (m \pm i)f$  for all  $f \in \mathcal{S}$ , then  $\int (m \mp i)\bar{g}f = 0$  for all  $f \in \mathcal{S}$ , so the locally integrable function  $(m \mp i)\bar{g}$  vanishes almost everywhere, and since  $m \mp i \neq 0$  we get  $g = 0$ .

The unitary groups are concrete:

$$(U(a)f)(x) = (e^{iaQ}f)(x) = e^{iax}f(x), \quad (V(b)f)(x) = (e^{ibP}f)(x) = f(x + b\hbar), \quad a, b \in \mathbb{R}. \quad (13.3)$$

The first is the spectral calculus of the multiplication operator. For the second,  $e^{ib\overline{P}} = \mathcal{F}^{-1}e^{ib\hbar k}\mathcal{F}$ , and multiplication of  $\hat{f}$  by  $e^{ib\hbar k}$  is the Fourier transform of the translate  $f(\cdot + b\hbar)$ . Both families are unitary by a change of variables and strongly continuous: for  $f \in C_c^\infty$ ,  $\|V(b)f - f\| \rightarrow 0$  by uniform continuity, and this extends to  $L^2$  by density and the uniform bound  $\|V(b)\| = 1$ ; similarly for  $U$ .

**Theorem 13.1** (Weyl relations and exact symmetric splitting). *For real  $a, b$ , the operator  $aQ + bP$  on  $\mathcal{S}(\mathbb{R})$  is essentially self-adjoint, and the unitary group of its closure acts by*

$$(e^{i(aQ+bP)}f)(x) = e^{ia(x+b\hbar/2)}f(x + b\hbar) =: (\mathcal{W}(a, b)f)(x). \quad (13.4)$$

Consequently, as identities between bounded operators on all of  $L^2(\mathbb{R})$ ,

$$e^{iaQ}e^{ibP} = e^{-iab\hbar/2}e^{i(aQ+bP)} = \exp(i(aQ + bP - \tfrac{1}{2}ab\hbar I)), \quad (13.5)$$

$$e^{i(aQ+bP)} = e^{iaQ/2}e^{ibP}e^{iaQ/2}, \quad (13.6)$$

$$e^{iaQ}e^{ibP} = e^{-iab\hbar}e^{ibP}e^{iaQ}, \quad (13.7)$$

$$\mathcal{W}(a, b)\mathcal{W}(c, d) = e^{i\hbar(bc-ad)/2}\mathcal{W}(a + c, b + d). \quad (13.8)$$

*Proof.* Let  $b \neq 0$  and let  $T$  be multiplication by  $\tau(x) = e^{iax^2/(2b\hbar)}$ , a unitary operator preserving  $\mathcal{S}$ . For  $f \in \mathcal{S}$ , since  $\tau' = (iax/(b\hbar))\tau$ ,

$$T^{-1}(bP)Tf = \tau^{-1}(-i\hbar b)(\tau f)' = (-i\hbar b)\left(\frac{iax}{b\hbar}f + f'\right) = axf - i\hbar bf' = (aQ + bP)f.$$

Thus  $aQ + bP$  on  $\mathcal{S}$  is unitarily equivalent to  $bP$  on  $\mathcal{S}$ , which is essentially self-adjoint with closure  $b\overline{P}$ ; hence  $aQ + bP$  is essentially self-adjoint with closure  $T^{-1}b\overline{P}T$ , and its unitary group is  $T^{-1}e^{itbP}T = T^{-1}V(tb)T$ . Evaluating,

$$\begin{aligned} (T^{-1}V(b)Tf)(x) &= e^{-iax^2/(2b\hbar)} e^{ia(x+b\hbar)^2/(2b\hbar)} f(x+b\hbar) \\ &= e^{ia(2xb\hbar+b^2\hbar^2)/(2b\hbar)} f(x+b\hbar) = e^{ia(x+b\hbar/2)} f(x+b\hbar), \end{aligned}$$

which is (13.4); for  $b = 0$  the statement is the multiplication case  $U(a)$ , also of the form (13.4). (One can also verify directly that  $\mathcal{W}(ta, tb)$  is a strongly continuous one-parameter unitary group whose derivative at  $t = 0$  on  $\mathcal{S}$  is  $i(aQ + bP)$ .)

Now compute with (13.3):  $(U(a)V(b)f)(x) = e^{iax}f(x+b\hbar) = e^{-iab\hbar/2}(\mathcal{W}(a, b)f)(x)$ , which is (13.5); the second form holds because the scalar  $-\frac{1}{2}iab\hbar I$  commutes with everything. Next,  $(U(a/2)V(b)U(a/2)f)(x) = e^{iax/2}e^{ia(x+b\hbar)/2}f(x+b\hbar) = e^{ia(x+b\hbar/2)}f(x+b\hbar)$ , which is (13.6). Further,  $(V(b)U(a)f)(x) = e^{ia(x+b\hbar)}f(x+b\hbar) = e^{iab\hbar}(U(a)V(b)f)(x)$ , which is (13.7). Finally,

$$(\mathcal{W}(a, b)\mathcal{W}(c, d)f)(x) = e^{ia(x+b\hbar/2)}e^{ic(x+b\hbar+d\hbar/2)}f(x+(b+d)\hbar),$$

and the phase of  $\mathcal{W}(a+c, b+d)f$  is  $e^{i(a+c)(x+(b+d)\hbar/2)}$ ; the difference of exponents is  $i\hbar(\frac{ab}{2} + cb + \frac{cd}{2} - \frac{(a+c)(b+d)}{2}) = i\hbar\frac{bc-ad}{2}$ , which proves (13.8). All four identities involve only bounded unitary operators and hold on all of  $L^2(\mathbb{R})$ .  $\square$

The formal central-commutator calculation agrees:  $[iaQ, ibP] = -ab \cdot i\hbar I = -iab\hbar I$  is central, so (9.3) predicts exactly the correction  $-\frac{1}{2}iab\hbar I$  in (13.5), and the symmetric splitting of theorem 9.2 becomes exact. The direct proof establishes that the formal calculation is legitimate for the chosen self-adjoint realizations; (13.5) has not been justified by substituting unbounded operators into a norm-convergent series. Induction in (13.8) gives the complete finite-factor formula

$$\prod_{j=1}^m \mathcal{W}(a_j, b_j) = \exp\left(\frac{i\hbar}{2} \sum_{j < k} (b_j a_k - a_j b_k)\right) \mathcal{W}\left(\sum_j a_j, \sum_j b_j\right), \quad (13.9)$$

which is (9.4) in its unitary Heisenberg representation, with no omitted higher-order terms.

## 13.2 Why the infinitesimal commutator is not enough

**Proposition 13.2.** *A relation  $[Q, P] = iI$  on a dense common invariant domain, even for self-adjoint  $Q$  and  $P$ , does not by itself imply the Weyl relations.*

*Proof.* On  $L^2(0, 1)$  let  $Q$  be multiplication by  $x$ , a bounded self-adjoint operator, and let  $P = -i d/dx$  with the periodic domain  $\{f \in H^1(0, 1) : f(0) = f(1)\}$ . Under the Fourier series isomorphism  $L^2(0, 1) \cong \ell^2(\mathbb{Z})$  given by the orthonormal basis  $e^{2\pi i n x}$ ,  $P$  becomes multiplication by the real sequence  $2\pi n$  on its maximal domain; so  $P$  is self-adjoint, with  $e^{iP}$  acting as multiplication by  $e^{2\pi i n} = 1$ , that is,  $e^{iP} = I$ . The dense subspace  $C_c^\infty(0, 1)$  lies in the domains of  $Q$  and  $P$ , is invariant under both, and satisfies  $[Q, P] = iI$  there by differentiation. If the Weyl relation (13.7) (with  $\hbar = 1$ ) held for all real  $a$  and  $b = 1$ , it would give  $e^{iaQ} = e^{-ia}e^{iaQ}$ , hence  $e^{-ia} = 1$ , which is false unless  $a \in 2\pi\mathbb{Z}$ .  $\square$

The subspace  $C_c^\infty(0, 1)$  is not a core for the periodic derivative, and the periodic domain is not preserved by multiplication by  $e^{iax}$ : that is exactly where a naive exponentiation argument breaks down. The failure is a domain and integrability issue, not an error in the formal central-commutator algebra.

There cannot even be bounded operators  $A, B$  on a Banach space with  $[A, B] = I$ . Indeed, induction gives  $[A, B^n] = nB^{n-1}$ : if it holds for  $n$ , then  $[A, B^{n+1}] = [A, B^n]B + B^n[A, B] =$

$nB^{n-1}B + B^n = (n+1)B^n$ . No power of  $B$  vanishes: if  $B^m = 0$  with  $m \geq 1$  minimal, then  $0 = [A, B^m] = mB^{m-1}$  forces  $B^{m-1} = 0$ , contradicting minimality unless  $m = 1$ , and  $B = 0$  contradicts  $[A, B] = I$ . Hence  $n\|B^{n-1}\| \leq 2\|A\|\|B^n\| \leq 2\|A\|\|B\|\|B^{n-1}\|$  with  $\|B^{n-1}\| \neq 0$ , so  $n \leq 2\|A\|\|B\|$  for every  $n$ , which is impossible. (For finite matrices, taking traces gives the contradiction  $0 = \operatorname{tr}[A, B] = \operatorname{tr} I = d$  immediately.) Some unboundedness is therefore intrinsic to the canonical relation, and the bounded-operator convergence theorems of section 7 cannot be applied directly to  $Q, P$ .

### 13.3 Creation and annihilation operators in Bargmann–Fock space

Let  $\mathcal{F}$  be the space of entire functions  $F$  on  $\mathbb{C}$  with

$$\|F\|_{\mathcal{F}}^2 = \frac{1}{\pi} \int_{\mathbb{C}} |F(z)|^2 e^{-|z|^2} d^2z < \infty. \quad (13.10)$$

Polar integration gives  $\langle z^m, z^n \rangle = \frac{1}{\pi} \int_0^\infty \int_0^{2\pi} \rho^{m+n} e^{i(n-m)\varphi} e^{-\rho^2} \rho d\varphi d\rho = \delta_{mn} 2 \int_0^\infty \rho^{2n+1} e^{-\rho^2} d\rho = \delta_{mn} n!$ , so

$$|n\rangle = e_n(z) = \frac{z^n}{\sqrt{n!}} \quad (n \geq 0) \quad (13.11)$$

is an orthonormal family. It is complete: expanding an entire  $F$  in its Taylor series  $\sum c_n z^n$ , which converges uniformly on each circle, integrating first over the angle (where the cross terms vanish) and then over the radius by monotone convergence gives  $\|F\|^2 = \sum_n n! |c_n|^2$ , so  $F = \sum_n c_n \sqrt{n!} |n\rangle$  in norm and polynomials are dense. Conversely  $\sum n! |c_n|^2 < \infty$  implies that  $\sum c_n z^n$  is entire, by Cauchy–Schwarz with  $\sum |z|^{2n}/n! = e^{|z|^2}$ ; the same estimate gives

$$|F(z)| \leq \left( \sum_n n! |c_n|^2 \right)^{1/2} \left( \sum_n \frac{|z|^{2n}}{n!} \right)^{1/2} = \|F\|_{\mathcal{F}} e^{|z|^2/2}, \quad (13.12)$$

so norm convergence in  $\mathcal{F}$  implies locally uniform pointwise convergence. On the dense polynomial domain  $\mathcal{P}$  define

$$\begin{aligned} aF &= F', & a^\dagger F &= zF, & [a, a^\dagger] &= I, & [a^\dagger, a] &= -I, \\ a|n\rangle &= \sqrt{n} |n-1\rangle, & a^\dagger|n\rangle &= \sqrt{n+1} |n+1\rangle. \end{aligned} \quad (13.13)$$

The ladder formulas are differentiation and multiplication of  $z^n/\sqrt{n!}$ ; the commutator is  $(zF)' - zF' = F$ . The ladder formulas show  $\langle a^\dagger F, G \rangle = \langle F, aG \rangle$  for  $F, G \in \mathcal{P}$ , which justifies the notation  $\dagger$  (the maximal closed realizations are Hilbert-space adjoints of each other).

**Theorem 13.3** (Displacement operators, generator, and normal ordering). *For  $v \in \mathbb{C}$  the formula*

$$(D(v)F)(z) = e^{-|v|^2/2 + vz} F(z - \bar{v}) \quad (13.14)$$

*defines a unitary operator on  $\mathcal{F}$ . For fixed  $v$ ,  $t \mapsto D(tv)$  ( $t \in \mathbb{R}$ ) is a strongly continuous one-parameter unitary group whose skew-adjoint generator is the closure of the operator  $T_v = va^\dagger - \bar{v}a$  defined on  $\mathcal{P}$ ; in this precise sense*

$$D(v) = e^{va^\dagger - \bar{v}a} = e^{va^\dagger} e^{-\bar{v}a} e^{-|v|^2/2}, \quad (13.15)$$

*where the right-hand product is defined on  $\mathcal{P}$  (the middle factor by a finite sum, the left factor by a norm-convergent series) and extends by continuity to the unitary operator (13.14).*

*Proof. Unitarity.* Substitute  $z = w + \bar{v}$  in (13.10): using  $\operatorname{Re}(v\bar{v}) = |v|^2$  and  $|w + \bar{v}|^2 = |w|^2 + 2\operatorname{Re}(v\bar{w}) + |v|^2$ ,

$$|e^{-|v|^2/2 + vz}|^2 e^{-|z|^2} = e^{-|v|^2 + 2\operatorname{Re}(v\bar{w}) + 2|v|^2 - |w|^2 - 2\operatorname{Re}(v\bar{w}) - |v|^2} = e^{-|w|^2}.$$

Hence  $\|D(v)F\| = \|F\|$ , and  $D(v)F$  is entire, so  $D(v)$  is an isometry of  $\mathcal{F}$ . Direct substitution shows  $D(-v)D(v) = I = D(v)D(-v)$ , so  $D(v)$  is unitary.

*Group law and continuity.* For real  $s, t$ , applying (13.14) twice,  $(D(tv)D(sv)F)(z) = e^{-t^2|v|^2/2+tvz}e^{-s^2|v|^2/2+sv(z-t\bar{v})}F(z - (t+s)\bar{v}) = e^{-(t+s)^2|v|^2/2+(t+s)vz}F(z - (t+s)\bar{v})$ , since  $-t^2|v|^2/2 - s^2|v|^2/2 - ts|v|^2 = -(t+s)^2|v|^2/2$ ; thus  $D(tv)D(sv) = D((t+s)v)$ . For a polynomial  $F$  of degree  $m$  and  $|t| \leq 1$ , the integrand of  $\|D(tv)F - F\|^2$  is bounded by  $C(1+|z|)^{2m}e^{2|v||z|-|z|^2}$ , which is integrable, and converges pointwise to 0 as  $t \rightarrow 0$ ; dominated convergence gives  $\|D(tv)F - F\| \rightarrow 0$ . For general  $F$ , approximate by polynomials and use  $\|D(tv)\| = 1$ . By Stone's theorem the group has a skew-adjoint generator  $G$ .

*The exponential series on polynomials.* On the span of  $|0\rangle, \dots, |j\rangle$ , the ladder formulas give  $\|a\| \leq \sqrt{j}$  and  $\|a^\dagger\| \leq \sqrt{j+1}$ , and  $T_v$  maps this span into the span of  $|0\rangle, \dots, |j+1\rangle$  with norm at most  $|v|(\sqrt{j} + \sqrt{j+1}) \leq 2|v|\sqrt{j+1}$ . Iterating from a polynomial  $F$  of degree at most  $m$ ,

$$\|T_v^k F\| \leq (2|v|)^k \sqrt{\frac{(m+k)!}{m!}} \|F\|, \quad (13.16)$$

so  $\sum_k |t|^k \|T_v^k F\|/k!$  converges for every  $t$  (the ratio of consecutive terms is  $2|v||t|\sqrt{m+k+1}/(k+1) \rightarrow 0$ ). Define  $E(t)F = \sum_k t^k T_v^k F/k!$ , a norm-convergent series. We claim  $E(t)F = D(tv)F$ . As a differential operator on entire functions,  $T_v = vz - \bar{v}\partial_z$ . For a polynomial  $F$  and fixed  $z$ , direct differentiation of (13.14) gives

$$\begin{aligned} \frac{d}{dt}(D(tv)F)(z) &= e^{-t^2|v|^2/2+tvz}[(vz - t|v|^2)F(z - t\bar{v}) - \bar{v}F'(z - t\bar{v})] \\ &= e^{-t^2|v|^2/2+tvz}[v(z - t\bar{v})F(z - t\bar{v}) - \bar{v}F'(z - t\bar{v})] = (D(tv)(T_v F))(z). \end{aligned}$$

Since  $T_v F$  is again a polynomial, induction gives  $\frac{d^k}{dt^k}(D(tv)F)(z) = (D(tv)T_v^k F)(z)$ , and at  $t = 0$  this is  $(T_v^k F)(z)$ . The function  $t \mapsto (D(tv)F)(z)$  is entire in  $t$  (a product of entire functions), so it equals its Taylor series:  $(D(tv)F)(z) = \sum_k t^k (T_v^k F)(z)/k!$ . By (13.12), the norm-convergent series  $E(t)F$  converges pointwise to the same sum. Two elements of  $\mathcal{F}$  that agree pointwise are equal; hence  $E(t)F = D(tv)F$ .

*Identification of the generator.* From the norm-convergent power series,  $t \mapsto D(tv)F = E(t)F$  is norm-differentiable at  $t = 0$  with derivative  $T_v F$ ; thus  $\mathcal{P} \subseteq \text{Dom } G$  and  $G|_{\mathcal{P}} = T_v$ , that is  $T_v \subseteq G$ . As  $G$  is closed,  $T_v$  is closable with  $\overline{T_v} \subseteq G$ . We show that  $(I - T_v)\mathcal{P}$  is dense. If  $h \perp (I - T_v)\mathcal{P}$ , then  $\langle h, T_v F \rangle = \langle h, F \rangle$  for every polynomial  $F$ , and applying this to  $T_v^{k-1}F$  gives  $\langle h, T_v^k F \rangle = \langle h, F \rangle$  for all  $k$ . Therefore

$$\langle h, D(tv)F \rangle = \langle h, E(t)F \rangle = \sum_k \frac{t^k}{k!} \langle h, T_v^k F \rangle = e^t \langle h, F \rangle \quad (t \in \mathbb{R}),$$

and the left side is bounded by  $\|h\| \|F\|$  for all  $t$ , so letting  $t \rightarrow +\infty$  forces  $\langle h, F \rangle = 0$ ; density of  $\mathcal{P}$  gives  $h = 0$ . The same argument with  $e^{-t}$  and  $t \rightarrow -\infty$  shows that  $(I + T_v)\mathcal{P}$  is dense. Now  $\overline{T_v}$  is skew-symmetric, so  $\|(I \mp \overline{T_v})x\|^2 = \|x\|^2 + \|\overline{T_v}x\|^2$ ; hence  $I - \overline{T_v}$  is bounded below and, being closed, has closed range, which contains the dense set  $(I - T_v)\mathcal{P}$  and is therefore all of  $\mathcal{F}$ . Given  $y \in \text{Dom } G$ , choose  $x \in \text{Dom } \overline{T_v}$  with  $(I - \overline{T_v})x = (I - G)y$ ; since  $\overline{T_v} \subseteq G$  this reads  $(I - G)x = (I - G)y$ , and  $I - G$  is injective because  $G$  is skew-adjoint ( $\|(I - G)u\|^2 = \|u\|^2 + \|Gu\|^2$ ). Thus  $y = x \in \text{Dom } \overline{T_v}$ , proving  $\overline{T_v} = G$ : the closure of  $T_v$  is exactly the generator, with  $\mathcal{P}$  as a core, and  $D(v) = e^{\overline{T_v}}$ .

*Normal ordering.* For a polynomial  $F$ , the series  $e^{-\bar{v}a}F = \sum_k (-\bar{v})^k F^{(k)}/k!$  is a finite sum equal to  $F(z - \bar{v})$  by Taylor's formula for polynomials. The series  $e^{va^\dagger}G = \sum_k v^k z^k G/k!$  converges in  $\mathcal{F}$  for every polynomial  $G$ : for  $G = \sum_j g_j z^j$ ,  $\|z^k z^j\| = \sqrt{(k+j)!}$  and  $\sum_k |v|^k \sqrt{(k+j)!}/k! <$

$\infty$ , and its sum is the entire function  $e^{vz}G(z)$ , again by (13.12). Hence  $e^{va^\dagger}e^{-\bar{v}a}e^{-|v|^2/2}F = e^{-|v|^2/2}e^{vz}F(z - \bar{v}) = D(v)F$  on  $\mathcal{P}$ , and the bounded extension is  $D(v)$ .  $\square$

Although the ordered product in (13.15) has a unitary extension, its two nonscalar factors are separately unbounded; the identity does not assert that each is everywhere defined.

**Theorem 13.4** (Composition law and all matrix elements). *For  $u, v \in \mathbb{C}$ ,*

$$D(v)D(u) = \exp\left(\frac{v\bar{u} - u\bar{v}}{2}\right)D(v+u), \quad (13.17)$$

and for any finite list  $v_1, \dots, v_m$ ,

$$\prod_{j=1}^m D(v_j) = \exp\left(\frac{1}{2} \sum_{j < k} (v_j \bar{v}_k - v_k \bar{v}_j)\right) D\left(\sum_j v_j\right). \quad (13.18)$$

The full normally ordered expansion, its number-state matrix elements, and the coherent states are

$$D(v) = e^{-|v|^2/2} \sum_{r,s \geq 0} \frac{v^r (-\bar{v})^s}{r! s!} (a^\dagger)^r a^s \quad \text{on } \mathcal{P}, \quad (13.19)$$

$$\langle m | D(v) | n \rangle = e^{-|v|^2/2} \sqrt{m! n!} \sum_{k=0}^{\min(m,n)} \frac{v^{m-k} (-\bar{v})^{n-k}}{k! (m-k)! (n-k)!}, \quad (13.20)$$

$$|v\rangle := D(v)|0\rangle = e^{-|v|^2/2} \sum_{n \geq 0} \frac{v^n}{\sqrt{n!}} |n\rangle, \quad \langle u | v \rangle = \exp\left(-\frac{|u|^2 + |v|^2}{2} + \bar{u}v\right), \quad \| |v\rangle \| = 1. \quad (13.21)$$

For  $m \geq n$ , equivalently,

$$\langle m | D(v) | n \rangle = e^{-|v|^2/2} \sqrt{\frac{n!}{m!}} v^{m-n} L_n^{(m-n)}(|v|^2), \quad L_n^{(\alpha)}(x) := \sum_{j=0}^n \binom{n+\alpha}{n-j} \frac{(-x)^j}{j!}, \quad (13.22)$$

and the case  $n \geq m$  follows from  $D(v)^* = D(-v)$ .

*Proof.* Applying (13.14) twice,

$$\begin{aligned} (D(v)D(u)F)(z) &= e^{-|v|^2/2+ vz} e^{-|u|^2/2+ u(z-\bar{v})} F(z - \bar{v} - \bar{u}), \\ (D(v+u)F)(z) &= e^{-|v+u|^2/2+ (v+u)z} F(z - \bar{v} - \bar{u}). \end{aligned}$$

The difference of the exponents is  $-\frac{|v|^2}{2} - \frac{|u|^2}{2} - \bar{u}\bar{v} + \frac{|v|^2+|u|^2+v\bar{u}+u\bar{v}}{2} = \frac{v\bar{u}-u\bar{v}}{2}$ , which is purely imaginary (it equals  $i \operatorname{Im}(v\bar{u})$ ), proving (13.17); the factor is a phase. Induction on  $m$ , exactly as in theorem 9.3, gives (13.18): the new phase from  $D(v_1 + \dots + v_{m-1})D(v_m)$  is  $\frac{1}{2} \sum_{j < m} (v_j \bar{v}_m - v_m \bar{v}_j)$ .

Expanding the two exponentials in (13.15) on a polynomial (the  $a$ -series is finite and the  $a^\dagger$ -series converges in norm) gives (13.19). Acting on  $|n\rangle$ : the term  $a^s$  with  $s = n - k$  lowers  $|n\rangle$  to  $\sqrt{n!/k!} |k\rangle$ , and  $(a^\dagger)^r$  with  $r = m - k$  raises  $|k\rangle$  to  $\sqrt{m!/k!} |m\rangle$ ; the product of the ladder factors is  $\sqrt{m!n!}/k!$ , and summing over the admissible intermediate levels  $0 \leq k \leq \min(m, n)$  gives (13.20). Setting  $n = 0$  (only  $k = 0$ ) gives  $\langle m | D(v) | 0 \rangle = e^{-|v|^2/2} v^m / \sqrt{m!}$ , which is the first formula in (13.21); equivalently  $D(v)|0\rangle = e^{-|v|^2/2} e^{vz}$  by (13.14). Its norm squared is  $e^{-|v|^2} \sum |v|^{2n}/n! = 1$ , and  $\langle u | v \rangle = e^{-(|u|^2+|v|^2)/2} \sum_n \bar{u}^n v^n / n!$  gives the overlap. For (13.22) with  $m \geq n$ , substitute  $j = n - k$  in (13.20):  $v^{m-k} (-\bar{v})^{n-k} = v^{m-n} (-|v|^2)^j$  and

$$\begin{aligned} \sqrt{\frac{n!}{m!}} L_n^{(m-n)}(|v|^2) &= \sqrt{\frac{n!}{m!}} \sum_{j=0}^n \frac{m!}{(n-j)! (m-n+j)!} \frac{(-|v|^2)^j}{j!} \\ &= \sqrt{m! n!} \sum_{j=0}^n \frac{(-|v|^2)^j}{(n-j)! (m-n+j)! j!}, \end{aligned}$$

which matches (13.20) term by term with  $k = n - j$ . Since  $D(v)^* = D(v)^{-1} = D(-v)$ ,  $\langle m|D(v)|n\rangle = \langle n|D(-v)|m\rangle$  handles  $n \geq m$ .  $\square$

Algebraically,  $[T_v, T_u] = (v\bar{u} - u\bar{v})I$  on  $\mathcal{P}$ , which is central and purely imaginary, and all triple commutators vanish: this is the class-two nilpotent Heisenberg structure of (8.7) behind (13.17). The representation-level proof above establishes the identity globally, without an unjustified interchange of unbounded-operator series. Finitely many independent modes are handled by tensoring these representations: all cross-mode commutators vanish, every scalar product  $v\bar{u}$  in the phase formulas is replaced by  $\sum_\ell v_\ell \bar{u}_\ell$ , and the vacuum and matrix-element formulas factor by mode; no further BCH terms appear. These formulas provide all terms of the oscillator applications on the source page, with no asymptotic truncation remaining.

## 14 Exact computation, additional terms, and reproducibility

The all-order assertions of this article are proved algebraically or analytically; a finite computation is not their substitute. Computation is nevertheless useful for certifying transcription, signs, bracket conventions, and the additional explicit terms, and for detecting errors in the finite displays. This section describes the exact computations that accompany the article.

### 14.1 Representation and independently organized calculations

An associative polynomial is stored as a dictionary from words in  $\{X, Y\}$  to exact rational numbers (Python’s `fractions.Fraction`); the empty word represents 1. Multiplication concatenates words and discards degrees beyond a requested cutoff. Exponential and logarithm are evaluated by their defining series, which become finite under this truncation. Because no relation is imposed on the two generators, equality of the resulting dictionaries is equality in the truncated free associative algebra, not a test on selected numerical matrices in which independent commutators might collapse.

The three source treatments each supplied a standard-library Python verifier, and all three are retained unchanged in the repository ([paper-1/code](#), [paper-2/code](#), [paper-3/code](#)). Between them they perform the following independently organized calculations, each compared coefficient by coefficient with the reference expansion  $\log(e^X e^Y)$ :

- the finite word-cut formula (2.5) evaluated by a dynamic program over all contiguous cuts of every word, including words with zero coefficient (through degree 10; 2046 words);
- the Poincaré operator-block expansion (6.3), assembled by repeated application of  $e^{\text{ad}_X} e^{t \text{ad}_Y} - I$  with exact integration of powers of  $t$  (through degree 10);
- the right Dynkin projection (4.2), verified to fix every homogeneous component (through degree 12);
- the total-degree Bernoulli recursion (6.16) (through degree 10), the single- $Y$  Bernoulli series (6.5) (through degree 12), and the quadratic kernel (6.10) (through degree 10);
- the primitive coproduct condition, by explicit computation of the reduced unshuffle coproduct (through degree 10);
- the exchange parity (4.5) and word-reversal parity (4.6) (through degree 12), and three-letter associativity (4.7) (through degree 6);
- the displayed terms (4.9)–(4.14), entered independently as nested brackets and expanded by (4.18);

- the Zassenhaus exponents computed three ways, by residual extraction (10.3), by the Lie recurrence (10.7), and by the forward partition recurrence (10.14), compared with each other and with the explicit enumeration of the weighted-tree formula (10.16) (through degree 10), followed by reconstruction of the ordered product and comparison with  $e^{t(X+Y)}$  (through degree 12), and by the displayed  $C_2, C_3, C_4$ ;
- the ordered-simplex expansion (10.19) (through degree 10), the symmetric splitting’s oddness and cubic coefficient (11.4) (through degree 12), Campbell’s conjugation series (5.1), the Bernoulli inverse identity  $\phi\beta = 1$ , and the invariant-metric even-series identity (12.14) (through degree 10).

The supplied runs report 20, 30, and 26 passed named checks respectively; the script `code/run_all_verifiers.py` in the article’s directory reruns all three and records their reports. These are exact finite algebraic verifications, not floating-point tolerance tests and not a claim of formal proof-assistant certification. Finite-degree agreement does not prove an all-orders identity, an analytic convergence theorem, or a domain statement for an unbounded operator; those are established by the proofs above.

## 14.2 Two commutator encodings

The right-nested data use exactly  $R(w)$  of (1.4), so that  $Z_n = \sum_{|w|=n} \frac{c(w)}{n} R(w)$ . This encoding is redundant: every word ending in two equal letters represents zero, and the remaining brackets are not claimed to be linearly independent.

For compact tables we use the standard Lyndon bracketing. Order the alphabet by  $X < Y$  and words lexicographically, a proper prefix preceding its extensions. A nonempty word is *Lyndon* if it is strictly smaller than each of its nonempty proper suffixes. Define  $L(X) = X$ ,  $L(Y) = Y$ , and for a Lyndon word  $w$  of length at least two write  $w = uv$  with  $v$  its longest proper suffix that is Lyndon, and set

$$L(w) = [L(u), L(v)]. \quad (14.1)$$

(The standard factorization theorem guarantees that  $u$  is again Lyndon; the table generator applies the recursion to  $u$  and  $v$  in any case, so every entry is an explicitly specified bracket polynomial.) For example  $L(XXY) = [X, [X, Y]]$  and  $L(XYY) = [[X, Y], Y]$ . Thus  $L(XYY)$  is *not*  $R(XYY)$ , which is zero; confusing the two conventions would corrupt many coefficients. No theorem about Lyndon bases is needed to verify the tables: the generator expands each tabulated bracket polynomial into words and requires the result to equal the associative polynomial exactly. A table is therefore a certificate for an explicit finite commutator identity whether or not one invokes the general basis theorem. The counts in table 5 are numbers of nonzero entries in these encodings, not dimensions of homogeneous free Lie algebras and not counts of terms in a minimal basis.

## 14.3 Data files and regeneration

The principal machine-readable files in the article’s `data/` directory are listed in table 1. In an associative file a row gives the coefficient of the literal word; in a `lyndon` file it multiplies  $L(w)$ , and in the right-commutator file it multiplies  $R(w)$ . Numerators and denominators are separate integers, and missing rows have coefficient zero. The tree-polynomial file gives each  $C_n$  as a polynomial in the  $A_j$  of (10.13); a key such as `2 3` denotes the ordered product  $A_2 A_3$ .

The coefficient appendix (section B) is generated from these files by `code/make_tables.py`, which independently recomputes every printed degree-one to degree-six coefficient by the cut formula (2.5) and by expansion of the displayed brackets, reconstructs every printed Lyndon table from its brackets, and refuses to write the appendix if any comparison fails. From the article’s directory:

| File                             | Meaning                                                                                 |
|----------------------------------|-----------------------------------------------------------------------------------------|
| bch_associative.csv              | Every nonzero BCH word coefficient $c(w)$ through degree 12.                            |
| bch_right_commutators.csv        | Dynkin-projection coefficients $c(w)/n$ , with identically zero right brackets omitted. |
| bch_lyndon.csv                   | Compact BCH commutator expansion through degree 12.                                     |
| zassenhaus_associative.csv       | Associative coefficients of each $C_n$ , $2 \leq n \leq 12$ .                           |
| zassenhaus_lyndon.csv            | Compact commutator expansions of the same $C_n$ .                                       |
| zassenhaus_tree_polynomials.json | Each $C_n$ , $2 \leq n \leq 10$ , as a polynomial in the ordered $A_j$ .                |
| strang_lyndon.csv                | $\log(e^{X/2}e^Ye^{X/2})$ through degree 12.                                            |
| bernoulli_plus.csv               | The coefficients $b_n^+ = B_n^+/n!$ through $n = 12$ .                                  |

Table 1: Machine-readable companion expansions (exact rationals).

```
python code/make_tables.py # regenerate tex/appendix_tables.tex with checks
python code/run_all_verifiers.py # rerun the three exact verifiers
latexmk -pdf -interaction=nonstopmode -halt-on-error bch_combined.tex
```

Python 3.10 or later with the standard library suffices; a standard TeX installation with the packages named in the preamble compiles the source without a bibliography processor, shell escape, or external files. The verifiers accept larger cutoffs, but the number of words grows exponentially with the degree; the finite coefficient formulas and recurrences of this article, not the chosen degree limit, specify all orders.

## A The ordered Poincaré–Birkhoff–Witt theorem, enveloping Hopf algebras, and the completed tensor product

This appendix proves the injectivity statement used in [section 3.3](#), constructs the Hopf structure on an arbitrary enveloping algebra, and shows concretely why the coproduct of an infinite series needs a completed tensor product.

### A.1 The universal enveloping algebra and its ordered normal forms

Let  $\mathfrak{h}$  be a Lie algebra over  $\mathbb{K}$ . Its universal enveloping algebra is

$$U(\mathfrak{h}) = T(\mathfrak{h}) / \langle u \otimes v - v \otimes u - [u, v] : u, v \in \mathfrak{h} \rangle, \quad (\text{A.1})$$

the quotient of the tensor algebra on the vector space  $\mathfrak{h}$  by the two-sided ideal generated by the displayed relations. A linear map from  $\mathfrak{h}$  to an associative algebra  $A$  extends uniquely to an algebra homomorphism on  $T(\mathfrak{h})$ , and this extension annihilates the ideal exactly when the map preserves brackets (with  $A$  given its commutator bracket); this is the universal property. It remains to see that  $\mathfrak{h} \rightarrow U(\mathfrak{h})$  is injective.

**Theorem A.1** (Ordered PBW theorem). *Let  $(e_i)_{i \in I}$  be a totally ordered basis of  $\mathfrak{h}$ . The ordered monomials  $e_{i_1}e_{i_2} \cdots e_{i_m}$  with  $i_1 \leq i_2 \leq \cdots \leq i_m$ , together with the empty monomial 1, form a vector-space basis of  $U(\mathfrak{h})$ . In particular the natural map  $\mathfrak{h} \rightarrow U(\mathfrak{h})$  is injective.*

*Proof.* Work in  $T(\mathfrak{h})$  with the basis of words in the letters  $e_i$ . Consider the rewriting rule, applicable to any adjacent inverted pair  $j > i$  inside a word,

$$\cdots e_j e_i \cdots \mapsto \cdots e_i e_j \cdots + \cdots [e_j, e_i] \cdots, \quad (\text{A.2})$$

where  $[e_j, e_i]$  is expanded as a finite linear combination of basis letters. Assign to a word the pair (length, number of inversions), ordered lexicographically with length first. The swapped word has the same length and one inversion fewer, and every word in the bracket term is shorter. So each application of the rule replaces a word by a linear combination of words of strictly smaller measure. Since the measure takes values in a well-ordered set, every sequence of rewritings of a fixed word terminates, and the result is a linear combination of ordered words (words with no inverted adjacent pair). This proves that ordered words span  $U(\mathfrak{h})$ .

We prove that the final result does not depend on the choices made (confluence), by induction on the measure of the starting word  $w$ . If two different first steps are available for  $w$  at disjoint adjacent pairs, performing the other one next in either case leads to the same linear combination, whose words all have smaller measure than  $w$ ; by the induction hypothesis each of those words has a unique normal form, hence so does the combination, and both first choices lead to it. If the two available first steps overlap, the overlap is a factor  $e_k e_j e_i$  with  $k > j > i$  inside  $w = p e_k e_j e_i q$ . Reducing the left pair first and then continuing to order the three original letters gives, before any normalization of the shorter terms,

$$p(e_i e_j e_k + [e_j, e_i] e_k + e_j [e_k, e_i] + [e_k, e_j] e_i) q,$$

and reducing the right pair first gives

$$p(e_i e_j e_k + e_i [e_k, e_j] + [e_k, e_i] e_j + e_k [e_j, e_i]) q.$$

(Each line records the sequence of swaps that brings  $e_k e_j e_i$  to  $e_i e_j e_k$ , collecting the bracket terms produced.) Their difference is  $p \Xi q$  with

$$\Xi = [e_j, e_i] e_k - e_k [e_j, e_i] + e_j [e_k, e_i] - [e_k, e_i] e_j + [e_k, e_j] e_i - e_i [e_k, e_j].$$

Every word in  $\Xi$  has length two, hence smaller measure than  $e_k e_j e_i$ , and the same holds after attaching  $p$  and  $q$ . Applying the rule to each length-two product  $ab - ba$  with  $a, b \in \mathfrak{h}$  (which, expanded in the basis and reduced, has normal form equal to the normal form of  $[a, b]$ ; this is the content of (A.2) for a single pair, applied bilinearly) shows that  $p \Xi q$  has the same normal form as

$$p([e_j, e_i], e_k) + [e_j, [e_k, e_i]] + [[e_k, e_j], e_i] q = p([e_j, e_i], e_k) + [[e_i, e_k], e_j] + [[e_k, e_j], e_i] q = 0,$$

by antisymmetry and the Jacobi identity in  $\mathfrak{h}$ , where all normal forms of the shorter words involved are unique by the induction hypothesis. Hence the two first choices lead to combinations with the same normal form. Since every rule has length two, there are no other overlapping patterns. Thus any two complete reductions of  $w$  agree: compare their first steps as above, then apply the induction hypothesis to all words below  $w$ . Extending by linearity yields a well-defined *normal-form map*  $\nu : T(\mathfrak{h}) \rightarrow T(\mathfrak{h})$  whose image is the span of ordered words and which fixes ordered words.

Each rewriting step changes an element by an element of the ideal  $J$  in (A.1): it replaces  $p e_j e_i q$  by  $p(e_i e_j + [e_j, e_i]) q$ , and the difference is  $p(e_j e_i - e_i e_j - [e_j, e_i]) q \in J$ . Hence  $x - \nu(x) \in J$  for every  $x$ . Conversely, every generator  $r = uv - vu - [u, v]$  of  $J$ , multiplied on either side by words, has normal form zero: by bilinearity it suffices to take  $u = e_j$ ,  $v = e_i$ , and for  $j > i$  the two sides  $e_j e_i$  and  $e_i e_j + [e_j, e_i]$  of the rule have the same normal form by construction, while for  $j = i$  the relation is 0 and for  $j < i$  it is the negative of a rule instance. Thus  $\nu(J) = 0$ , and  $\nu$  descends to a linear map  $U(\mathfrak{h}) = T(\mathfrak{h})/J \rightarrow \text{span}(\text{ordered words})$  that is surjective (ordered words are fixed) and injective (if  $\nu(x) = 0$  then  $x = x - \nu(x) \in J$ ). Therefore the ordered words form a basis of  $U(\mathfrak{h})$ . Words of length one are ordered, so the basis vectors  $e_i$  remain linearly independent in  $U(\mathfrak{h})$ , which proves injectivity of  $\mathfrak{h} \rightarrow U(\mathfrak{h})$ .  $\square$

The proof uses only bilinearity, antisymmetry, and the Jacobi identity; it is independent of the BCH theorem and of any matrix realization, so its use in section 3.3 introduces no circularity.

## A.2 The free Lie algebra and the free associative algebra

Let  $F(X, Y)$  be the abstract free Lie algebra on two generators, the quotient of the free nonassociative algebra on bracketed words by the ideal generated by antisymmetry and Jacobi. A Lie homomorphism out of  $F(X, Y)$  is the same as a choice of two elements of the target: substitution defines the map, and the imposed relations are precisely what makes it well defined. Hence the universal property of  $U(F(X, Y))$  says that an associative homomorphism from it to any associative algebra is exactly a choice of two elements, which is also the universal property of the free associative algebra  $\mathbb{K}\langle X, Y \rangle = T(V)$ ; the mutually inverse homomorphisms exchanging the generators identify the two, as in the proof of [theorem 3.7](#). By [theorem A.1](#),  $F(X, Y)$  injects into  $U(F(X, Y)) \cong T(V)$ , and its image is the Lie subalgebra  $\mathcal{L}(X, Y)$  generated by  $X, Y$ . This identifies the concrete  $\mathcal{L}(X, Y)$  used throughout the article with the abstract free Lie algebra, and shows that an equality of bracket polynomials verified as an equality of words in  $T(V)$  is an identity in every Lie algebra.

## A.3 The enveloping Hopf structure

On the tensor algebra  $T(\mathfrak{h})$  set, for  $u \in \mathfrak{h}$ ,

$$\Delta u = u \otimes 1 + 1 \otimes u, \quad \varepsilon(u) = 0, \quad S(u) = -u, \quad (\text{A.3})$$

and extend  $\Delta, \varepsilon$  as algebra homomorphisms and  $S$  as an antihomomorphism. For a defining relation  $r = uv - vu - [u, v]$  of [\(A.1\)](#), direct expansion gives

$$\Delta r = (uv - vu) \otimes 1 + 1 \otimes (uv - vu) - [u, v] \otimes 1 - 1 \otimes [u, v] = r \otimes 1 + 1 \otimes r,$$

because the mixed terms  $u \otimes v + v \otimes u - v \otimes u - u \otimes v$  cancel; also  $\varepsilon(r) = 0$  and  $S(r) = vu - uv + [u, v] = -r$ . Hence  $\Delta(J) \subseteq J \otimes T(\mathfrak{h}) + T(\mathfrak{h}) \otimes J$ ,  $\varepsilon(J) = 0$ , and  $S(J) \subseteq J$ , so the three maps descend to  $U(\mathfrak{h})$ . Coassociativity and the counit identities hold on generators, hence on all of  $U(\mathfrak{h})$  by multiplicativity; the antipode identities  $m(S \otimes \text{id})\Delta = \eta\varepsilon = m(\text{id} \otimes S)\Delta$  hold on generators ( $-u + u = 0 = \varepsilon(u)$ ) and propagate to products by the inductive argument of [theorem 3.1](#), which used only that  $\Delta$  is multiplicative and  $S$  antimultiplicative. Thus  $U(\mathfrak{h})$  is a Hopf algebra, and under  $U(\mathcal{L}(X, Y)) \cong T(V)$  this is exactly the unshuffle Hopf algebra of [section 3](#); degree completion gives its formal power-series version. For a general  $\mathfrak{h}$ , tensor length is not a grading of  $U(\mathfrak{h})$ , since a bracket relation changes length; formal exponentials are then taken in  $U(\mathfrak{h})[[t]]$  with arguments  $tx, ty$ , which preserves every coefficient identity while avoiding an inappropriate grading.

## A.4 Why the coproduct needs a completed tensor product

Restrict to the one-generator subalgebra and consider  $F = \sum_{n \geq 0} X^n$ . By [\(3.2\)](#),

$$[X^p \otimes X^q] \Delta F = \binom{p+q}{p} \quad (p, q \geq 0),$$

since  $X^{p+q}$  has  $\binom{p+q}{p}$  subsets of positions of size  $p$ . The leading  $(N+1) \times (N+1)$  block of this coefficient matrix is  $C = ((\binom{p+q}{p})_{0 \leq p, q \leq N})$ . By Vandermonde's identity,  $\binom{p+q}{p} = \sum_k \binom{p}{k} \binom{q}{k}$  (compare coefficients of  $t^p$  in  $(1+t)^p(1+t)^q = (1+t)^{p+q}$ , using  $\binom{q}{p-k} = \binom{q}{q-p+k}$  and reindexing), so  $C = PP^\top$  with  $P_{pk} = \binom{p}{k}$  for  $k \leq p$  and  $P_{pk} = 0$  otherwise. Since  $P$  is lower triangular with diagonal entries 1,  $\det C = (\det P)^2 = 1$ , so  $C$  has rank  $N+1$  for every  $N$ . On the other hand, if  $\Delta F$  were a finite sum  $\sum_{j=1}^m A_j \otimes B_j$  of separated tensors with  $A_j, B_j \in \mathbb{K}[[X]]$ , its coefficient matrix would be  $\sum_j \alpha_j \beta_j^\top$  with  $\alpha_j, \beta_j$  the coefficient vectors of  $A_j, B_j$ , of rank at most  $m$ . Taking

$N \geq m$  gives a contradiction. Thus  $\Delta F$  does not lie in the algebraic tensor product  $\hat{T} \otimes \hat{T}$ , while it does lie in the completed tensor product used in section 3, whose elements are arbitrary families of coefficients indexed by pairs of words.

## B Coefficient certificates and additional complete degrees

Every table in this appendix was regenerated from the exact data files by `code/make_tables.py`, which recomputed each printed degree-one to degree-six coefficient by the finite cut formula (2.5), expanded the displayed brackets (4.9)–(4.14) and (10.8)–(10.10) by (4.18), reconstructed every Lyndon table below from its brackets, and expanded the tree polynomials of theorem 10.5 into words, comparing all results exactly before writing the file.

### B.1 All nonzero associative BCH coefficients through degree six

For each listed word  $w$  the table gives  $c(w)$  of (2.5); every unlisted word of degree at most six has coefficient zero, for both the cut formula and the expansion of (4.9)–(4.14). Among the  $2 + 4 + 8 + 16 + 32 + 64 = 126$  words there are 72 nonzero entries. This is a complete finite certificate for the displayed polynomial through degree six.

Table 2: All nonzero word coefficients  $c(w)$  of  $Z_1, \dots, Z_6$ .

| Word $w$        | $c(w)$            | Word $w$ | $c(w)$           | Word $w$ | $c(w)$           |
|-----------------|-------------------|----------|------------------|----------|------------------|
| <i>Degree 1</i> |                   |          |                  |          |                  |
| X               | 1                 | Y        | 1                |          |                  |
| <i>Degree 2</i> |                   |          |                  |          |                  |
| XY              | $\frac{1}{2}$     | YX       | $-\frac{1}{2}$   |          |                  |
| <i>Degree 3</i> |                   |          |                  |          |                  |
| XXY             | $\frac{1}{12}$    | XYY      | $\frac{1}{12}$   | YXY      | $-\frac{1}{6}$   |
| XYX             | $-\frac{1}{6}$    | YXX      | $\frac{1}{12}$   | YYX      | $\frac{1}{12}$   |
| <i>Degree 4</i> |                   |          |                  |          |                  |
| XXYY            | $\frac{1}{24}$    | YXYX     | $\frac{1}{12}$   |          |                  |
| XYXY            | $-\frac{1}{12}$   | YYXX     | $-\frac{1}{24}$  |          |                  |
| <i>Degree 5</i> |                   |          |                  |          |                  |
| XXXXY           | $-\frac{1}{720}$  | XYXYX    | $-\frac{1}{120}$ | YXYXY    | $\frac{1}{30}$   |
| XXXYY           | $\frac{1}{180}$   | XYXXY    | $-\frac{1}{120}$ | YXYXX    | $-\frac{1}{120}$ |
| XXYYX           | $-\frac{1}{120}$  | XYXXY    | $-\frac{1}{120}$ | YXYYY    | $\frac{1}{180}$  |
| XXYXX           | $-\frac{1}{120}$  | XYYYX    | $\frac{1}{180}$  | YYXXX    | $\frac{1}{180}$  |
| XXYXY           | $-\frac{1}{120}$  | XYYYY    | $-\frac{1}{720}$ | YYXXY    | $-\frac{1}{120}$ |
| XXYYX           | $-\frac{1}{120}$  | YXXXX    | $-\frac{1}{720}$ | YYXYX    | $-\frac{1}{120}$ |
| XXYYY           | $\frac{1}{180}$   | YXXXY    | $\frac{1}{180}$  | YYXYX    | $-\frac{1}{120}$ |
| XYXXX           | $\frac{1}{180}$   | YXXYX    | $-\frac{1}{120}$ | YYYXX    | $\frac{1}{180}$  |
| XYXXY           | $-\frac{1}{120}$  | YXXYY    | $-\frac{1}{120}$ | YYYXY    | $\frac{1}{180}$  |
| XYXYX           | $\frac{1}{30}$    | YXYXX    | $-\frac{1}{120}$ | YYYYX    | $-\frac{1}{720}$ |
| <i>Degree 6</i> |                   |          |                  |          |                  |
| XXXXXY          | $-\frac{1}{1440}$ | XYXYXY   | $\frac{1}{360}$  | YXYXYX   | $-\frac{1}{360}$ |
| XXXYYX          | $\frac{1}{360}$   | XYXXXY   | $-\frac{1}{240}$ | YYXXXX   | $\frac{1}{1440}$ |
| XXYYXY          | $\frac{1}{360}$   | XYXXYX   | $-\frac{1}{240}$ | YYXXXY   | $\frac{1}{240}$  |
| XXYXXY          | $-\frac{1}{240}$  | XYYYXY   | $\frac{1}{360}$  | YYXXXX   | $\frac{1}{240}$  |
| XXYXXY          | $-\frac{1}{240}$  | YXXXXX   | $-\frac{1}{360}$ | YYXYXX   | $\frac{1}{240}$  |
| XXYYXX          | $-\frac{1}{240}$  | YXXYXX   | $\frac{1}{240}$  | YYYXXX   | $-\frac{1}{360}$ |
| XXYYYX          | $-\frac{1}{1440}$ | YXXYYX   | $\frac{1}{240}$  | YYYXYX   | $-\frac{1}{360}$ |
| XYXXXY          | $\frac{1}{360}$   | YXYXXX   | $-\frac{1}{360}$ | YYYYXX   | $\frac{1}{1440}$ |

Table 2 (continued)

| Word $w$ | $c(w)$           | Word $w$ | $c(w)$          | Word $w$ | $c(w)$ |
|----------|------------------|----------|-----------------|----------|--------|
| YXXXXY   | $-\frac{1}{240}$ | YXYXXY   | $-\frac{1}{60}$ |          |        |
| XYXXXY   | $\frac{1}{60}$   | YXYXXY   | $\frac{1}{240}$ |          |        |

## B.2 Complete seventh and eighth BCH degrees

The entries specify exactly  $Z_n = \sum_w c_w L(w)$  for  $n = 7, 8$  in the standard Lyndon bracketing (14.1); there are no omitted nonzero terms in this convention. The CSV file continues these expansions through degree twelve.

Table 3: Complete BCH homogeneous polynomials of degrees seven and eight in Lyndon bracketing.

| $Z_7$    |                       | $Z_8$    |                       |
|----------|-----------------------|----------|-----------------------|
| $w$      | coefficient of $L(w)$ | $w$      | coefficient of $L(w)$ |
| XXXXXXY  | $\frac{1}{30240}$     | XXXXXXYY | $\frac{1}{60480}$     |
| XXXXXXYY | $-\frac{1}{5040}$     | XXXXXXXY | $-\frac{1}{15120}$    |
| XXXXXYXY | $\frac{1}{10080}$     | XXXXXYYY | $-\frac{1}{10080}$    |
| XXXXYXYX | $\frac{1}{3780}$      | XXXXYXXY | $\frac{1}{20160}$     |
| XXXXYXXY | $\frac{1}{10080}$     | XXXXYXYY | $-\frac{1}{20160}$    |
| XXXXYXYX | $\frac{1}{1680}$      | XXXXYXXY | $\frac{1}{2520}$      |
| XXXXYXYX | $\frac{1}{1260}$      | XXXXYYYY | $\frac{23}{120960}$   |
| XXXXYYYY | $\frac{1}{3780}$      | XXXXXXYY | $\frac{1}{4032}$      |
| XXYXXXY  | $\frac{1}{2016}$      | XXYXXXY  | $-\frac{1}{10080}$    |
| XXYXXXY  | $-\frac{1}{5040}$     | XXYXXYY  | $\frac{13}{30240}$    |
| XXYXXYY  | $\frac{13}{15120}$    | XXYXXXY  | $\frac{1}{20160}$     |
| XXYXXYY  | $\frac{1}{10080}$     | XXYXXYY  | $-\frac{1}{3024}$     |
| XXYXXXY  | $-\frac{1}{1512}$     | XXYXXYY  | $-\frac{1}{10080}$    |
| XXYXXYY  | $-\frac{1}{5040}$     | XXYXXYY  | $\frac{1}{2520}$      |
| XYXXXY   | $\frac{1}{1260}$      | XYXXYY   | $-\frac{1}{4032}$     |
| XYXXYY   | $-\frac{1}{2016}$     | XYXXYY   | $-\frac{1}{10080}$    |
| XYXXYY   | $-\frac{1}{5040}$     | XYXXYY   | $\frac{1}{60480}$     |
| XYXXYY   | $\frac{1}{30240}$     | —        | —                     |

## B.3 Complete fifth and sixth Zassenhaus exponents

These are the next two complete exponents after the terms displayed on the source page. A coefficient multiplies  $L(w)$ , not  $R(w)$ . Every entry follows from (10.3), from (10.7), and from (10.16); the three constructions agree exactly.

Table 4: Complete fifth and sixth Zassenhaus exponents in Lyndon bracketing.

| $C_5$ |                       | $C_6$  |                       |
|-------|-----------------------|--------|-----------------------|
| $w$   | coefficient of $L(w)$ | $w$    | coefficient of $L(w)$ |
| XXXXY | $\frac{1}{120}$       | XXXXXY | $-\frac{1}{720}$      |
| XXXYY | $-\frac{1}{30}$       | XXXXY  | $\frac{1}{144}$       |
| XXYXY | $-\frac{1}{60}$       | XXXYYY | $-\frac{1}{72}$       |
| XXYYY | $\frac{1}{20}$        | XXYYYY | $\frac{1}{72}$        |
| XYXY  | $-\frac{1}{20}$       | XYXYYY | $-\frac{1}{72}$       |
| XYYYY | $-\frac{1}{30}$       | XYYYYY | $-\frac{1}{144}$      |

#### B.4 Tree polynomials for the Zassenhaus exponents

Each  $C_n$  below is written as a polynomial in the ordered raw polynomials  $A_j$  of (10.13), obtained by summing the weights  $\omega(T)$  of theorem 10.5 over trees with the same leaf sequence. Products are noncommutative and are read from left to right;  $A_2^2 A_3$  and  $A_3 A_2^2$  are different polynomials.

| $n$ | $C_n$ in terms of the $A_j$                                                                                                                                                                                                                                                                                                                                                  |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2   | $A_2$                                                                                                                                                                                                                                                                                                                                                                        |
| 3   | $A_3$                                                                                                                                                                                                                                                                                                                                                                        |
| 4   | $A_4 - \frac{1}{2}A_2^2$                                                                                                                                                                                                                                                                                                                                                     |
| 5   | $A_5 - A_2 A_3$                                                                                                                                                                                                                                                                                                                                                              |
| 6   | $A_6 + \frac{1}{3}A_2^3 - A_2 A_4 - \frac{1}{2}A_3^2$                                                                                                                                                                                                                                                                                                                        |
| 7   | $A_7 + \frac{1}{2}A_2^2 A_3 - A_2 A_5 - A_3 A_4 + \frac{1}{2}A_3 A_2^2$                                                                                                                                                                                                                                                                                                      |
| 8   | $A_8 - \frac{1}{4}A_2^4 + \frac{3}{4}A_2^2 A_4 - A_2 A_6 - A_3 A_5 + A_3 A_2 A_3 - \frac{1}{2}A_4^2 + \frac{1}{4}A_4 A_2^2$                                                                                                                                                                                                                                                  |
| 9   | $A_9 - \frac{2}{3}A_2^3 A_3 + A_2^2 A_5 - A_2 A_7 + \frac{1}{3}A_3^3 - A_3 A_6 - \frac{1}{3}A_3 A_2^3 + A_3 A_2 A_4 - A_4 A_5 + A_4 A_2 A_3$                                                                                                                                                                                                                                 |
| 10  | $A_{10} + \frac{1}{5}A_2^5 - \frac{2}{3}A_2^3 A_4 - \frac{1}{4}A_2^2 A_3^2 + A_2^2 A_6 + \frac{1}{2}A_2 A_3 A_5 - \frac{1}{2}A_2 A_3 A_2 A_3 - A_2 A_8 + \frac{1}{2}A_3^2 A_4 - \frac{1}{4}A_3^2 A_2^2 - A_3 A_7 - \frac{1}{2}A_3 A_2^2 A_3 + A_3 A_2 A_5 - A_4 A_6 - \frac{1}{3}A_4 A_2^3 + A_4 A_2 A_4 + \frac{1}{2}A_4 A_3^2 - \frac{1}{2}A_5^2 + \frac{1}{2}A_5 A_2 A_3$ |

#### B.5 Sizes of the exported expansions

| Degree | BCH words | BCH Lyndon entries | $C_n$ words | $C_n$ Lyndon entries |
|--------|-----------|--------------------|-------------|----------------------|
| 1      | 2         | 2                  | 0           | 0                    |
| 2      | 2         | 1                  | 2           | 1                    |
| 3      | 6         | 2                  | 6           | 2                    |
| 4      | 4         | 1                  | 12          | 3                    |
| 5      | 30        | 6                  | 30          | 6                    |
| 6      | 28        | 5                  | 42          | 6                    |
| 7      | 126       | 18                 | 126         | 18                   |
| 8      | 124       | 17                 | 234         | 29                   |
| 9      | 390       | 55                 | 510         | 56                   |
| 10     | 388       | 55                 | 954         | 95                   |
| 11     | 2046      | 186                | 2046        | 186                  |
| 12     | 2044      | 185                | 3986        | 326                  |

Table 5: Numbers of nonzero coefficients in the supplied encodings. The degree-one Zassenhaus factor  $e^Y$  is separated from the exponents  $C_n$ ,  $n \geq 2$ .

## C Claim-by-claim coverage of the source page

The following concordance refers to revision 1368909113 of [1]. It tracks mathematical content rather than historical prose; repeated displays of the same identity are grouped, but different hypotheses and convergence interpretations are distinguished. An entry marked *corrected* or *qualified* gives the valid statement and identifies why the unqualified original cannot be asserted; it does not mean that a false unrestricted assertion has been proved. Theorems appearing only as names in links are not statements made by the page; the Trotter and higher-order splitting results are included as additional consequences.

Table 6: Claim-by-claim mathematical coverage.

| Source location              | Formula or assertion                                                                                                                                 | Proof or qualification here                                                                  |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Introduction                 | Exponential and logarithm series;<br>$e^X e^Y = e^Z$ with $Z$ a formal series of iterated commutators; leading terms.                                | <a href="#">Theorems 2.1, 2.2, 3.6 and 4.1</a> ; (4.9)–(4.11).                               |
| Introduction                 | Commutator interpretation, antisymmetry, Jacobi identity.                                                                                            | <a href="#">Section 1</a> , (1.3), by associative expansion.                                 |
| Introduction                 | Small elements give a convergent BCH series; the exponential gives coordinates near the identity; local multiplication is determined by the bracket. | <a href="#">Theorems 7.1 and 8.3</a> .                                                       |
| Introduction; explicit forms | Lie correspondence; integration of Lie algebra homomorphisms and representations.                                                                    | <a href="#">Theorems 8.4 and 8.5</a> ; simple connectedness stated as the global hypothesis. |
| Dynkin formula               | Complete block sum, factorial and total-degree denominators, right-nested convention.                                                                | <a href="#">Theorem 4.1</a> , (4.1), (1.4).                                                  |
| Dynkin formula               | Vanishing of summands whose final block ends in repeated letters.                                                                                    | Last assertion of <a href="#">theorem 4.1</a> , $[a, a] = 0$ .                               |
| Dynkin formula               | Every displayed BCH term through degree six.                                                                                                         | (4.9)–(4.14), <a href="#">theorem 4.3</a> , certificate <a href="#">table 2</a> .            |
| Dynkin formula               | All nonlinear terms lie in the ideal generated by $[X, Y]$ .                                                                                         | <a href="#">Theorem 3.6</a> .                                                                |
| Dynkin formula               | $Z(Y, X) = -Z(-X, -Y)$ ; homogeneous interchange signs.                                                                                              | <a href="#">Theorem 4.2</a> , (4.4), (4.5).                                                  |

Table 6 (continued)

| Source location                     | Formula or assertion                                                                                                              | Proof or qualification here                                                                      |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Integral form                       | Poincaré integral and the function $\psi(x) = x \log x / (x - 1)$ .                                                               | Theorem 6.2, (6.2), (5.14).                                                                      |
| Integral form; note                 | $\psi(e^y)$ and the Bernoulli coefficients with $B_1 = +\frac{1}{2}$ .                                                            | (5.15), theorem 5.3.                                                                             |
| Matrix illustration                 | Tangent Lie algebra, matrix commutator, matrix exponential.                                                                       | Opening of section 8.                                                                            |
| Matrix illustration                 | Full associative logarithmic block sum.                                                                                           | (2.3), theorem 2.2; absolute convergence in theorem 7.1.                                         |
| Matrix illustration                 | Associative terms $z_1, \dots, z_4$ and their expression by brackets.                                                             | (4.15)–(4.17), theorem 4.3.                                                                      |
| Matrix illustration; note           | Conditions $\ X\  + \ Y\  < \log 2$ and $\ Z\  < \log 2$ ; Hilbert–Schmidt norm.                                                  | Theorem 7.1; discussion after (7.6); submultiplicativity of the HS norm proved in section 7.     |
| Matrix illustration                 | Trace of the BCH logarithm; higher terms traceless.                                                                               | Theorem 7.2; <i>qualified</i> for other branches by (7.8).                                       |
| Convergence issues                  | The $\mathfrak{sl}_2(\mathbb{C})$ matrices, their product, and absence of a traceless logarithm; divergence of BCH.               | Theorem 7.3; all logarithms classified, even conditional convergence excluded.                   |
| Convergence issues                  | Nonisomorphic groups with the same Lie algebra.                                                                                   | The $\mathbb{R}$ versus $S^1$ example after theorem 8.3.                                         |
| Convergence issues                  | Bound $\ X\  + \ Y\  < \frac{1}{2} \log 2$ in any submultiplicative norm.                                                         | Theorem 7.1(iii), distinguished from the grouped bound.                                          |
| Special cases                       | Commuting variables.                                                                                                              | Theorem 9.1.                                                                                     |
| Special cases; Campbell application | Central commutator: truncation and disentangling without smallness.                                                               | Theorem 9.2, including the differential proof of global validity.                                |
| Special cases                       | Expansion linear in $Y$ , hyperbolic cotangent form, Bernoulli coefficients of $\text{ad}_X^n Y$ .                                | Theorem 6.4; all higher $Y$ -orders in (6.4), (6.7), (6.10).                                     |
| Special cases                       | $[X, Y] = sY$ : no terms with two or more $Y$ 's; closed form; excluded poles; removable $s = 0$ .                                | Theorem 9.4, theorem 9.5; Taylor convergence versus global resummation distinguished.            |
| Special cases                       | Eigenvector braiding and adjoint dilation.                                                                                        | (9.7), theorem 5.1.                                                                              |
| Existence results                   | Formal exp/log bijection; homogeneous grouping.                                                                                   | Theorems 2.1 and 2.2.                                                                            |
| Existence results                   | Coproduct on noncommuting series.                                                                                                 | (3.1), (3.2); <i>qualified</i> : completed tensor product, with the counterexample in section A. |
| Existence results                   | Exponential is group-like iff exponent is primitive; group-like elements form a group.                                            | Theorem 3.3, with constant term 1 in the definition.                                             |
| Existence results                   | Friedrichs' characterization of primitive elements; BCH follows.                                                                  | Theorems 3.4 to 3.6.                                                                             |
| Existence results                   | Free enveloping algebra is the free associative algebra; natural Hopf structure; validity in any characteristic-zero Lie algebra. | Theorem 3.7, section 3.3, section A.                                                             |
| Existence results                   | Existence of a recursive Lie-only construction.                                                                                   | Theorem 6.8, independent of the Hopf proof.                                                      |
| Zassenhaus formula                  | Ordered product; displayed $C_2, C_3, C_4$ including the left-nested $C_4$ .                                                      | Theorem 10.1, theorem 10.3.                                                                      |

Table 6 (continued)

| Source location      | Formula or assertion                                                                                    | Proof or qualification here                                                                                        |
|----------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Zassenhaus formula   | Compact product $e^{-tX}e^{t(X+Y)} = \prod e^{t^n C_n}$ .                                               | Convention $C_1 = Y$ , after theorem 10.1.                                                                         |
| Zassenhaus formula   | All later exponents are homogeneous Lie polynomials; recursive determination; the ellipsis.             | Theorem 10.1, theorem 10.2, theorem 10.5, table 4; convergence in theorem 10.7.                                    |
| Zassenhaus formula   | Suzuki–Trotter consequence.                                                                             | Section 11: theorems 11.2 to 11.4 with complete error series.                                                      |
| Campbell identity    | $\text{Ad}_{e^X} = e^{\text{ad}_X}$ , all iterated commutators, and the defining differential equation. | Theorem 5.1, (5.2).                                                                                                |
| Campbell identity    | General conjugation and braiding identities; central reduction.                                         | (5.3), (5.4), (9.1).                                                                                               |
| Campbell application | Differentiation of $e^{sX}e^{sY}$ and its solution.                                                     | Proof of theorem 9.2.                                                                                              |
| Infinitesimal case   | Full series for $e^{-X}de^X$ .                                                                          | (5.6), (12.2).                                                                                                     |
| Infinitesimal case   | Structure-constant expansion; matrices $M$ and $W$ ; quotient notation.                                 | (12.3)–(12.5); <i>qualified</i> : $(I - e^{-M})M^{-1}$ means $\phi(M)$ .                                           |
| Infinitesimal case   | Identification of $M$ as a vielbein.                                                                    | <i>Corrected</i> by theorem 12.1: $W$ is the coframe and $M$ is always singular.                                   |
| Infinitesimal case   | Pullback metric $g_{ij} = W^a_i W^b_j B_{ab}$ and the Killing form as metric.                           | Theorem 12.2, (12.11); <i>qualified</i> : Jacobian factors and nondegeneracy required; all orders in theorem 12.3. |
| Quantum mechanics    | $[Q, P] = i\hbar I$ and centrality.                                                                     | (13.1), (13.2) on the stated invariant core.                                                                       |
| Quantum mechanics    | Exponentiated canonical relation; exact symmetric three-factor form.                                    | Theorem 13.1; theorem 13.2 shows a bare dense-domain commutator is insufficient.                                   |
| Quantum mechanics    | $[a^\dagger, a] = -I$ ; normal ordering of the displacement operator.                                   | (13.13), theorem 13.3, with explicit domain and closure.                                                           |
| Quantum mechanics    | Product of two displacements, its phase, Heisenberg nilpotence.                                         | Theorem 13.4, (13.17), (8.7).                                                                                      |
| Quantum mechanics    | Expansions hidden in the displacement exponentials.                                                     | (13.19)–(13.22).                                                                                                   |

## Summary of essential qualifications

The tensor product in the formal existence proof must be completed, and group-like elements have constant term one. The trace identity refers to the BCH logarithm, not to an arbitrary branch. Smallness is a sufficient convergence condition, not a necessary one, and resummed validity of a closed form does not imply convergence of its Taylor series; at nonzero resonances  $2\pi i k$  the closed form of shape  $X + cY$  fails although a logarithm exists. In exponential coordinates  $M = \text{ad}_X$  is singular and is not the coframe;  $W = \phi(M)$  is the relevant matrix, and its inverse exists exactly off the resonances. Neither an arbitrary pullback nor the Killing form is automatically a metric. The alternate Zassenhaus product starting at  $n = 1$  needs  $C_1 = Y$ . Finally, an unbounded commutator identity requires domain and exponentiation hypotheses; concrete realizations, not formal substitution alone, justify the quantum formulas.

## Beyond the finite displays

Every BCH coefficient is given by (2.5) and (4.3); every bidegree by (6.4); every power of  $Y$  by (6.7) and (6.10); every Zassenhaus exponent by (10.16); every Maurer–Cartan component by

(12.4)–(12.5); every invariant-metric coefficient by (12.14); every splitting error by (2.8); and every oscillator matrix element by (13.20). These statements specify coefficients at arbitrary order; the accompanying tables are finite illustrations of unlimited formulas, not their claimed endpoint.

## References

- [1] Wikipedia contributors, *Baker–Campbell–Hausdorff formula*, English Wikipedia, revision 1368909113 (11 August 2026), consulted 16 September 2026. [Permanent revision 1368909113](#). This is the coverage target; every proof in the present article is supplied independently.
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